Topological Structural Optimization Framework for Solids of Revolution Applied to an Aeronautical Component

Fernando Londoño Palacio
flondonop@eafit.edu.co

Undergraduate thesis submitted in partial fulfillment of the requirements
for the degree of Engineering Physicist

Advisor
Ph.D. Ronald Mauricio Martinod Restrepo

Engineering Physics
School of Applied Sciences and Engineering
EAFIT University
Medellín, Colombia
2026

Contents

LIST OF FIGURES 3

LIST OF TABLES 5

Resumen 7

Abstract 8

Acknowledgments 9

1 Introduction 10
- 1.1 Context . . . 10
- 1.2 Problem Statement . . . 12
- 1.3 Justification . . . 12
- 1.4 Objectives . . . 13
  - 1.4.1 General Objective . . . 13
  - 1.4.2 Specific Objectives . . . 13
- 1.5 State of the Art . . . 14
  - 1.5.1 Density-Based Topology Optimization . . . 15
  - 1.5.2 Stress-Constrained Formulations . . . 15
  - 1.5.3 Optimization of Solids of Revolution . . . 15
  - 1.5.4 Topology Optimization with Manufacturing-Induced Anisotropy . . . 16
  - 1.5.5 Experimental Assessment of Optimized Printed Parts . . . 16
  - 1.5.6 Gap Addressed by This Work . . . 17
- 1.6 Scope . . . 17
- 1.7 Limitations . . . 17
- 1.8 Structure of the Document . . . 18

2 Theoretical Foundations 19
- 2.1 Notation and Conventions . . . 19
- 2.2 Physical Phenomena in Parts Made by Fused Filament Fabrication . . . 19
  - 2.2.1 Mesostructure and Origin of Anisotropy . . . 20
  - 2.2.2 Infill, Homogenization and Scale Separation . . . 20
  - 2.2.3 Other Phenomena and Their Treatment . . . 21
- 2.3 Linear Elasticity of Solids of Revolution . . . 21
  - 2.3.1 Kinematics and Axisymmetric Reduction . . . 21
  - 2.3.2 Boundary Conditions and Weak Form . . . 23

2.3.3 Strain Energy, Potential Energy and Compliance . . . 24
2.3.4 Displacement-Controlled Loading and Monotonicity . . . 25
2.4 Anisotropic Constitutive Theory . . . 25
2.4.1 Orthotropy and Transverse Isotropy . . . 25
2.4.2 Compatibility Between Material Symmetry and Axisymmetry . . . 26
2.4.3 Closed-Form Stiffness Matrix . . . 27
2.4.4 Admissibility of the Elastic Constants . . . 27
2.4.5 Estimation of the Constants Not Measured . . . 28
2.5 Strength Criteria and Stress Evaluation . . . 28
2.5.1 Yield Criteria . . . 28
2.5.2 Stress Singularities and the Percentile Measure . . . 29
2.6 Axisymmetric Finite Element Model . . . 30
2.6.1 Isoparametric Q4 Element and Element Stiffness . . . 30
2.6.2 Nodal Loads, Global System and Element Volumes . . . 31
2.7 Topology Optimization by the Density Method . . . 31
2.7.1 The Material Distribution Problem . . . 31
2.7.2 SIMP Interpolation . . . 32
2.7.3 Numerical Instabilities and Helmholtz Filtering . . . 32
2.7.4 Heaviside Projection . . . 34
2.7.5 Sensitivity Analysis by the Adjoint Method . . . 34
2.7.6 Optimality Criteria Method . . . 35
2.7.7 Local Constraints and Binary Designs . . . 35
2.8 Multi-Load and Multi-Objective Formulation . . . 36
2.9 Elastoplasticity and Limit Analysis . . . 36
2.9.1 Elastic-Perfectly Plastic Material . . . 37
2.9.2 Limit Loads by Elastic Compensation . . . 37
2.10 Geometric Relations Used in the Reconstruction . . . 38

3 Formulation of the Method **39**
3.1 Design Problem . . . 39
3.2 Constitutive Model of the Printed Material . . . 41
3.3 Axisymmetric Finite Element Model . . . 41
3.3.1 Meridian Section, Mesh and Element Volumes . . . 41
3.3.2 Element Stiffness and Solution . . . 42
3.4 Utilization Index . . . 43
3.5 SIMP Topology Optimization . . . 43
3.5.1 Filtering and Projection . . . 43
3.5.2 Sensitivities and Optimality Criteria Update . . . 45
3.5.3 Binarization and Engine Settings . . . 45
3.5.4 Benchmark of the Engine . . . 47
3.6 Search for the Minimum-Mass Design . . . 47
3.7 Geometric Reconstruction and STL Export . . . 48
3.7.1 Meridian Contours . . . 48
3.7.2 Revolution and Closed Meshes . . . 49
3.8 Virtual Compression Test . . . 49

3.8.1 Model and Solution . . . 50
3.8.2 Metrics . . . 50
3.9 Verification Procedures . . . 51

4 Application to the Selected Aeronautical Component **52**
4.1 Component Definition and Design Requirements . . . 52
4.1.1 Reference Component and Supports . . . 52
4.1.2 Design Load Cases . . . 53
4.1.3 Critical Dimensions and Admissible Displacements . . . 54
4.2 Additive Manufacturing Setup . . . 55
4.2.1 Printer, Modifications and Cost . . . 55
4.2.2 Materials and Printing Parameters . . . 55
4.3 Experimental Material Characterization . . . 56
4.3.1 Specimens and Tests . . . 56
4.3.2 Data Reduction and Apparent Density . . . 58
4.4 Material Constants . . . 60
4.5 Discretization of the Component . . . 61
4.6 Optimization Studies . . . 61
4.7 Experimental Validation with Full-Scale Sectors . . . 62
4.7.1 Purpose of the Test and Conversion of the Predictions . . . 62
4.7.2 Test Procedure and Data Reduction . . . 63
4.8 Summary of Modeling Assumptions . . . 63

5 Verification and Results **65**
5.1 Verification of the Implementation . . . 65
5.2 Mesh Study . . . 67
5.3 Benchmark of the Optimization Engine . . . 68
5.4 Response of the Original Component . . . 68
5.5 Minimum-Mass Designs . . . 70
5.6 Search on the Volume Fraction . . . 72
5.7 Comparison Between PA6 and PA6-CF . . . 73
5.8 Cross-Utilization . . . 75
5.9 Strength Checks . . . 76
5.10 Sensitivity to the Interlaminar Shear Modulus . . . 76
5.11 Virtual Compression Tests . . . 77
5.12 Predictions for the Sector Tests . . . 78
5.13 Experimental Results . . . 78
5.13.1 Stiffness and Relative Design Response . . . 80
5.13.2 Manufacturing Conditions and Scope of the Comparison . . . 81

6 Conclusions **83**
6.1 Limitations and Future Work . . . 84
6.1.1 Outputs of the Work . . . 86

List of Figures

1.1.1 CAD model of one 180° half of the aeronautical component. 11

2.2.1 Mesostructure (a) and transversely isotropic symmetry (b) of an FFF part, schematic. 21
2.3.1 Hoop strain (a) and radial resultant of the hoop stresses (b) in a solid of revolution, top views. 22
2.3.2 Axisymmetric reduction: solid of revolution (a) and two-dimensional model on the meridian section Ω (b). 23
2.3.3 Two-dimensional idealizations. Only the axisymmetric model keeps the hoop strain and hoop stress. 24
2.3.4 Kinematic restrictions and stiffness: ring sector (a), free sector (b) and bonded platens (c). 25
2.4.1 Printing orientation and axisymmetry: axis along the build direction (a) and part lying on its side (b). 27
2.5.1 Stress singularity (a) and distribution of element stresses with the 99.5th percentile (b), schematic. 29
2.6.1 Q4 element: physical element in the meridian plane and reference square in natural coordinates. 30
2.7.1 Schematic of density-based topology optimization on the half MBB beam. . . 32
2.7.2 SIMP interpolation: volume and stiffness factor as functions of the physical density. 33
2.7.3 Checkerboards without filtering (a) and filtered members with gray boundaries (b), schematic. 33
2.7.4 Filter kernels with equal variance (a) and Heaviside projection for increasing β, with $\eta_H = 0.5$ (b). 34
2.8.1 Mass-utilization trade-off and selected design, schematic. 36
2.9.1 Elastic-perfectly plastic material (a) and force-displacement curve under displacement control (b), schematic. 37

3.1.1 Workflow of the framework. Shaded boxes are physical or experimental stages, and white boxes are blocks of the notebook. 40
3.5.1 SIMP loop with axisymmetric Helmholtz filter, Heaviside projection with β-continuation and volume-weighted OC update. 44
3.6.1 Search for the minimum-mass feasible design along the volume fraction. . . . 48
3.7.1 Reconstruction of a printable solid from a binary design, schematic. 48
3.8.1 Boundary conditions of the virtual compression test. 50

4.1.1 Supports of the component on the meridian section. 53
4.1.2 Supports and loaded surfaces of the four design load cases on the meridian section. 54
4.2.1 Modified Flashforge Adventurer 5M used for all printed specimens and sectors. 55
4.3.1 PA6-CF tensile specimens (a) and compression prisms (b) after testing, and tensile specimen mounted in the testing machine (c). 58
4.3.2 Engineering stress-strain curves in tension (left) and compression (right). Dashed: premature fillet failures. 59
4.3.3 Weighing of a PA6 compression prism. 59
4.7.1 Top view of the ring with the tested 7.5° sector (a) and printed sectors (b). . 62
4.7.2 Test setup (a) (the figure is rotated 90 degrees) and sector after testing (b). . 64

5.2.1 Mesh refinement of the original solid under the axial reference load. Left: maximum von Mises stress and 99.5th percentiles weighted by element count and volume. Right: compliance. 67
5.3.1 Benchmark at $V_f = 0.60$: material layouts, displacement and stress fields, and ratios to the solid. The dotted threshold applies to the displacement ratio only. 69
5.4.1 Original component under the four load cases: normalized critical-wall displacement, normalized remaining-boundary displacement and total utilization. 70
5.5.1 Selected topologies in PA6 and PA6-CF. The dark band is the 2 mm passive shell, the dashed line is the original contour, and green marks the critical walls. 71
5.6.1 Ten volume-fraction searches. The dashed line marks $U = 1$. Lines connect the non-dominated evaluated points, while stars mark the lightest feasible designs found. 73
5.7.1 Calculated masses of the selected designs and savings relative to the original solid of each material. 74
5.7.2 PA6 and PA6-CF search results for each problem, using absolute mass in the upper row and mass normalized by each material's original solid in the lower row. 74
5.7.3 Constraint ratios of the selected designs in their governing load case. Critical-wall displacement governs all selections. 75
5.8.1 Cross-utilization of the optimized designs. Rows identify designs and columns identify evaluated load cases. Values above 1 violate a design limit. 75
5.13.1 Experimental force-displacement curves of the PA6 and PA6-CF sectors after shifting the displacement origin by the stiffness-fit intercept. Circles identify the measured maxima. 80
5.13.2 Measured stiffness against the sector prediction on logarithmic axes. The solid line indicates equality and dotted lines indicate constant ratios. The predicted upper bound presumes corresponding material, geometry and loading conditions. 81
5.13.3 PA6-CF prints showing moisture-related defects: (a) overview and (b) detail. The defective pieces were rejected before testing and were not used to determine the material properties. 82

List of Tables

1.5.1 Works closest to this thesis. A dash means not addressed. 14

2.2.1 Physical phenomena and their treatment in the model. 22

3.5.1 Parameters of the optimization engine and of the search for the minimum-mass design. 46
3.7.1 Angular discretization and purpose of the STL exports. 49

4.1.1 Design load cases and utilization of the solid component, from Table `T10_02`. . 53
4.2.1 Approximate cost of the manufacturing setup, in USD, consulted on 2026-10-28. 56
4.2.2 FFF printing parameters, common to PA6 and PA6-CF. 57
4.4.1 Material constants used in the model, from Table `T10_01`. 60
4.6.1 Optimization studies and tests carried out on the component. 62

5.1.1 Verification of the implementation. Relative quantities are dimensionless. . . . 66
5.4.1 Response of the solid component to the design load cases: largest displacement of the critical walls and of the remaining boundary, 99.5th-percentile von Mises stress, safety factor, utilization and governing limit (crit.: critical walls; rem.: remaining boundary) (F10_02; T10_03). 68
5.5.1 Minimum-mass feasible designs. Displacements and safety factor in the governing case, which is C4 for the multi-load designs and the design case for the single-case designs. 72
5.10.1 Governing utilization of the selected designs for different values of $\alpha_G = G_{rz}/E_z$. Values above 1 are infeasible. 76
5.11.1 Virtual compression test between rigid platens with bonded contact. Collapse values are indicative. 77
5.12.1 Predictions for the 7.5° sectors. K_{pred} is a conditional elastic upper bound, and F_{pred} is indicative. 78
5.13.1 Sector tests and predictions. Stiffness and force are given in kN/mm and kN. Displacements are shifted by the fitted intercept. One record is available per design. 79

Resumen

Los componentes aeronáuticos deben ser livianos sin perder desempeño mecánico. Muchos son sólidos de revolución dimensionados de forma conservadora, lo que deja material innecesario. Este trabajo busca la topología de mínima masa de un componente de este tipo, inspirado en una pieza del eje de una turbina y fabricado con filamento fundido (FFF).

Se desarrolló en Python un marco de optimización topológica basado en el método SIMP (*Solid Isotropic Material with Penalization*). Usa un modelo de elementos finitos axisimétrico de un material transversalmente isótropo, caracterizado sobre poliamida 6 (PA6) y PA6 reforzada con fibra de carbono (PA6-CF) impresas. Una búsqueda sobre la fracción de volumen encuentra el diseño más liviano que cumple las tolerancias geométricas y un factor de seguridad de von Mises de 1.5 bajo cuatro casos de carga. Sectores impresos se ensayaron a compresión.

Los diseños pesan 964.7 g en PA6 (35.7 % menos que la pieza maciza) y 537.0 g en PA6-CF (61.9 % menos). Los diseños de un solo caso violan los límites de los demás. Los ensayos confirmaron el orden de rigidez predicho para los diseños de PA6, pero midieron cerca de 3.5 veces la rigidez predicha. La humedad del filamento de PA6-CF impidió su validación.

Por lo tanto, los componentes con varios casos de carga requieren una formulación multicarga. Los ensayos sugieren además medir las propiedades a la escala de las paredes impresas delgadas y secar el PA6-CF antes de imprimir.

Su valor está en combinar SIMP axisimétrico, isotropía transversal y una búsqueda de mínima masa con límites geométricos, para diseñadores de piezas aeronáuticas livianas fabricadas por FFF.

Palabras clave: Optimización topológica, Método SIMP, Isotropía transversal, Fabricación con filamento fundido, Validación experimental.

Abstract

Aeronautical components must be light without losing mechanical performance. Many are solids of revolution sized conservatively, which leaves unnecessary material. This work seeks the minimum-mass topology of such a component, inspired by a turbine shaft part and made by Fused Filament Fabrication (FFF).

A topology optimization framework based on the Solid Isotropic Material with Penalization (SIMP) method was developed in Python. It uses an axisymmetric finite element model of a transversely isotropic material, characterized on printed polyamide 6 (PA6) and carbon-fiber-reinforced PA6 (PA6-CF). A search on the volume fraction finds the lightest design that meets the geometric tolerances and a von Mises safety factor of 1.5 under four load cases. Printed sectors were tested in compression.

The designs weigh 964.7 g in PA6 (35.7% less than the solid part) and 537.0 g in PA6-CF (61.9% less). Single-case designs violate the limits of the other cases. The tests confirmed the predicted stiffness ranking of the PA6 designs, but measured about 3.5 times the predicted stiffness. Moisture in the PA6-CF filament prevented its validation.

Components under several load cases therefore need a multi-load formulation. The tests also suggest measuring properties at the scale of thin printed walls and drying PA6-CF before printing.

Its value lies in combining axisymmetric SIMP, transverse isotropy and a minimum-mass search with geometric limits, for designers of lightweight aeronautical parts made by FFF.

Keywords: Topology optimization, SIMP method, Transverse isotropy, Fused filament fabrication, Experimental validation.

Acknowledgments

There are many people I would like to mention here, and I want each of you to know that the order of your names does not reflect your importance in my life or the gratitude I feel towards you. Each of you played a different role and contributed in ways you may never fully realize.

First, I would like to thank Mafe, my girlfriend, who, with so much love and patience, helped illuminate the darkness of the cave I had found myself in. Thank you for staying by my side, bringing light into uncertain moments, and walking with me on the difficult way out. Your presence made the hardest days easier to bear, and I am deeply grateful to have had you beside me.

To my best friends, Thomas and Juan Fernando, who went looking for me in that cave until they found me, even when I seemed lost beyond reach. Thank you for never giving up, for being there when I could not find my way back, and for reminding me that I was never truly alone. Having people willing to look for you, even when you have lost sight of yourself, is something I will never take for granted.

To Professor Ronald, thank you for not giving up on me and for giving me the chance to move forward when I finally climbed out of that pit. Your willingness to welcome me at such a difficult point made a difference beyond the academic scope of this project. I am sincerely grateful for your trust and for the opportunity to bring this work to completion.

To Professor Camilo, thank you for being unconditionally present throughout this process and supporting me until the very end. Your commitment, dedication, and genuine interest in this project were invaluable. I deeply appreciate all the time and effort you devoted to it.

Above all, I want you to know that each of you was essential in completing this degree project. Although this thesis bears my name, the journey that made it possible was never mine alone. Each of you helped me in a different way, at the moment I needed it most, and I will carry that gratitude with me long after this chapter of my life has ended.

Fernando Londoño Palacio
Medellín, 2026

Chapter 1

Introduction

1.1 Context

This document[1] is the author's undergraduate thesis in Physics Engineering. It develops a topology optimization framework for solids of revolution, based on the Solid Isotropic Material with Penalization (SIMP) method and implemented in Python. The framework is applied to the redesign of an aeronautical component made by Fused Filament Fabrication (FFF), a form of additive manufacturing.

Topology optimization[2] (TO) is a computational method for structural design. It decides where to place material within a design domain so that the structure performs its function as efficiently as possible [2, 3]. The problem seeks a material indicator field $\rho(\boldsymbol{x})$, equal to 1 where material is present and 0 where it is absent, that minimizes an objective functional Φ. The displacement field $\boldsymbol{u}(\rho)$ must satisfy the equations of elasticity, and a set of performance constraints must hold:

$$\begin{aligned} \min_{\rho} \quad & \Phi\big(\boldsymbol{u}(\rho), \rho\big) \\ \text{subject to} \quad & \mathcal{L}(\rho)\,\boldsymbol{u} = \boldsymbol{f} \quad \text{in } \Omega, \\ & G_j\big(\boldsymbol{u}(\rho), \rho\big) \le 0, \quad j = 1, \dots, n_g, \\ & \rho(\boldsymbol{x}) \in \{0, 1\} \quad \forall\, \boldsymbol{x} \in \Omega, \end{aligned} \tag{1.1}$$

where Ω is the design domain, $\mathcal{L}(\rho)$ is the elasticity operator for the material layout, $\boldsymbol{f}$ is the applied load, and G_j are n_g constraints on quantities such as volume, stress, or displacement. The number and location of holes are not fixed in advance, so TO can reach layouts that a designer would not propose by intuition.

[1] This is a hyper-document with embedded links. Reading the digital version is recommended. The project repository is available at github.com/Progyt08/to-for-aeronautical-component.

[2] Structural optimization problems fall into three families [1]. In *sizing* optimization, the design variables are dimensions such as thicknesses or cross-sectional areas. In *shape* optimization, the boundary of a domain with fixed connectivity can move. In *topology* optimization, the number, location, and shape of holes, and therefore the connectivity of the domain, are also unknown. This work refers only to the last family.

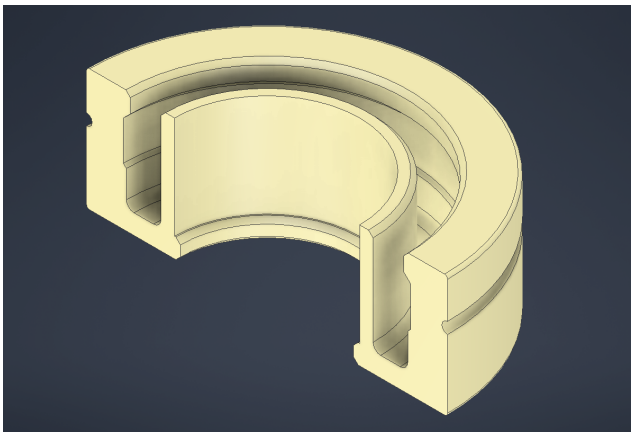


Figure 1.1.1: CAD model of one 180° half of the aeronautical component.

Here, this problem is posed for the component shown in Figure 1.1.1. It is a polyamide solid of revolution inspired by a turbine shaft part. It is a ring with a U-shaped meridian section, a central bore and a radial cavity, made of two identical 180° halves that are modeled as one continuous ring. Its geometry, material, and loads do not vary with the circumferential angle. Under this symmetry, the three-dimensional problem reduces to a two-dimensional one on the meridian plane. The part does not become a plane structure. Each point of the meridian section represents a ring of radius r, so the hoop strain u_r/r and the factor $2\pi r$ in the volume integrals preserve its three-dimensional behavior, as formulated in Chapter 2.

Several TO approaches exist. They include the homogenization method of Bendsøe and Kikuchi [2], density-based methods [4], evolutionary structural optimization [5], and level-set methods [6, 7]. Comparative reviews are given in [8, 9]. This work uses SIMP[3]. It assigns a continuous density to each finite element and penalizes intermediate values. This turns a combinatorial problem into a gradient-based one that couples directly with the finite element method [10].

Additive manufacturing (AM) can produce the complex geometries that TO generates [11, 12]. FFF builds the part from extruded beads stacked in layers. The bond between layers is weaker and more compliant than the material within each layer, so FFF parts are stiffer and stronger in the layer plane than across it [13, 14]. A design method for such parts must model this anisotropy. In the coupons tested here, the in-plane modulus, measured in tension, is 2.9 times (PA6) and 3.0 times (PA6-CF) the out-of-plane modulus, measured in compression across the layers, as reported in Chapter 4.

This work combines these elements. The meridian section of the component is optimized with an axisymmetric finite element model. The printed material is modeled as transversely isotropic, with moduli measured on printed coupons of polyamide 6 (PA6) and of polyamide 6 reinforced with short carbon fibers (PA6-CF). For each material, the framework searches for the lightest design that keeps the critical surfaces within the drawing tolerances and the stresses within an admissible safety factor under four static load cases. These limits are

[3]Here, SIMP designates both the power-law interpolation scheme and the optimization method built on it.

checked on the binary design of each optimization run. The designs are verified numerically and exported as printable geometry. Full-scale angular sectors, printed on a modified low-cost desktop printer, are compressed between flat platens and compared with the predictions. The main contributions are:

- An axisymmetric SIMP implementation in Python with a transversely isotropic constitutive matrix, an axisymmetric Helmholtz filter, and a volume constraint weighted by the volume of revolution of each element.
- A minimum-mass search on the volume fraction under displacement and stress limits in four load cases, with a cross-utilization analysis that evaluates the design for each load case under all the others.
- A complete workflow from the STL file of the component to closed STL files of the optimized rings and sectors, with independent verification and a virtual compression test whose initial stiffness is compared with the physical tests.

1.2 Problem Statement

Removing mass while keeping mechanical performance is a main driver of aerospace engineering [15, 16]. Many aeronautical components are solids of revolution, such as bushings, spacers, housings, and shaft parts. They are usually produced by turning and sized with conservative criteria, which leaves excess material in the part. Turning also limits the geometry to profiles a cutting tool can reach, so internal cavities and complex layouts fall outside the design space [11, 12]. Sizing and shape optimization can refine such a design, but only within the topology first proposed [17].

TO can remove this excess by placing material only where the loads require it [2, 3], and AM can produce the resulting geometries [11]. Two obstacles remain for this class of components. First, TO has been developed mostly for general two- and three-dimensional domains [8, 9]. Its axisymmetric formulations, including recent ones for AM, target turbomachinery disks [18, 19]. Second, FFF produces anisotropic parts with weak interlayer bonds [13, 14]. A design based on an isotropic material may therefore not predict the response of the printed part [20]. Few studies combine TO, numerical verification, and physical tests of such parts [11, 21]. This work therefore addresses the following question:

> *How much can a SIMP-based topology optimization framework reduce the mass of an axisymmetric aeronautical component made by FFF, while keeping its displacements within the drawing tolerances and its von Mises stress within the admissible safety factor in every load case?*

1.3 Justification

The problem matters because mass drives the operating cost and emissions of an aircraft. Less mass requires less lift and thrust, and therefore less fuel [15]. Fuel is a large share of the operating cost, so weight reduction is one of the main goals of the aerospace industry [22]. Removing the excess material of conventionally designed parts reduces both cost and

environmental impact.

SIMP is used because density methods are the most mature and robust family of TO methods [8], and SIMP is the most widespread among them owing to its simplicity [9]. Its continuous element densities couple directly with a finite element model and lead to compact codes that are easy to verify [10, 23]. The same power law can scale a whole anisotropic constitutive matrix, so the transversely isotropic model of the printed material enters without changing the method. Stress-constrained extensions are also available if strength becomes governing [24, 25].

This work is important because it fills the gap identified in Section 1.5. The axisymmetric reduction keeps the three-dimensional behavior of the part with a two-dimensional model [26]. The complete component has 77 309 free degrees of freedom, and one SIMP iteration with the four load cases takes between 0.4 and 1.2 s, as verified in Section 5.1, which allows wide searches on the volume fraction. Modeling FFF anisotropy makes the method usable with a widely adopted AM process [13]. The modified desktop printer used here costs roughly 13 to 19 times less than a commercial printer marketed for fiber-reinforced polyamides, according to the cost analysis in Section 4.2.1. Finally, physical tests are needed because predicted and measured responses of optimized AM parts often differ [20, 27].

1.4 Objectives

1.4.1 General Objective

Develop a SIMP topology optimization framework implemented in Python for an axisymmetric aeronautical component, integrating numerical simulation and experimental assessment of mechanical performance for optimized designs intended for additive manufacturing.

1.4.2 Specific Objectives

1. Implement a density-based SIMP (Solid Isotropic Material with Penalization) framework for axisymmetric domains in Python, defining the minimization of mass as the objective function subject to global stiffness and stress constraints.
2. Characterize the original aeronautical component through numerical simulation (FEA) to establish baseline performance metrics, including mass, maximum displacement, and Von Mises stress distribution.
3. Validate optimized designs via Finite Element Method (FEM) using the Von Mises yield criterion to ensure structural integrity and verify that maximum displacements remain within the defined acceptance threshold relative to the original component.
4. Fabricate optimized Nylon prototypes using additive manufacturing, selecting designs through an efficiency index that balances mass reduction with uniform stress distribution and ensuring remaining displacements within defined acceptance thresholds.
5. Verify the topology optimization framework's reliability by comparing numerical predic-

tions with strain-stress experimental results, analyzing the influence of manufacturing-related effects on the mechanical performance of the optimized component.

1.5 State of the Art

Related work falls into five lines: density-based topology optimization, stress-constrained formulations, optimization of solids of revolution, TO with AM-induced anisotropy, and experimental assessment of optimized printed parts. Table 1.5.1 summarizes the works closest to this thesis.

Table 1.5.1: Works closest to this thesis. A dash means not addressed.

Work	Structure	Material model	Manufacturing	Physical tests
[28]	Solid of revolution, shape	Isotropic elastic	-	-
[18]	Axisymmetric solids, turbine disk	Isotropic elastic	-	-
[19]	Axisymmetric turbine disks	Thermoelastic	AM, self-supporting cavities	-
[29]	Rotary parts, compressor disk	Anisotropic, Tsai-Hill	Multi-axis AM, build direction	-
[30]	Axisymmetric aero-engine disks	Anisotropic, Tsai-Wu	AM-induced anisotropy	-
[31]	General	Transversely isotropic, SAMP	Build direction, SLA	-
[32]	General	Transversely isotropic	Build direction	-
[33]	2D	Orthotropic, SOMP	FDM deposition paths	-
[34]	General	Transversely isotropic and isotropic phases, Tsai-Wu	Material extrusion	Stiffness and strength tests
[35]	Beam	Isotropic	3D printing	Three-point bending
[21]	Flange test structure	Isotropic	PBF, PA2200	Mechanical test, DIC
[27]	Functional parts	Isotropic	SLM and DED, SS316L	Tension and compression zones
[20]	Beam with lattice	Isotropic	FDM, PLA	Three-point bending
This work	Solid of revolution, axisymmetric	Transversely isotropic, coupon moduli	FFF, PA6 and PA6-CF	Compression of 7.5° sectors

1.5.1 Density-Based Topology Optimization

Bendsøe and Kikuchi [2] founded TO of continua. They turned shape design into a material distribution problem on a fixed domain, solved by homogenization and without remeshing. Bendsøe [4] then replaced the microstructure with a continuous density and a power-law stiffness interpolation that penalizes intermediate densities. This scheme became known as SIMP.

The density approach brought numerical issues. Sigmund and Petersson [36] reviewed checkerboards, mesh dependence, and local minima, and the filters and continuation methods that control them. Wang et al. [37] added a smoothed Heaviside projection and showed how robust formulations impose a minimum length scale. Compact educational codes made the method widely accessible [10, 23].

Comparative reviews rate density methods as the most mature and robust family [8]. A 3D benchmark of SIMP, level-set, BESO, and VARTOP found that no method dominates in every criterion, while SIMP remains the most widespread [9]. Recent reviews cover nonlinear extensions [3] and machine-learning acceleration [38].

1.5.2 Stress-Constrained Formulations

Stress constraints raise three difficulties: singular optima, one constraint per element, and strong nonlinearity. Cheng and Guo [39] removed singular optima in trusses with an ε-relaxation. Duysinx and Bendsøe [24] extended stress constraints to SIMP continua with a density-scaled local stress and the same relaxation. Le et al. [25] grouped local stresses into regional p-norm measures with adaptive normalization. Their designs round off the re-entrant corners that concentrate stress in minimum-compliance layouts. Such constraints are usually handled with the method of moving asymptotes [40]. SIMP also extends to damage criteria, where it converges better than BESO from a homogeneous initial design [41]. This work does not impose stress constraints inside the optimizer. It checks a percentile von Mises stress on the binary designs, so these formulations are the natural extension if strength becomes governing.

1.5.3 Optimization of Solids of Revolution

Axisymmetric finite elements reduce a solid of revolution to its meridian section while keeping the hoop strain [26]. Structural optimization of such bodies began with shape optimization. Chun and Haug [28] minimized the weight of an elastic solid of revolution under stress and displacement limits by moving its boundary. Liu, Parks and Clarkson [18] extended the problem to topology and shape. Their metamorphic development method grows and degenerates simple axisymmetric structures to minimize mass. It handles self-weight, centrifugal, and thermal loads and was applied to a turbine disk.

Recent work brings AM into this line. Wang et al. [19] combined stress constraints with self-supporting enclosed cavities in axisymmetric turbine disks under thermomechanical loads. Yan et al. [29] optimized the build direction and the topology of rotary parts for multi-axis AM, with a Tsai-Hill strength constraint. Guo et al. [30] built an anisotropic material model

for axisymmetric structures and minimized the mass of aero-engine disks under a Tsai-Wu strength constraint. They report mass reductions above 20% relative to designs based on the von Mises criterion. These works address turbomachinery parts, and none models a polymer printed by FFF with constants measured on printed coupons.

1.5.4 Topology Optimization with Manufacturing-Induced Anisotropy

AM is the natural partner of TO because it can build the complex layouts that TO produces [11]. Early AM-oriented formulations addressed geometric limits. Langelaar [42] added a layer-by-layer filter that yields self-supporting 3D designs at a small cost in compliance.

Material anisotropy is a second limit. MJF and SLS give PA12 parts with comparable properties [43, 44]. FFF parts are clearly anisotropic because of their weak interlayer bonds [13], and their layers behave as orthotropic laminae [14].

Several works include anisotropy in the optimization. Li et al. [31] coupled a transversely isotropic model of SLA material with the build direction through a SAMP interpolation. Wang et al. [32] made the build direction angle a design variable of a transversely isotropic model. For FDM, Xu et al. [33] used a SOMP model with contour paths at the boundary and zig-zag paths in the substrate. Yu et al. [45] split each layer into patches and optimized the topology, the patch layout, and the deposition directions together. Kundu and Zhang [34] combined transversely isotropic and isotropic infill phases with Tsai-Wu and von Mises constraints and tested the designs made by material extrusion. They note that most anisotropic TO studies for AM consider constitutive anisotropy but not anisotropic strength. Shin et al. [46] optimized density and build orientation together for multi-axis AM and showed that path-dependent anisotropy changes the optimal design. All of these works use general 2D or 3D domains.

1.5.5 Experimental Assessment of Optimized Printed Parts

Few studies combine optimization, numerical verification, and physical tests of AM prototypes [11, 21]. Grinde [35] optimized a beam with SIMP, printed the designs, and compared their load-deflection curves in three-point bending. Kyun Sung et al. [21] printed a flange-type test structure in PA2200 and measured it with a universal testing machine and digital image correlation (DIC). Simulated and printed responses differed because of AM fidelity.

Rankouhi et al. [27] tested optimized SS316L parts made by SLM and DED and found the finite element predictions conservative. They traced the differences to process-dependent microstructure. Bastos Júnior et al. [20] optimized a PLA beam, printed it by FDM, and tested it in three-point bending. Simulation and test agreed in the elastic range but diverged after yield, which the authors attributed to printing-induced anisotropy, geometric imperfections, and a simplified constitutive model. Full-field DIC maps can be compared with simulations through moment-based validation metrics [47], and performance indices rank optimized topologies by mass and stress [48, 49].

1.5.6 Gap Addressed by This Work

Axisymmetric TO now includes AM constraints and anisotropic strength, but only for turbomachinery parts, none of them printed by FFF. Anisotropic TO for FFF has been developed for general 2D and 3D domains. Experimental studies of optimized printed parts mostly use isotropic material models. No reviewed work combines an axisymmetric reduction, a transversely isotropic model built from moduli measured on FFF coupons, a minimum-mass search with displacement tolerances and a stress limit under several load cases, and physical tests of the printed designs. This combination is the subject of the present work.

1.6 Scope

This work develops a Python SIMP framework for axisymmetric domains and applies it to a representative aeronautical component in PA6 and PA6-CF. It seeks minimum-mass designs under four static load cases, using nodal-displacement limits motivated by the drawing and a von Mises safety factor based on the 99.5th stress percentile.

The scope includes coupon stress–strain characterization, finite element characterization of the original component, numerical verification, virtual compression, closed STL export, and fabrication and testing of full-scale 7.5° sectors. The efficiency index evaluates the benchmark; prototype selection uses minimum mass subject to $U \leq 1$. Sector force–displacement tests provide a partial experimental assessment of predicted stiffness.

The work does not address general three-dimensional or non-axisymmetric domains, nor dynamic, fatigue, thermal, creep, or moisture effects. It considers no materials other than the two polyamides and no manufacturing process other than FFF, so no machined prototypes are made for comparison. Other TO methods, such as BESO or level-set approaches, and multi-material or lattice optimization are outside its scope. The complete rings are exported but not printed or tested, and no certification-level qualification for aeronautical service is attempted.

1.7 Limitations

The results of this work must be read within the assumptions listed in Section 4.8, which Chapter 5 discusses with each result.

The component is inspired by an aeronautical part but does not reproduce it exactly. Its drawing gives no service loads or mounting conditions. The supports, a sliding base and a clamped outer wall, and the four static load cases are therefore assumed, and the designs are optimal only for them. The two 180° halves are modeled as one continuous ring whose joint transmits the hoop forces. The whole tolerance band of the drawing is assigned to elastic deformation, so manufacturing deviations are not included.

The printed material is described by a homogenized, linear-elastic, and transversely isotropic model. Its two Young's moduli, its yield strengths, and its apparent density were measured on printed coupons. The Poisson's ratios and the interlaminar shear modulus were estimated,

and the designs are sensitive to the latter, as shown in Section 5.10. The same moduli are used in tension and compression, and only one in-plane direction was tested. Thin members printed with the five perimeters may be stiffer and denser than the homogenized material.

The strength check uses the isotropic von Mises criterion on the 99.5th percentile of the element stresses. Direction-independent criteria may not suit anisotropic printed parts [34], and the interlaminar tensile and shear strengths were not characterized.

The displacement and stress limits are checked on the binary designs rather than imposed inside the optimizer. The problem is non-convex, so the selected designs are the lightest feasible designs found, not proven global optima. Several of them are within 1% of the feasibility limit. The formulation includes no overhang, support, or enclosed-cavity constraint. In a complete ring, the internal voids form closed annular cavities from which support material could not be removed.

The virtual compression test is geometrically linear and has no contact formulation, so only its initial stiffness is used quantitatively. The prototypes are full-scale 7.5° sectors with free cut faces. The converted prediction is an elastic upper bound conditional on equivalent geometry, material properties and prescribed loading. The tests assess structural response and relative design trends under platen compression rather than service conditions.

1.8 Structure of the Document

Chapter 2 presents the physical and mathematical foundations of the work. It covers the mechanics of parts made by FFF, the linear elasticity of solids of revolution, anisotropic constitutive theory, strength criteria, the finite element method, density-based topology optimization, multi-objective optimization, and limit analysis. Chapter 3 formulates the design framework in general terms and its Python implementation and the procedures used to verify it.

Chapter 4 applies the framework to the component. It defines the supports, the load cases, and the admissible displacements, describes the modified printer and the characterization of the two materials, and presents the optimization studies and the sector tests. Chapter 5 verifies the implementation and reports the response of the original component, the minimum-mass designs, the comparison between materials, the cross-utilization, the sensitivity to the estimated constants, and the virtual and physical compression tests.

Chapter 2

Theoretical Foundations

This chapter builds the theory behind the framework, from the printed material to the optimized geometry. The framework itself is formulated in Chapter 3, implemented in Chapter **??** and applied in Chapter 4. The chapter starts with the phenomena that make parts printed by Fused Filament Fabrication (FFF) anisotropic, which are covered in Section 2.2. It then develops the elasticity of solids of revolution in Section 2.3 and the transversely isotropic material model in Section 2.4, and it defines how stresses are judged in Section 2.5. With these ingredients, Section 2.6 builds the axisymmetric finite element model, and Sections 2.7 and 2.8 use it for density-based topology optimization under several load cases. Finally, Section 2.9 covers the limit analysis of the virtual test, and Section 2.10 the geometric relations used to turn the designs into printable solids.

2.1 Notation and Conventions

A cylindrical coordinate system (r, θ, z) is used. The axis z is the axis of revolution of the component and the build direction of the printer. Bold lowercase symbols denote vectors, bold uppercase symbols denote matrices, and italic symbols are scalars. Stresses and strains are written in Voigt notation [50, 51]. In the axisymmetric problem the components are ordered as

$$\boldsymbol{\varepsilon} = [\varepsilon_r,\ \varepsilon_z,\ \varepsilon_\theta,\ \gamma_{rz}]^\top, \qquad \boldsymbol{\sigma} = [\sigma_r,\ \sigma_z,\ \sigma_\theta,\ \tau_{rz}]^\top, \tag{2.1}$$

where $\gamma_{rz} = 2\varepsilon_{rz}$ is the engineering shear strain. The index e refers to elements, i to nodes and c to load cases. The symbol $\delta(\cdot)$ denotes a virtual variation, whereas δ alone is the displacement of the upper platen in the compression tests.

2.2 Physical Phenomena in Parts Made by Fused Filament Fabrication

The material model of this work follows from how an FFF part is built. This section explains why a printed part is anisotropic, how its sparse infill is turned into effective properties, and which other phenomena are left out of the model.

2.2.1 Mesostructure and Origin of Anisotropy

In FFF, a thermoplastic filament is melted in a heated nozzle and deposited as a bead, also called a road or raster, along a programmed path [52, 53]. Each layer consists of closed perimeters that follow the contour of the part and an infill pattern [33], and the layers are stacked along the build direction z. The printed solid is therefore a mesostructure of partially fused beads separated by voids, whose size depends on the line width, the layer height and the overlap between beads [14, 54]. Interfaces form between adjacent beads of a layer and between consecutive layers [13], and all interlayer interfaces are perpendicular to z.

These interlayer interfaces govern the anisotropy. A new bead bonds to the layer below by thermal welding, through contact and wetting, neck growth, and interdiffusion of polymer chains. In the reptation theory of Wool and O'Connor [55], the interface strength grows as $\sigma_{\mathrm{int}}/\sigma_{\infty} = (t/t_r)^{1/4}$ for a contact time t shorter than the reptation time t_r, which drops strongly as the temperature rises. In FFF the interface stays hot only briefly, so the bond quality depends on the processing temperatures and the cooling conditions [56]. In a semicrystalline polymer such as polyamide 6 the available time is even shorter, because chain mobility drops sharply below the crystallization temperature. This is why the nozzle temperature is high and the part-cooling fan is disabled in Table 4.2.2.

Two consequences follow. First, the interlayer interfaces are weaker and more compliant than the beads, and the voids reduce the load-bearing area across the layers, so stiffness and strength along z are lower than in the deposition plane [13, 53]. In this work, $E_r/E_z = 2.852$ for PA6 and 3.005 for PA6-CF. Second, the flow in the nozzle aligns short fibers with the bead [57]. The fibers stiffen each layer but do not bridge the interfaces, so they raise the in-plane stiffness more than the out-of-plane stiffness.

The resulting material behaves like a stack of plies with a common weak normal z, which can be modeled with orthotropic layers [14]. If the properties inside each layer do not depend on direction, the material is *transversely isotropic*, with the deposition plane as the plane of isotropy and z as the axis of symmetry, as illustrated in Figure 2.2.1. Here the gyroid infill provides the in-plane isotropy, because it does not concentrate the beads along one or two directions. The concentric perimeters of a ring are the exception. They run along θ and may stiffen the outer shell circumferentially, which would make the material cylindrically orthotropic. That model is also compatible with axisymmetry, as discussed in Section 2.4.2, but it would require measuring E_θ independently of E_r.

2.2.2 Infill, Homogenization and Scale Separation

Besides its layers, the part has a sparse infill, which makes it a cellular solid. Its effective modulus scales as $E^*/E_s = C(\rho_{\mathrm{mat}}/\rho_s)^n$, where $\rho_{\mathrm{mat}}/\rho_s$ is the relative density and C and n depend on the cell architecture and on whether the walls deform by bending or stretching [58]. Neither constant is known for the printed gyroid with its perimeters, so the effective properties are measured on specimens printed with the same infill and parameters, as described in Section 4.3. For the same reason, the apparent density ρ_{mat} is the mass of a specimen divided by its external volume.

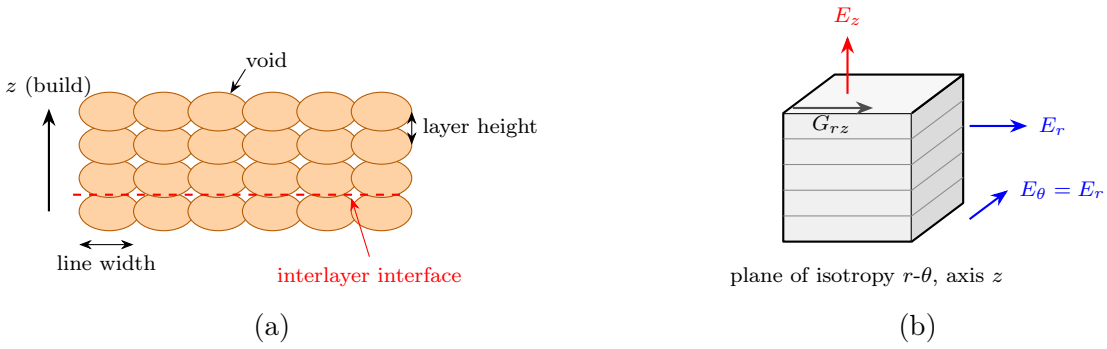


Figure 2.2.1: Mesostructure (a) and transversely isotropic symmetry (b) of an FFF part, schematic.

Using these effective properties requires scale separation [59]. The line width, the layer height and the infill cell must be small compared with the structural features and with the length over which the stresses vary. This holds in the bulk of the component, but not in members about as thick as the five perimeters, close to 2.2 mm, which print almost fully dense. These regions are stiffer and heavier than the homogenized material predicts, as discussed in Section 4.8.

2.2.3 Other Phenomena and Their Treatment

Anisotropy and infill are modeled explicitly. Table 2.2.1 lists the other phenomena that affect polymer parts and how this work treats them. Self-weight, for instance, can be estimated directly. A column of printed material of height H produces an axial stress $\rho_{\mathrm{mat}}\, g\, H \approx 790\ \mathrm{kg/m^3} \times 9.81\ \mathrm{m/s^2} \times 0.081\ \mathrm{m} \approx 6.3 \times 10^{-4}\,\mathrm{MPa}$, about 0.3% of the smallest design pressure of 0.225 MPa, so body forces are neglected.

2.3 Linear Elasticity of Solids of Revolution

With the material described, the next step is the mechanics of the component. Because the part is a solid of revolution printed along its axis, its elasticity problem can be reduced to two dimensions without losing its three-dimensional behavior.

2.3.1 Kinematics and Axisymmetric Reduction

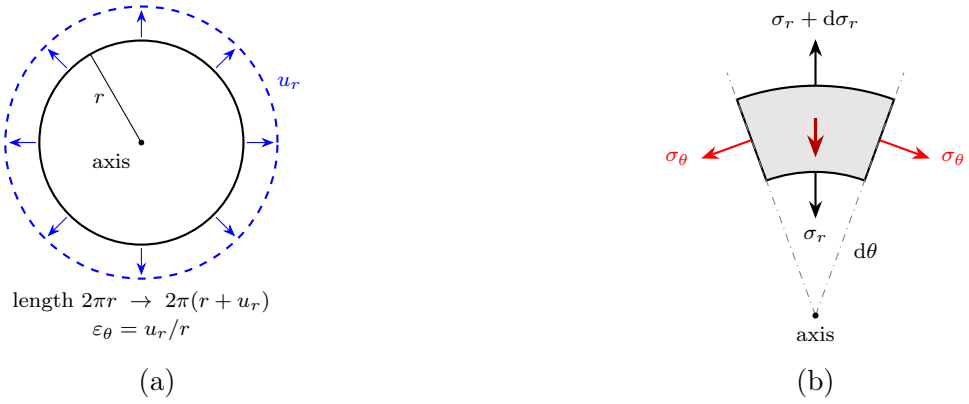
Under small displacements and small displacement gradients, the strain tensor is the symmetric part of $\nabla \boldsymbol{u}$. In a solid of revolution, the circumferential strain contains a term that planar problems lack. A material circle of radius r that moves radially by u_r changes its length from $2\pi r$ to $2\pi(r + u_r)$, so

$$\varepsilon_\theta = \frac{2\pi(r + u_r) - 2\pi r}{2\pi r} = \frac{u_r}{r}. \tag{2.2}$$

Any radial motion therefore stretches or shortens the rings of material, as shown in Figure 2.3.1a.

Table 2.2.1: Physical phenomena and their treatment in the model.

Phenomenon	Effect	Treatment in this work
Weak interlayer bonding	Lower stiffness and strength along z [13, 53]	Transversely isotropic model with measured E_z
Fiber orientation along the beads	Higher in-plane stiffness of PA6-CF [57]	Included through the measured E_r
Sparse infill	Effective properties depend on relative density and cell architecture [58]	Properties and density measured on specimens with the same infill
Perimeters along θ	Possible $E_\theta \neq E_r$ in the shell	Neglected, $E_\theta = E_r$, listed as an assumption
Tension-compression asymmetry	Different responses in tension and compression [60]	One elastic matrix, with E_r from tension and E_z from compression
Viscoelasticity and creep	Rate-dependent response of polymers [61]	Quasi-static loading, time effects not modeled
Moisture absorption	Loss of stiffness and strength of polyamides [61]	Filament dried before printing, dry properties assumed
Temperature and residual stresses	Thermal strains, warping, process-induced stresses [62]	Room-temperature analysis, residual stresses neglected
Self-weight	Body force $\rho_{\text{mat}}\,\boldsymbol{g}$	Neglected, about 0.3% of the smallest pressure



(a) (b)

Figure 2.3.1: Hoop strain (a) and radial resultant of the hoop stresses (b) in a solid of revolution, top views.

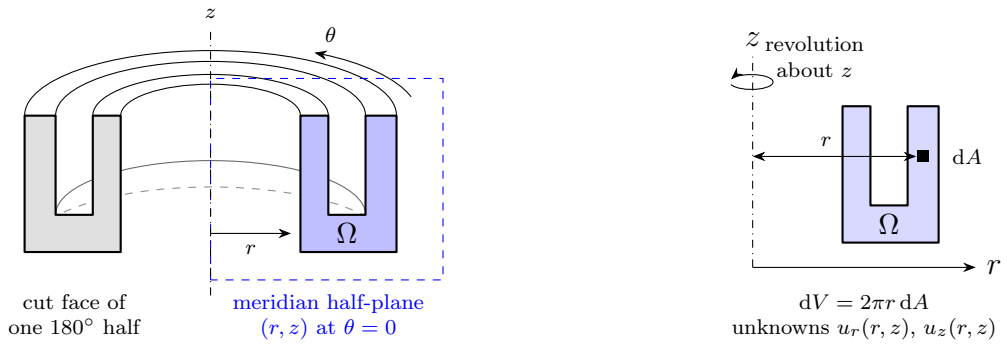
This hoop term is kept when the problem is reduced to two dimensions. The reduction is valid when the geometry is a solid of revolution about z, the material properties in the cylindrical basis do not depend on θ, a condition examined in Section 2.4.2, the supports and loads do not depend on θ, and there are no circumferential loads. The solution is then independent of θ, $u_\theta = 0$, and the problem is solved on the meridian section Ω alone, where each area element $\mathrm{d}A$ at radius r represents a ring of volume $\mathrm{d}V = 2\pi r\,\mathrm{d}A$, as depicted in Figure 2.3.2. The strains reduce to

$$\begin{aligned} \varepsilon_r &= \frac{\partial u_r}{\partial r}, \qquad \varepsilon_z = \frac{\partial u_z}{\partial z}, \qquad \varepsilon_\theta = \frac{u_r}{r}, \\ \gamma_{rz} &= \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}, \qquad \gamma_{r\theta} = \gamma_{\theta z} = 0, \end{aligned} \tag{2.3}$$

so $\tau_{r\theta} = \tau_{\theta z} = 0$, and without body forces, a point treated in Section 2.2.3, the equilibrium equations become [51, 63]

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \tau_{rz}}{\partial z} + \frac{\sigma_r - \sigma_\theta}{r} = 0, \qquad \frac{\partial \tau_{rz}}{\partial r} + \frac{\partial \sigma_z}{\partial z} + \frac{\tau_{rz}}{r} = 0. \tag{2.4}$$

The terms divided by r come from the curvature of the coordinate lines. The term $(\sigma_r - \sigma_\theta)/r$, in particular, is the radial resultant of the hoop stress acting on a curved ring, as shown in Figure 2.3.1b.



(a) (b)

Figure 2.3.2: Axisymmetric reduction: solid of revolution (a) and two-dimensional model on the meridian section Ω (b).

The unknowns are only $u_r(r, z)$ and $u_z(r, z)$, yet four stress components remain, including σ_θ. The reduction is not an approximation. If its conditions hold, the two-dimensional solution is the exact three-dimensional one [26, 51]. This is what separates it from plane strain, where ε_θ would vanish, and from plane stress, where σ_θ would vanish, as compared in Figure 2.3.3.

2.3.2 Boundary Conditions and Weak Form

To solve these equations numerically they are written in weak form. On the part Γ_u of the boundary the displacements are prescribed, $\boldsymbol{u} = \bar{\boldsymbol{u}}$, and on the part Γ_t the tractions are prescribed, $\boldsymbol{t} = \bar{\boldsymbol{t}}$. The traction follows from the stress and the outward unit normal $\boldsymbol{n}$

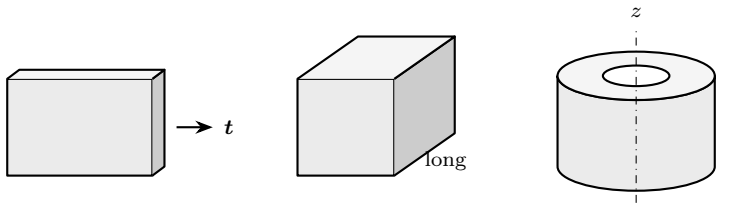


Figure 2.3.3: Two-dimensional idealizations. Only the axisymmetric model keeps the hoop strain and hoop stress.

through Cauchy's relation [63], and a uniform pressure P produces $\boldsymbol{t} = -P\,\boldsymbol{n}$. The supports of this work are of the first type and the design loads of the second.

Multiplying Eqs. (2.4) by virtual displacements that vanish on Γ_u, integrating over the volume and integrating by parts gives the principle of virtual work [51, 64],

$$\int_\Omega \delta\boldsymbol{\varepsilon}^\top \boldsymbol{\sigma}\; 2\pi r \,\mathrm{d}A = \int_{\Gamma_t} \delta\boldsymbol{u}^\top \boldsymbol{t}\; 2\pi r \,\mathrm{d}s \qquad \forall\, \delta\boldsymbol{u} \text{ with } \delta\boldsymbol{u} = \boldsymbol{0} \text{ on } \Gamma_u, \tag{2.5}$$

with $\delta\boldsymbol{\varepsilon} = [\delta\varepsilon_r, \delta\varepsilon_z, \delta\varepsilon_\theta, \delta\gamma_{rz}]^\top$. In the radial equation, the identity $\partial(r\sigma_r)/\partial r = r\,\partial\sigma_r/\partial r + \sigma_r$ makes the hoop stress do work through the virtual hoop strain $\delta u_r/r$. With the linear law $\boldsymbol{\sigma} = \boldsymbol{C}_0\boldsymbol{\varepsilon}$, Eq. (2.5) is the weak form solved in Section 2.6. Every integral carries the factor $2\pi r$, so material far from the axis contributes more to stiffness, mass and external work.

2.3.3 Strain Energy, Potential Energy and Compliance

The weak form is also the stationarity condition of an energy, which gives the stiffness measure used by the optimizer. For a kinematically admissible field $\boldsymbol{v}$, one with $\boldsymbol{v} = \bar{\boldsymbol{u}}$ on Γ_u, the strain energy and total potential energy are

$$W(\boldsymbol{v}) = \frac{1}{2}\int_\Omega \boldsymbol{\varepsilon}(\boldsymbol{v})^\top \boldsymbol{C}_0\,\boldsymbol{\varepsilon}(\boldsymbol{v})\; 2\pi r \,\mathrm{d}A, \qquad \Pi(\boldsymbol{v}) = W(\boldsymbol{v}) - \int_{\Gamma_t} \boldsymbol{v}^\top \boldsymbol{t}\; 2\pi r \,\mathrm{d}s. \tag{2.6}$$

If $\boldsymbol{C}_0$ is positive definite, a condition established in Section 2.4.4, and the supports prevent rigid-body motion, the solution is the unique minimizer of Π [51]. In discrete form, $\Pi(\boldsymbol{U}) = \frac{1}{2}\boldsymbol{U}^\top\boldsymbol{K}\boldsymbol{U} - \boldsymbol{F}^\top\boldsymbol{U}$ is minimized by $\boldsymbol{K}\boldsymbol{U} = \boldsymbol{F}$, and at the solution

$$J \equiv \boldsymbol{F}^\top\boldsymbol{U} = \boldsymbol{U}^\top\boldsymbol{K}\boldsymbol{U} = 2W, \qquad \Pi^* = -\tfrac{1}{2}J. \tag{2.7}$$

The compliance J is twice the stored strain energy. A stiffer structure has a smaller compliance under the same loads, which makes compliance minimization the standard measure of stiffness maximization [4, 10].

2.3.4 Displacement-Controlled Loading and Monotonicity

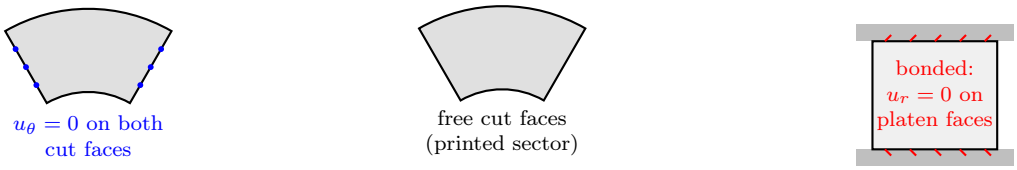
The same energy argument bounds the stiffness measured in a compression test. For a body loaded only by a prescribed displacement δ, such as a platen, the minimum potential energy is the minimum strain energy,

$$W^*(\delta) = \min_{\boldsymbol{v}\in\mathcal{A}} W(\boldsymbol{v}) = \frac{1}{2}K\delta^2, \qquad F = \frac{\mathrm{d}W^*}{\mathrm{d}\delta} = K\delta, \tag{2.8}$$

where $\mathcal{A}$ is the set of admissible fields, K the stiffness and F the reaction. A minimum over a larger set cannot be larger, so

$$\mathcal{A}_1 \subseteq \mathcal{A}_2 \quad\Longrightarrow\quad W_1^* \geq W_2^* \quad\Longrightarrow\quad K_1 \geq K_2, \tag{2.9}$$

and adding kinematic restrictions never decreases the elastic stiffness. This result is used twice, as summarized in Figure 2.3.4. Bonded platens add $u_r = 0$ on the platen faces, so their stiffness bounds from above that of frictionless platens, and a real frictional contact lies between the two. A sector of angle Θ cut from the axisymmetric solution has $u_\theta = 0$ on its cut faces and $\Theta/360°$ times the energy of the ring. A printed sector has free cut faces and a larger admissible set, so its stiffness cannot exceed $(\Theta/360°)\,K_{\mathrm{ini}}$, where K_{ini} is the initial stiffness of the ring. The difference is the hoop stress that confines the ring radially, as quantified in Section 4.7.1.



(a) sector of the ring (b) free sector (c) bonded platens

Figure 2.3.4: Kinematic restrictions and stiffness: ring sector (a), free sector (b) and bonded platens (c).

2.4 Anisotropic Constitutive Theory

The elasticity of the previous section holds for any linear material through $\boldsymbol{C}_0$. This section builds $\boldsymbol{C}_0$ for the transversely isotropic printed material, checks that it is compatible with the axisymmetric reduction, and estimates the constants that were not measured.

2.4.1 Orthotropy and Transverse Isotropy

A linear elastic material relates stress and strain through a fourth-order stiffness tensor, which in Voigt notation becomes a symmetric 6×6 matrix $\boldsymbol{C}_6$ with at most 21 independent constants [50, 63]. The stored energy must be positive for any non-zero strain, so $\boldsymbol{C}_6$ and its inverse $\boldsymbol{S}_6$ must be positive definite [50, 65]. Material symmetry reduces the number of constants. An orthotropic material has nine, and a transversely isotropic one, in which every

direction of one plane is equivalent, has even fewer. With the plane of isotropy r-θ, the axis z, and the order $[\varepsilon_r, \varepsilon_\theta, \varepsilon_z, \gamma_{\theta z}, \gamma_{rz}, \gamma_{r\theta}]$, its compliance matrix is [50]

$$\boldsymbol{S}_6 = \begin{bmatrix} 1/E_r & -\nu_{r\theta}/E_r & -\nu_{zr}/E_z & 0 & 0 & 0 \\ -\nu_{r\theta}/E_r & 1/E_r & -\nu_{zr}/E_z & 0 & 0 & 0 \\ -\nu_{rz}/E_r & -\nu_{rz}/E_r & 1/E_z & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/G_{rz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/G_{rz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/G_{r\theta} \end{bmatrix}. \tag{2.10}$$

The Poisson's ratios follow the convention $\nu_{ij} = -\varepsilon_j/\varepsilon_i$ under a uniaxial stress σ_i. Under σ_z alone, for example, $\varepsilon_r = -(\nu_{zr}/E_z)\sigma_z$, which is the entry S_{13}. Conventions differ between authors, so the meaning of each index must be checked before taking values from the literature.

Two relations leave five independent constants, E_r, E_z, $\nu_{r\theta}$, ν_{rz} and G_{rz}. The symmetry of $\boldsymbol{S}_6$ gives reciprocity, and in-plane isotropy fixes the in-plane shear modulus [50]:

$$\frac{\nu_{zr}}{E_z} = \frac{\nu_{rz}}{E_r}, \tag{2.11}$$

$$G_{r\theta} = \frac{E_r}{2(1+\nu_{r\theta})}. \tag{2.12}$$

2.4.2 Compatibility Between Material Symmetry and Axisymmetry

The axisymmetric reduction of Section 2.3.1 requires a constitutive matrix that does not depend on θ in the cylindrical basis. A transversely isotropic material whose axis coincides with the axis of revolution meets this condition, because the radial and circumferential directions at any point both lie in the plane of isotropy. A cylindrically orthotropic material, with $E_\theta \neq E_r$, meets it too, because its principal directions follow the cylindrical basis. If the part were printed lying on its side, the layers would be parallel to the axis, the components of $\boldsymbol{C}_6$ would vary with θ, and the problem would no longer be axisymmetric. Printing the part with its axis along the build direction is therefore what makes FFF anisotropy compatible with the axisymmetric reduction (Figure 2.4.1).

With this orientation, $\gamma_{r\theta} = \gamma_{\theta z} = 0$ (Eq. (2.3)), and shear and normal components are uncoupled in Eq. (2.10). Removing the fourth and sixth rows and columns of $\boldsymbol{S}_6$ and reordering as in Eq. (2.1) gives the 4×4 compliance matrix of the axisymmetric problem,

$$\boldsymbol{S} = \begin{bmatrix} 1/E_r & -\nu_{zr}/E_z & -\nu_{r\theta}/E_r & 0 \\ -\nu_{rz}/E_r & 1/E_z & -\nu_{rz}/E_r & 0 \\ -\nu_{r\theta}/E_r & -\nu_{zr}/E_z & 1/E_r & 0 \\ 0 & 0 & 0 & 1/G_{rz} \end{bmatrix}. \tag{2.13}$$

Its inverse is the stiffness matrix $\boldsymbol{C}_0 = \boldsymbol{S}^{-1}$ used in Eqs. (2.5) and (2.6). The modulus E_r acts in the radial and circumferential directions, both within the layers, E_z acts across the layers, and G_{rz} governs the shear between them. The in-plane shear modulus $G_{r\theta}$ does not appear, because $\gamma_{r\theta} = 0$.

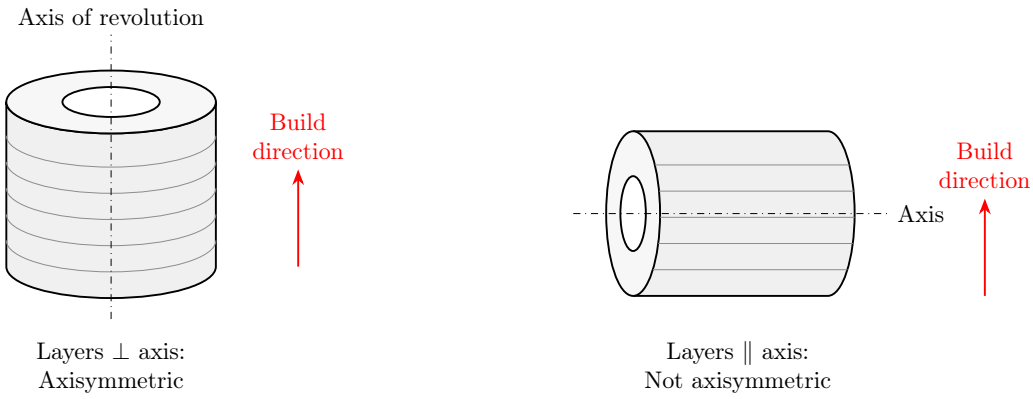


Figure 2.4.1: Printing orientation and axisymmetry: axis along the build direction (a) and part lying on its side (b).

2.4.3 Closed-Form Stiffness Matrix

For the implementation, $\boldsymbol{C}_0$ is written in closed form by inverting Eq. (2.13) [50]. With the admissibility determinant

$$\Delta = (1+\nu_{r\theta})\,(1-\nu_{r\theta}-2\nu_{rz}\nu_{zr})\,, \tag{2.14}$$

the non-zero entries are

$$\begin{aligned} C_{rr} = C_{\theta\theta} &= \frac{E_r\,(1-\nu_{rz}\nu_{zr})}{\Delta}, & C_{r\theta} &= \frac{E_r\,(\nu_{r\theta}+\nu_{rz}\nu_{zr})}{\Delta}, \\ C_{zz} &= \frac{E_z\,(1-\nu_{r\theta}^2)}{\Delta}, & C_{44} &= G_{rz}, \\ C_{rz} = C_{\theta z} &= \frac{E_r\,\nu_{zr}\,(1+\nu_{r\theta})}{\Delta} = \frac{E_z\,\nu_{rz}\,(1+\nu_{r\theta})}{\Delta}. \end{aligned} \tag{2.15}$$

The coupling term C_{rz} links an axial strain to radial and hoop stresses, and produces the confinement stresses that appear when the radial expansion of a compressed part is restrained. For an isotropic material, Eq. (2.15) reduces to the classical Lamé values. With the constants of Table 4.4.1 it gives the matrices of Eqs. (4.1) and (4.2).

2.4.4 Admissibility of the Elastic Constants

These matrices are only physical if the energy stays positive. Matrix (2.13) is positive definite if and only if its leading principal minors are positive, which with Eq. (2.11) gives the conditions of Lempriere [65],

$$E_r, E_z, G_{rz} > 0, \qquad |\nu_{r\theta}| < 1, \qquad |\nu_{rz}| < \sqrt{\frac{E_r}{E_z}}, \qquad \Delta > 0. \tag{2.16}$$

For an anisotropic material the bound on a Poisson's ratio depends on the ratio of the moduli and is not 0.5. With $E_r/E_z = 2.852$ for PA6, ν_{rz} may reach $\sqrt{2.852} = 1.689$. The condition

$\Delta > 0$ is stricter, because it couples ν_{rz} with $\nu_{r\theta}$. For $\nu_{rz} = 0.6586$, $\nu_{rz}\nu_{zr} = \nu_{rz}^2 E_z/E_r = 0.152$, so $\Delta > 0$ requires $\nu_{r\theta} < 0.696$. The constants of Table 4.4.1 are therefore admissible even though ν_{rz} exceeds 0.5.

2.4.5 Estimation of the Constants Not Measured

Of the five constants, the tests measure only E_r and E_z, so the other three are estimated. The first relation below is an additional Poisson closure adopted in this work. The second is Huber's reference estimate for orthotropic shear [66, 67]:

$$\nu_{rz}\nu_{zr} = \nu_{r\theta}^2, \qquad G_{rz}^{\mathrm{H}} = \frac{\sqrt{E_r E_z}}{2\left(1 + \sqrt{\nu_{rz}\nu_{zr}}\right)}. \tag{2.17}$$

With reciprocity, the first relation gives $\nu_{rz} = \nu_{r\theta}\sqrt{E_r/E_z}$ and $\nu_{zr} = \nu_{r\theta}\sqrt{E_z/E_r}$ (Eq. (3.3)). Substituting it in Eqs. (2.14) and (2.15) gives

$$\begin{gathered} \Delta = (1+\nu_{r\theta})^2(1-2\nu_{r\theta}), \\ C_{ii} = \frac{E_i\,(1-\nu_{r\theta})}{(1+\nu_{r\theta})(1-2\nu_{r\theta})}, \qquad C_{ij} = \frac{\sqrt{E_i E_j}\;\nu_{r\theta}}{(1+\nu_{r\theta})(1-2\nu_{r\theta})} \quad (i \neq j), \qquad i,j \in \{r,z,\theta\}, \end{gathered} \tag{2.18}$$

with $E_\theta = E_r$. The normal block is the isotropic block of Poisson's ratio $\nu_{r\theta}$, scaled on both sides by $\mathrm{diag}(\sqrt{E_r}, \sqrt{E_z}, \sqrt{E_r})$. This congruence preserves positive definiteness, so the constants are admissible for any $-1 < \nu_{r\theta} < 0.5$, whatever E_r/E_z.

For $\nu_{r\theta} \geq 0$, Huber's estimate becomes $G_{rz}^{\mathrm{H}} = \sqrt{E_r E_z}/[2(1+\nu_{r\theta})]$, giving $\alpha_G^{\mathrm{H}} = G_{rz}^{\mathrm{H}}/E_z = 0.6075$ for PA6 and 0.6235 for PA6-CF. The adopted value $\alpha_G = 1/3$ is a modeling assumption, while Huber's estimate provides the upper endpoint of the sensitivity study.

2.5 Strength Criteria and Stress Evaluation

Once the stresses are known, the design needs a way to judge them. This section selects the yield criterion, explains why the largest element stress cannot be used, and defines the stress measure and the utilization index that replace it.

2.5.1 Yield Criteria

The von Mises criterion states that yielding begins when the second invariant of the deviatoric stress reaches a critical value [68]. The equivalent stress is $\sigma_{\mathrm{VM}} = \sqrt{3J_2}$, and for the axisymmetric stress state

$$J_2 = \frac{1}{6}\left[(\sigma_r - \sigma_z)^2 + (\sigma_z - \sigma_\theta)^2 + (\sigma_\theta - \sigma_r)^2\right] + \tau_{rz}^2. \tag{2.19}$$

The equivalent stress is positively homogeneous of degree one, so multiplying all stresses by $\kappa > 0$ multiplies σ_{VM} by κ.

The printed material is anisotropic, so an anisotropic criterion such as Hill's quadratic criterion [69] would seem more natural. It is not used for two reasons. It needs the interlaminar shear yield strength, which was not measured. It is also symmetric in tension and compression, whereas the available strengths come from tension in the plane, $S_{y,T}$, and from compression along z, S_y. The design therefore uses the von Mises criterion with S_y, and the verification adds a directional check with the two measured strengths.

2.5.2 Stress Singularities and the Percentile Measure

Applying the criterion element by element raises a difficulty. Near a re-entrant corner, or where a clamped boundary meets a free or loaded one, the linear-elastic stresses are singular [70]. At a distance ϱ from the corner, $\sigma_{\mathrm{VM}} \sim \varrho^{\lambda_W - 1}$, with $0 < \mathrm{Re}(\lambda_W) < 1$ for singular configurations, so the stress is unbounded while the strain energy remains finite. An element at the corner samples a stress that grows without limit as the mesh is refined, so the largest element stress cannot be used as a design criterion (Figure 2.5.1a).

The design uses a percentile instead. With n_e solid elements, the empirical distribution of the element stresses and its 99.5th percentile are

$$\mathcal{F}(\hat{\sigma}) = \frac{1}{n_e} \#\left\{e : \sigma_{\mathrm{VM},e} \le \hat{\sigma}\right\}, \qquad \sigma_{\mathrm{VM},p99.5} = \min\left\{\hat{\sigma} : \mathcal{F}(\hat{\sigma}) \ge 0.995\right\}. \tag{2.20}$$

All elements of the structured grid have the same meridian area, so $\mathcal{F}$ is the fraction of the meridian area, not of the volume of revolution, in which $\sigma_{\mathrm{VM}} \le \hat{\sigma}$. The percentile ignores the 0.5% of the area with the highest stresses, which contains the singular corners (Figure 2.5.1b). As the mesh is refined, the singular zones shrink and the quantiles are expected to converge to the finite values of the exact solution, which the verification checks with three meshes.

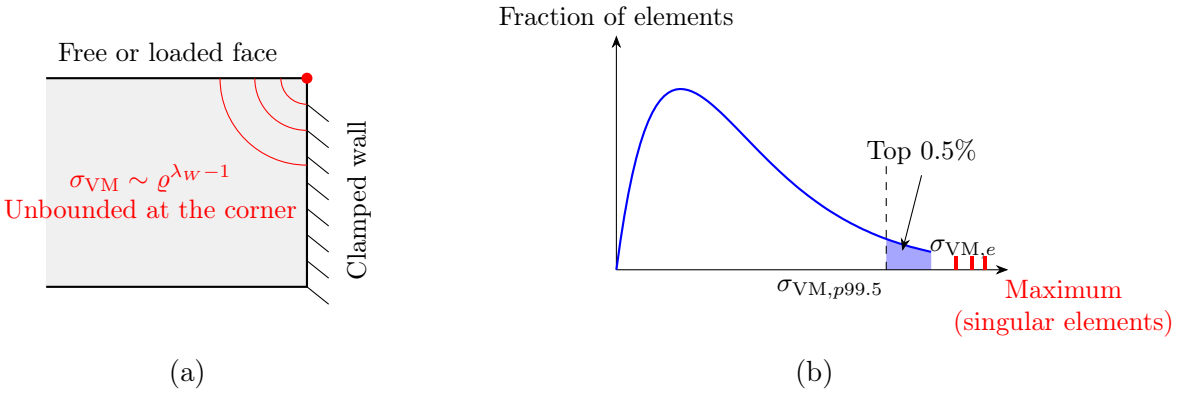


Figure 2.5.1: Stress singularity (a) and distribution of element stresses with the 99.5th percentile (b), schematic.

This percentile and the displacements of the critical walls are combined into the utilization of Eq. (3.8), which has a simple meaning in linear elasticity. Replacing $\boldsymbol{F}$ by $\kappa\boldsymbol{F}$ with $\kappa > 0$ scales the displacements and stresses by κ. Displacement magnitudes, the von Mises stress, quantiles of positive values and maxima of such functions are all positively homogeneous of degree one, so

$$U_c(\kappa\,\boldsymbol{F}_c) = \kappa\, U_c(\boldsymbol{F}_c), \tag{2.21}$$

and $1/U_c$ is the factor by which the loads of case c can grow before the governing limit is reached, as defined in Section 3.4.

2.6 Axisymmetric Finite Element Model

The weak form of Section 2.3.2 and the material of Section 2.4 are now discretized on the meridian section. This section defines the element, its stiffness, the nodal loads and the assembled system.

2.6.1 Isoparametric Q4 Element and Element Stiffness

The meridian domain is divided into four-node bilinear quadrilaterals, Q4 [51, 71]. Each element is mapped onto the reference square $(\xi, \eta) \in [-1, 1]^2$, with ξ along r and η along z, so every element uses the same shape functions and integration points (Figure 2.6.1).

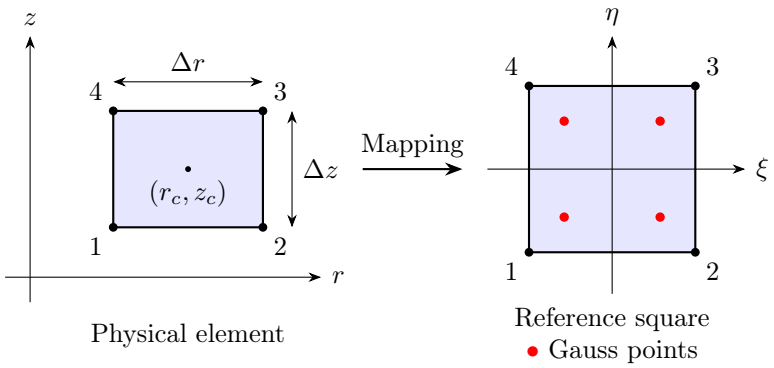


Figure 2.6.1: Q4 element: physical element in the meridian plane and reference square in natural coordinates.

The shape functions

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta), \qquad (\xi_i, \eta_i) \in \{(-1, -1), (1, -1), (1, 1), (-1, 1)\} \tag{2.22}$$

interpolate both the geometry and the displacements [51, 72]. For the rectangular elements of the structured grid, $r = r_c + \xi\, \Delta r/2$ and $z = z_c + \eta\, \Delta z/2$, so the mapping is linear with the constant Jacobian determinant $\Delta r\, \Delta z/4$, and $\partial N_i/\partial r = (2/\Delta r)\, \partial N_i/\partial \xi$ and $\partial N_i/\partial z = (2/\Delta z)\, \partial N_i/\partial \eta$.

Inserting this interpolation into Eq. (2.3) gives $\boldsymbol{\varepsilon} = \boldsymbol{B}\boldsymbol{u}_e$, with the nodal blocks of Eq. (3.6), where the third row, N_i/r, is the discrete hoop strain. The weak form (2.5) then gives the stiffness matrix of a solid element,

$$\boldsymbol{k}_e^0 = \int_{A_e} \boldsymbol{B}^\top \boldsymbol{C}_0\, \boldsymbol{B}\; 2\pi r \,\mathrm{d}A = \frac{\Delta r\, \Delta z}{4} \int_{-1}^{1}\int_{-1}^{1} \boldsymbol{B}^\top \boldsymbol{C}_0\, \boldsymbol{B}\; 2\pi r \,\mathrm{d}\xi\, \mathrm{d}\eta, \tag{2.23}$$

evaluated with a 2×2 Gauss-Legendre rule, which integrates exactly polynomials up to degree three [73]. All terms of the integrand are such polynomials except the hoop-hoop

term, proportional to $N_i N_j / r$. Expanding $1/r$ in powers of $\xi\,\Delta r/(2r_c)$ shows that its relative error is of order $(\Delta r/2r_c)^2$, negligible for the meshes of Chapter 4.

2.6.2 Nodal Loads, Global System and Element Volumes

The pressures of the load cases enter the right-hand side as consistent nodal forces. For a loaded edge of length L between nodes a and b, with arc length $\ell \in [0, L]$, edge functions $N_a = 1 - \ell/L$ and $N_b = \ell/L$, and radius $r(\ell) = r_a N_a + r_b N_b$, a uniform pressure P gives

$$\boldsymbol{F}_a = -\boldsymbol{n} \int_0^L N_a\, P\, 2\pi r(\ell)\, \mathrm{d}\ell = \frac{2\pi L P}{6}\,(2r_a + r_b)\,(-\boldsymbol{n}), \tag{2.24}$$

and symmetrically $\boldsymbol{F}_b = \frac{2\pi LP}{6}(r_a + 2r_b)(-\boldsymbol{n})$. The node at the larger radius receives more force because it represents a larger ring. For a horizontal edge, $L = r_b - r_a$, the two forces add up to $\pi P(r_b^2 - r_a^2)$, the exact force on the annulus, which the code checks.

Assembling element matrices and forces gives $\boldsymbol{K}\boldsymbol{U} = \boldsymbol{F}$, with $\boldsymbol{K}$ symmetric and positive semi-definite. Its only rigid-body mode is a translation along z. Since the domain does not reach the axis, $\varepsilon_\theta = u_r/r = 0$ forces $u_r = 0$, and then $\varepsilon_z = \gamma_{rz} = 0$ leaves only a constant u_z. A uniform radial translation is not rigid, because it stretches the rings. Restraining u_z at one or more nodes makes the reduced matrix positive definite. Stresses are evaluated at the element centroid, the most accurate sampling point of the bilinear element [51].

The same grid also gives exact element volumes, needed for the mass and the volume constraint. By the theorem of Pappus, revolving a plane region about an axis that does not cross it generates a volume equal to its area times the length of the circle described by its centroid [74]. A rectangular element with radii r_1 and r_2 and centroid at r_c therefore has $V_e = 2\pi r_c\,\Delta r\,\Delta z = \pi(r_2^2 - r_1^2)\Delta z$ (Eq. (3.4)).

2.7 Topology Optimization by the Density Method

The finite element model evaluates a given design. Topology optimization uses it to decide where material should be. This section states the problem, introduces the SIMP interpolation and the regularization it needs, and derives the gradients and the update used by the optimizer.

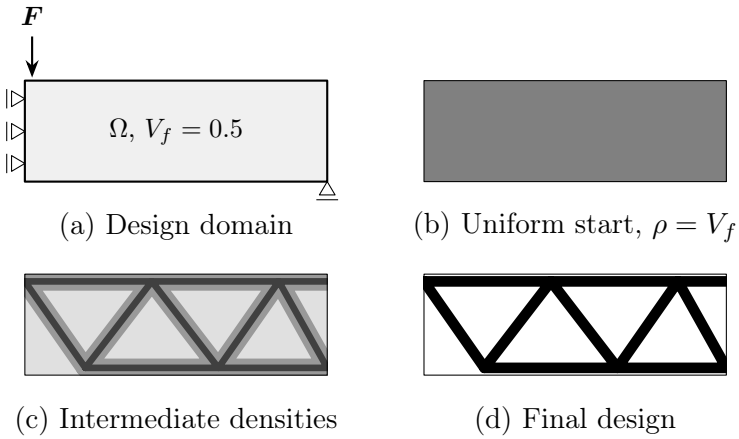
2.7.1 The Material Distribution Problem

Topology optimization seeks the distribution of material within a fixed domain that minimizes an objective under constraints [17]. For minimum compliance, the general problem of Eq. (1.1) takes $\Phi = J$, with J from Eq. (2.7), and the single constraint $\int_\Omega \rho\, 2\pi r\, \mathrm{d}A \le V_f V_0$, with $\rho \in \{0, 1\}$ at every point of Ω.

This binary problem is in general ill-posed. A minimizing sequence can keep improving with ever finer mixtures of solid and void, so an optimal design may not exist [2, 36]. Relaxation enlarges the design space with composite microstructures, as in the homogenization

method [2], whereas restriction limits the complexity of the designs, for example with a filter that imposes a minimum length scale [36, 75].

SIMP, whose choice was justified in Section 1.3, combines a continuous interpolation with a filter and a projection. Figure 2.7.1 sketches the process on the half MBB beam. From a uniform density, the optimizer removes material where it contributes little to the stiffness and concentrates it along the load paths, until a truss-like design emerges.



(a) Design domain (b) Uniform start, $\rho = V_f$

(c) Intermediate densities (d) Final design

Figure 2.7.1: Schematic of density-based topology optimization on the half MBB beam.

2.7.2 SIMP Interpolation

The first ingredient of SIMP relaxes $\rho \in \{0, 1\}$ to $\rho \in [0, 1]$ and interpolates the stiffness with a power law [4]. In this work the whole transversely isotropic matrix is scaled,

$$\boldsymbol{C}(\bar{\rho}) = [E_{\min} + \bar{\rho}^p(1 - E_{\min})]\, \boldsymbol{C}_0, \tag{2.25}$$

so the anisotropy ratios and the Poisson's ratios are the same at every density. The word isotropic in SIMP refers to the original formulation and is not a restriction. The volume is linear in $\bar{\rho}$, whereas the stiffness grows as $\bar{\rho}^p$, as illustrated in Figure 2.7.2. For $p = 3$, a density of 0.5 costs half the volume of a solid element but gives only 12.5% of its stiffness, so intermediate densities are uneconomical and the optimizer is driven towards 0-1 designs. The small relative stiffness $E_{\min} = 10^{-9}$ keeps $\boldsymbol{K}$ non-singular in void regions.

Bendsøe and Sigmund [76] showed that intermediate densities can be read as real composites if p is large enough, which for an isotropic two-dimensional material with $\nu_0 = 1/3$ requires $p \geq 3$. For the anisotropic, axisymmetric material of this work the bound is only indicative. This work adopts $p = 3$, the most common value [10, 17, 23], and evaluates the final designs only after binarization (Section 3.5.3), so the interpretation of intermediate densities does not affect the results. Alternative interpolations exist, such as RAMP [77].

2.7.3 Numerical Instabilities and Helmholtz Filtering

Penalization alone does not give usable designs, because density methods suffer from three numerical problems [36]. Checkerboards of alternating solid and void elements appear because

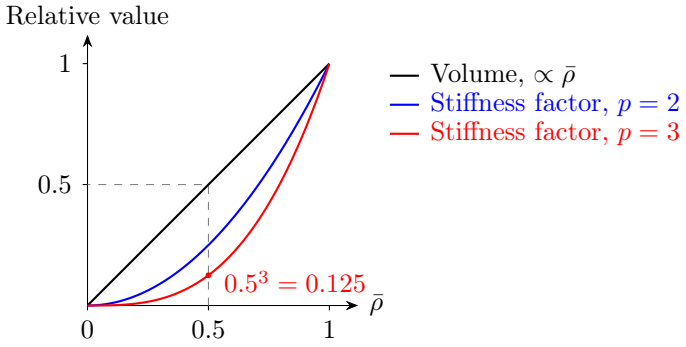
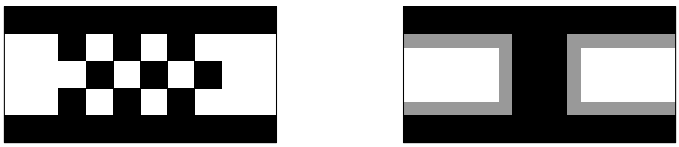


Figure 2.7.2: SIMP interpolation: volume and stiffness factor as functions of the physical density.

bilinear elements overestimate their stiffness. Mesh dependence makes a finer mesh produce more and thinner members, a consequence of the ill-posedness discussed in Section 2.7.1. Local minima appear because the penalized problem is non-convex. Filtering addresses the first two, but leaves a gray transition zone at the member boundaries, as shown in Figure 2.7.3, which the projection described in Section 2.7.4 then sharpens.



(a) No filter (b) Density filter

Figure 2.7.3: Checkerboards without filtering (a) and filtered members with gray boundaries (b), schematic.

The density filter [75] replaces each density by a weighted average over a neighborhood of radius $R_{\min}$,

$$\tilde{\rho}_e = \frac{\sum_i H_{ei}\, V_i\, \rho_i}{\sum_i H_{ei}\, V_i}, \qquad H_{ei} = \max\left(0,\ R_{\min} - \|\boldsymbol{x}_e - \boldsymbol{x}_i\|\right), \tag{2.26}$$

which imposes a minimum length scale of order $R_{\min}$ but costs more as the number of neighbors grows. Lazarov and Sigmund [78] obtained a similar smoothing from the Helmholtz-type equation

$$-r_H^2\, \nabla^2 \tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \qquad \frac{\partial \tilde{\rho}}{\partial n} = 0 \quad \text{on } \partial\Omega, \tag{2.27}$$

which needs only one sparse factorization. Matching the variances of the two kernels in one dimension, $R_{\min}^2/6$ for the hat kernel and $2r_H^2$ for the Helmholtz Green's function $g_H(x) = e^{-|x|/r_H}/(2r_H)$, gives [78]

$$r_H = \frac{R_{\min}}{2\sqrt{3}}, \tag{2.28}$$

so $R_{\min} = 2.6\,\text{mm}$ gives $r_H = 0.751\,\text{mm}$. The Helmholtz kernel has no finite support, so the equivalence holds only for the length scale, as shown in Figure 2.7.4a.

In the axisymmetric problem the weak form of Eq. (2.27) carries the same factor $2\pi r$ as the rest of the model,

$$\int_\Omega \left(r_H^2 \, \nabla\tilde{\rho} \cdot \nabla v + \tilde{\rho}\, v \right) 2\pi r \, \mathrm{d}A = \int_\Omega \rho \, v \, 2\pi r \, \mathrm{d}A \qquad \forall\, v, \tag{2.29}$$

so the filter smooths the density as a three-dimensional filter on the solid of revolution. Taking $v \equiv 1$ removes the gradient term, which shows that the filter conserves the volume of material exactly. In discrete form this is the property $\sum_i T_{ie} = V_e$ of Eq. (3.10), and the filtered density of an element is the volume average of the nodal field over it (Eq. (3.9)).

2.7.4 Heaviside Projection

The projection removes the gray zone left by the filter by mapping the filtered density to a nearly binary physical density [37, 79, 80]. The smooth function of Eq. (3.11) satisfies

$$\bar{\rho}(0) = 0, \qquad \bar{\rho}(1) = 1, \qquad \frac{\partial \bar{\rho}}{\partial \tilde{\rho}} > 0, \qquad \lim_{\beta \to 0} \bar{\rho} = \tilde{\rho}, \qquad \lim_{\beta \to \infty} \bar{\rho} = \begin{cases} 0, & \tilde{\rho} < \eta_H \\ 1, & \tilde{\rho} > \eta_H \end{cases}, \tag{2.30}$$

so increasing β turns the identity into a step at the threshold η_H, as illustrated in Figure 2.7.4b. The projection does not preserve volume, so the volume constraint is imposed on the projected densities $\bar{\rho}$, which represent the physical design. A large β from the start would make the problem nearly discrete and strongly non-convex, so β is increased gradually by continuation, which also mitigates the local minima mentioned above.

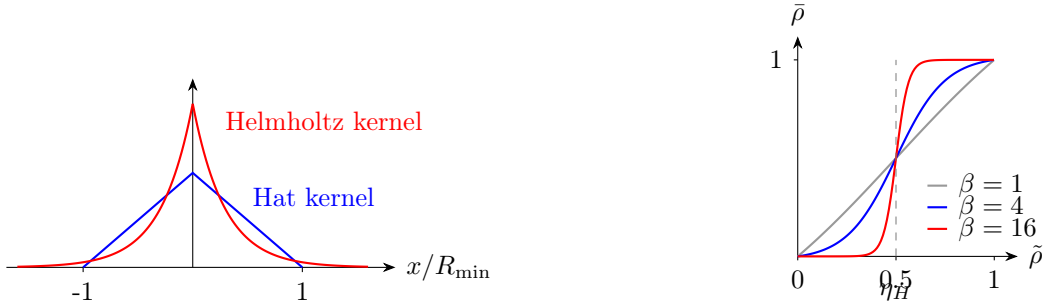


Figure 2.7.4: Filter kernels with equal variance (a) and Heaviside projection for increasing β, with $\eta_H = 0.5$ (b).

2.7.5 Sensitivity Analysis by the Adjoint Method

With the design parameterized by ρ, filtered and projected, the optimizer needs the gradient of each response. For a function $\Psi(\boldsymbol{U}, \bar{\boldsymbol{\rho}})$, differentiating $\boldsymbol{K}\boldsymbol{U} = \boldsymbol{F}$ gives [17]

$$\frac{\mathrm{d}\Psi}{\mathrm{d}\bar{\rho}_e} = \frac{\partial \Psi}{\partial \bar{\rho}_e} + \boldsymbol{\lambda}^\top \left(\frac{\partial \boldsymbol{K}}{\partial \bar{\rho}_e} \boldsymbol{U} - \frac{\partial \boldsymbol{F}}{\partial \bar{\rho}_e} \right), \qquad \boldsymbol{K}\boldsymbol{\lambda} = -\left(\frac{\partial \Psi}{\partial \boldsymbol{U}} \right)^\top, \tag{2.31}$$

so one adjoint system is solved per function, whatever the number of design variables. For the compliance, $\boldsymbol{K}\boldsymbol{\lambda} = -\boldsymbol{F}$ gives $\boldsymbol{\lambda} = -\boldsymbol{U}$, the problem is self-adjoint, and no extra solution is needed:

$$\frac{\mathrm{d}J}{\mathrm{d}\bar{\rho}_e} = -p\,\bar{\rho}_e^{\,p-1}(1 - E_{\min})\,\boldsymbol{u}_e^{\top}\boldsymbol{k}_e^0\,\boldsymbol{u}_e. \tag{2.32}$$

The load term vanishes because the loads act on the passive shell and do not depend on the design, as established in Section 3.5. Since $\boldsymbol{k}_e^0$ is positive semi-definite, adding material never increases the compliance. For the weighted multi-load compliance J_w, the sensitivities of the cases are added with the weights w_c, as given by Eq. (3.13), and the chain rule through the diagonal projection Jacobian and the filter gives Eq. (3.14).

2.7.6 Optimality Criteria Method

These sensitivities drive the update of the design. For minimum compliance with one volume constraint, the Lagrangian is

$$\mathcal{L}_{\mathrm{OC}} = J(\boldsymbol{\rho}) + \Lambda\,(V(\boldsymbol{\rho}) - V_f V_0) + \sum_e \mu_e^-(0 - \rho_e) + \sum_e \mu_e^+(\rho_e - 1), \tag{2.33}$$

with multipliers $\Lambda, \mu_e^{\pm} \geq 0$, and for a variable strictly between its bounds the Karush-Kuhn-Tucker condition is

$$B_e \equiv \frac{-\partial J/\partial \rho_e}{\Lambda\,\partial V/\partial \rho_e} = 1. \tag{2.34}$$

The Optimality Criteria (OC) method uses the fixed-point update $\rho_e \leftarrow \rho_e B_e^{1/2}$, limited by the bounds and by the move limit $\Delta\rho_{\lim}$ [10, 17, 23]. Elements with $B_e > 1$, where material is worth more than its volume cost, gain density, and those with $B_e < 1$ lose it. A fixed point satisfies Eq. (2.34), the exponent $1/2$ damps the iteration, and Λ is found by bisection so that the updated design meets the volume constraint [10].

2.7.7 Local Constraints and Binary Designs

The OC update handles a single global constraint, whereas the requirements of the component are local, namely the displacements of the critical walls and a stress limit. Each constrained displacement $u_k = \boldsymbol{l}^{\top}\boldsymbol{U}$ needs its own adjoint system $\boldsymbol{K}\boldsymbol{\lambda} = -\boldsymbol{l}$, so with 1520 boundary control nodes and four load cases the constraints would have to be aggregated. Stress constraints are harder still. They are very numerous, and the feasible space has degenerate regions in which vanishing members keep finite stresses [39], so they require relaxation and aggregation [24, 25], usually with the method of moving asymptotes [40].

This work therefore minimizes compliance, a global stiffness measure, and checks the local constraints on each binary design. The outer search on the volume fraction, described in Section 3.6, then finds the lightest design that satisfies them. This keeps the self-adjointness of compliance, at the cost that the local constraints do not steer the material distribution directly, which the adaptive weights of the multi-load problem partly compensate.

Checking the binary designs requires a sound definition of connectivity. Connectivity is defined through shared edges, because two elements that share only a corner node form a

one-node hinge, a numerical artifact rather than a real member [36, 37] that cannot be printed as a sound connection. Components not connected through edges to the passive shell or the supports carry no load in the model and are removed.

2.8 Multi-Load and Multi-Objective Formulation

The previous section assumed a single compliance, but the component has several load cases, each with its own compliance $J_c = \boldsymbol{F}_c^\top \boldsymbol{U}_c$. Minimizing all of them is a multi-objective problem, in which a design is Pareto optimal when no other design improves one objective without worsening another [81]. The weighted sum

$$J_w = \sum_c w_c \, J_c, \qquad w_c > 0, \qquad \sum_c w_c = 1, \tag{2.35}$$

gives a Pareto-optimal design at its minimum, although it only reaches the convex part of the Pareto front [82]. Because the compliances of different cases can differ by orders of magnitude, the initial weights are $w_c \propto 1/J_{c,\text{solid}}$, which makes every term $w_c J_c$ equal for the solid component. Since J_w is linear in the J_c, the problem remains self-adjoint.

A second trade-off appears between the mass m and the governing utilization U. A design a dominates a design b if $m_a \le m_b$ and $U_a \le U_b$, with at least one strict inequality, and the non-dominated designs form the mass-utilization front, as plotted in Figure 2.8.1. In the figure, open markers are infeasible designs with $U > 1$, filled markers are feasible, and gray markers are dominated. The selected design solves the epsilon-constraint problem [81]

$$\min \; m \quad \text{subject to} \quad U \le 1. \tag{2.36}$$

The search on the volume fraction assumes that U increases locally as V_f decreases. SIMP is non-convex, so this is expected but not guaranteed, and all evaluated designs are kept and the non-dominated ones reported.

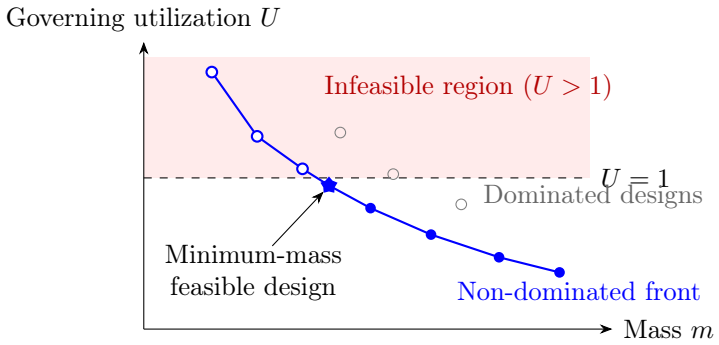


Figure 2.8.1: Mass-utilization trade-off and selected design, schematic.

2.9 Elastoplasticity and Limit Analysis

The optimization and the design checks are linear. The virtual compression test goes further and loads the designs into the plastic range, which requires the elements of limit analysis

presented here.

2.9.1 Elastic-Perfectly Plastic Material

The virtual test uses an elastic-perfectly plastic material. While $\sigma_{\mathrm{VM}} < S_y$ the response is linear elastic, as described in Section 2.4, the stress cannot exceed the yield surface $\sigma_{\mathrm{VM}} = S_y$, and there is no hardening. A structure of such a material reaches a collapse load F_{col}, at which the plastic zones form a mechanism and the deformation grows at constant load. Under displacement control, the force-displacement curve is linear up to the onset of yielding, then softens as the plastic zones spread and approaches F_{col}, as shown in Figure 2.9.1.

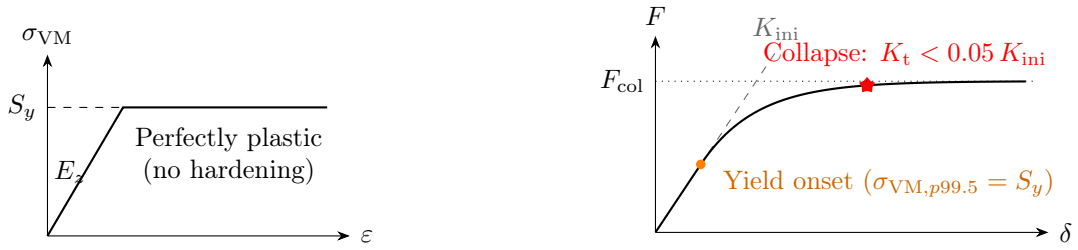


Figure 2.9.1: Elastic-perfectly plastic material (a) and force-displacement curve under displacement control (b), schematic.

2.9.2 Limit Loads by Elastic Compensation

The collapse load is bounded by the theorems of limit analysis, which hold for a perfectly plastic material with a convex yield surface and associated flow under small deformations [83]. By the static theorem, a load in equilibrium with a stress field that satisfies the traction conditions and nowhere exceeds the yield surface is a lower bound of F_{col}. By the kinematic theorem, the load obtained by equating the external work rate to the plastic dissipation of any admissible mechanism is an upper bound. The von Mises surface is convex, so both theorems apply.

The elastic compensation method of Mackenzie and Boyle [84] exploits the static theorem with a sequence of linear analyses. After each analysis, the modulus of every element whose stress exceeds the yield limit is reduced in proportion to the excess, as given by Eq. (**??**), which redistributes the stress to less loaded regions. When no element exceeds S_y, the stress field is approximately statically admissible and the load estimates a lower bound of the collapse load. The estimate is approximate, because the yield condition is checked only at the centroids within a tolerance of 1% and finite element equilibrium holds only in the weak sense. The secant reduction does not integrate the plastic flow incrementally. It describes a non-linear elastic material, which under monotonic, proportional loading corresponds to the deformation theory of plasticity of Hencky [85]. The curve $F(\delta)$ rises until a mechanism forms and then flattens, and the virtual test detects this point when the tangent stiffness K_{t} falls below 5% of K_{ini}, as defined in Section 3.8.2.

These collapse loads inherit the limits of the small-strain kinematics of Eq. (2.3), which hold only for small strains and rotations [73]. For PA6, the linear model reaches the compressive yield strength at $S_y/E_z = 18.1\%$, well beyond that range. The virtual test therefore flags every collapse with $\delta/H > 5\%$, and the initial stiffness, computed at vanishing strain, is the main quantity compared with the experiments.

2.10 Geometric Relations Used in the Reconstruction

The last step turns the meridian sections into printable solids, which relies on four geometric relations. The first places the component on the grid. A ray cast from a point crosses a simple closed curve an odd number of times if the point is inside and an even number of times if it is outside [86], and this even-odd rule is used to voxelize the meridian section.

The second checks the volume of the reconstructed contours. The volume generated by a region of the meridian plane, $V = \iint 2\pi r \, \mathrm{d}r \, \mathrm{d}z$, becomes after integration in r a sum of washer areas πr^2 along the boundary [74],

$$V = \oint \pi r^2 \, \mathrm{d}z, \tag{2.37}$$

for a counterclockwise boundary. A straight edge from (r_a, z_a) to (r_b, z_b) contributes $\frac{\pi}{3}(r_a^2 + r_a r_b + r_b^2)(z_b - z_a)$, and cavities, oriented clockwise, contribute with a negative sign.

The third extracts and simplifies those contours. Marching squares, the two-dimensional analogue of marching cubes [87], locates the iso-contour of the density field by linear interpolation along the cell edges. The Douglas-Peucker algorithm [88] then removes vertices while keeping every removed vertex within a prescribed distance of the simplified polyline, which bounds the geometric error by that tolerance.

The fourth controls the faceting of the revolved surface. A facet spanning an angle $\Delta\theta$ on a circle of radius r is a chord whose midpoint lies at $r\cos(\Delta\theta/2)$ from the axis, so the largest gap between facet and circle is $r\,[1 - \cos(\Delta\theta/2)] \approx r\,\Delta\theta^2/8$, which gives Eq. (3.18). Halving the angular step reduces this error by a factor of four.

Chapter 3

Formulation of the Method

This chapter turns the theory of Chapter 2 into a design method and describes how it is implemented. It is written in general terms, for any solid of revolution made of a transversely isotropic material under axisymmetric supports and loads. The values specific to the component, such as its geometry, load cases, tolerances, material constants and mesh, are given in Chapter 4.

The chapter follows the path of a design through the framework. Section 3.1 states the design problem and the strategy used to solve it. Sections 3.2 and 3.3 build the material model and the finite element model, and Section 3.4 defines how a design is judged. Sections 3.5 and 3.6 describe the optimizer and the search for the lightest feasible design. Finally, Sections 3.7 to 3.9 turn the designs into printable geometry, test them virtually and verify the implementation.

3.1 Design Problem

The design domain Ω is the meridian section of the component. It lies in the rectangle $[R_{\text{in}}, R_{\text{out}}] \times [0, H]$ of the half-plane $r > 0$, where R_{in} and R_{out} are the inner and outer radii and H is the height. Its boundary has supported parts Γ_u, loaded parts Γ_t and free parts. A layer of thickness t_{s} along the whole contour, the passive shell, stays solid and is excluded from the design.

The component must withstand n_c static load cases, each defined by uniform pressures. With the density $\rho(\boldsymbol{x})$ of Eq. (1.1), the mass of a design is

$$m(\rho) = \rho_{\text{mat}} \int_{\Omega} \rho \, 2\pi r \, \mathrm{d}A. \tag{3.1}$$

In every load case, three functional limits must hold. The displacement magnitude of every node of the critical surfaces Γ_{crit} must not exceed $u_{\text{adm,crit}}$, and that of every node of the remaining boundary Γ_{rem} must not exceed $u_{\text{adm,rem}}$. The yield safety factor $\text{FS}_{p99.5} = S_y/\sigma_{\text{VM},p99.5}$, based on the percentile of Eq. (2.20), must be at least FS_{min}. The utilization

index U_c of Section 3.4 combines the three limits and does not exceed 1 when all of them hold.

The design problem is therefore the particular case of Eq. (1.1) with $\Phi = m$ and one constraint $G_c = U_c - 1$ per load case

$$\begin{aligned} \min_{\rho} \quad & m(\rho) \\ \text{subject to} \quad & \boldsymbol{K}(\rho)\,\boldsymbol{U}_c = \boldsymbol{F}_c, \qquad U_c(\rho) \le 1, \qquad c = 1, \dots, n_c, \\ & \rho(\boldsymbol{x}) \in \{0, 1\} \quad \forall\, \boldsymbol{x} \in \Omega. \end{aligned} \tag{3.2}$$

This is the minimization of mass under displacement and stress limits stated in Specific Objective 1 of Section 1.4.

Problem (3.2) is not solved directly. Its constraints are local, and each displacement constraint would need its own adjoint solution, as explained in Section 2.7.7. The framework solves it instead in two nested levels. At the inner level, SIMP minimizes the weighted compliance J_w of Eq. (2.35), or a single compliance, for a prescribed volume fraction V_f, following the formulation of Section 3.5. At the outer level, a search on V_f evaluates $U_c \le 1$ on each binary design and selects the lightest feasible one, which solves the epsilon-constraint problem of Eq. (2.36), as developed in Section 3.6.

Two kinds of problem are solved with this strategy. The *multi-load* problem considers all n_c cases at once and gives the design proposed for the component. The *single-case* problems consider one case at a time and show whether the multi-load formulation is necessary.

The framework is implemented in Python, in a single notebook with ten blocks that share their state in memory. Figure 3.1.1 shows how the blocks connect with the physical and experimental stages of the work. The rest of the chapter describes each stage of the method together with the block that implements it.

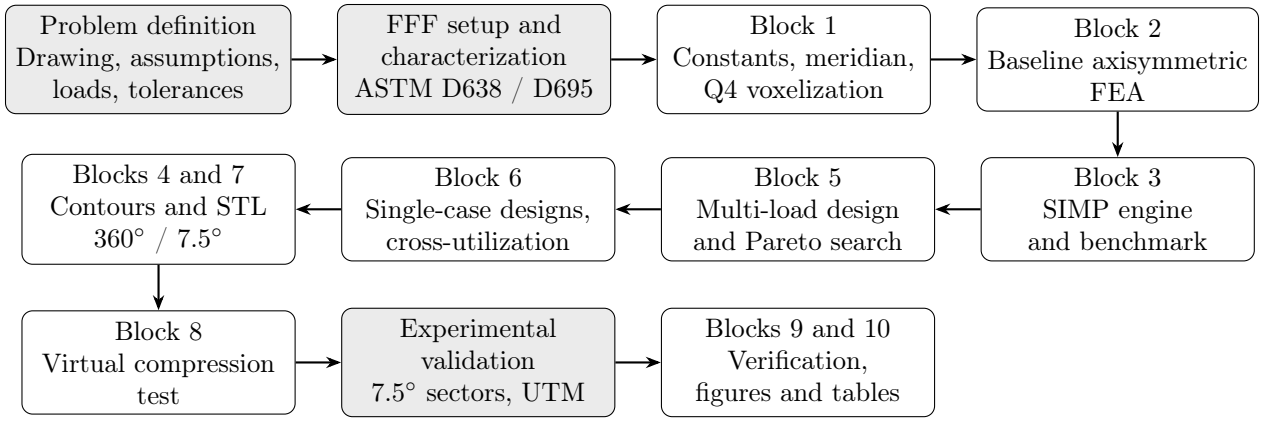


Figure 3.1.1: Workflow of the framework. Shaded boxes are physical or experimental stages, and white boxes are blocks of the notebook.

3.2 Constitutive Model of the Printed Material

The first block defines the material. The part is printed with its axis of revolution along the build direction z, and its in-plane properties are assumed equal, $E_\theta = E_r$, as justified in Section 4.2.2. The printed material is therefore the homogenized, transversely isotropic continuum of Eq. (2.13), with five independent constants.

Only two of these constants are measured on printed specimens, as described in Section 4.3. The in-plane modulus E_r comes from tension parallel to the layers, and the out-of-plane modulus E_z from compression across them. Biaxial strains and shear were not measured, so the other three constants are estimated. The in-plane Poisson's ratio $\nu_{r\theta}$ is taken from the bulk polymer. The out-of-plane ratios follow from reciprocity, Eq. (2.11), and Huber's closure, Eq. (2.17)

$$\nu_{rz} = \nu_{r\theta}\sqrt{\frac{E_r}{E_z}}, \qquad \nu_{zr} = \nu_{r\theta}\sqrt{\frac{E_z}{E_r}}. \tag{3.3}$$

The interlaminar shear modulus is $G_{rz} = \alpha_G E_z$. The factor α_G is chosen below the isotropic and Huber values to reflect imperfect bonding, and it is varied in the sensitivity analysis, as discussed in Sections 4.4 and 5.10.

Before assembly, the constants must satisfy the Lempriere conditions of Eq. (2.16) [65]. With Huber's closure they hold for any $-1 < \nu_{r\theta} < 0.5$, whatever E_r/E_z, because the normal block is a scaled isotropic block (Eq. (2.18)). The constitutive matrix is then $\boldsymbol{C}_0 = \boldsymbol{S}^{-1}$ in the closed form of Eq. (2.15), and the values of the two materials are checked in Section 4.4.

3.3 Axisymmetric Finite Element Model

With the material defined, the framework builds the finite element model of the meridian section. It starts from the STL file of the component, extracts the meridian contour, meshes it on a structured grid, and assembles the axisymmetric stiffness of Section 2.6.

3.3.1 Meridian Section, Mesh and Element Volumes

The input geometry is a closed STL surface mesh. If its axis lies along x, a rigid rotation first aligns it with z, and the part is translated so that its base lies on $z = 0$. The mesh is then cut by the meridian half-plane $x = 10^{-4}\,\mathrm{mm}$, $y < 0$, where the small offset avoids vertices lying exactly on the plane. The two edge crossings of each intersected triangle are interpolated linearly and mapped to $r = \sqrt{x^2 + y^2}$ and z. The result is the meridian contour $\partial\Omega$ as a closed polygon.

The rectangle $[R_{\mathrm{in}}, R_{\mathrm{out}}] \times [0, H]$ is divided into a structured grid of $n_r \times n_z$ Q4 elements of size $\Delta r \times \Delta z$, numbered row by row with counterclockwise connectivity. The grid makes the Jacobian equal in every element, which allows a fully vectorized integration. It also turns the density field into an $n_z \times n_r$ image for connected-component labeling and marching squares, and gives the Helmholtz filter a uniform length scale. Since $R_{\mathrm{in}} > 0$, the domain does not touch the axis and the hoop term $1/r$ stays bounded.

Each element is classified as solid or void with the even-odd rule of Section 2.10. A ray in the $+r$ direction is cast from its centroid, and its crossings with the contour are counted. Elements with an odd count form Ω, and the others are excluded from the model.

The element volumes follow from the theorem of Pappus, as presented in Section 2.6.2. The element between the radii r_1 and r_2 generates a ring of exact volume

$$V_e = \pi(r_2^2 - r_1^2)\,\Delta z = 2\pi r_c\,\Delta r\,\Delta z, \tag{3.4}$$

and the discrete form of Eq. (3.1) is

$$m(\boldsymbol{\rho}) = \rho_{\text{mat}} \sum_{e\in\Omega} \rho_e V_e. \tag{3.5}$$

As a check of the voxelization, the voxelized volume $V_0 = \sum_e V_e$ is compared with the volume enclosed by the STL surface, computed with the divergence theorem.

3.3.2 Element Stiffness and Solution

On this mesh, each element uses the strains of Eq. (2.3), discretized with the Q4 element of Section 2.6.1. The nodal blocks of the strain-displacement matrix are

$$\boldsymbol{B}_i(\xi,\eta) = \begin{bmatrix} \partial N_i/\partial r & 0 \\ 0 & \partial N_i/\partial z \\ N_i/r(\xi,\eta) & 0 \\ \partial N_i/\partial z & \partial N_i/\partial r \end{bmatrix}, \qquad r(\xi,\eta) = \sum_{i=1}^{4} N_i(\xi,\eta)\, r_i, \tag{3.6}$$

where the third row is the discrete hoop strain. The element matrix of Eq. (2.23) is integrated with the 2×2 Gauss-Legendre rule, with points $\xi_k, \eta_l = \pm 1/\sqrt{3}$ and unit weights:

$$\boldsymbol{k}_e^0 \approx \frac{\Delta r\,\Delta z}{4} \sum_{k=1}^{2}\sum_{l=1}^{2} \boldsymbol{B}^\top(\xi_k,\eta_l)\,\boldsymbol{C}_0\,\boldsymbol{B}(\xi_k,\eta_l)\; 2\pi r(\xi_k,\eta_l). \tag{3.7}$$

This rule is accurate when the radius varies little within an element, $\Delta r \ll R_{\text{in}}$, as derived in Section 2.6.1, which the mesh of Section 4.5 satisfies. Volumetric locking is not expected either. With Huber's closure the normal block is a scaled isotropic block with $\nu_{r\theta}$ well below 0.5, as shown in Eq. (2.18), even though ν_{rz} exceeds 0.5 [51].

Supports are imposed by eliminating the restrained degrees of freedom (DOFs). A condition $u_z = 0$ on a supported surface removes the only rigid-body mode, the axial translation, as explained in Section 2.6.2. Nodes outside Ω, and in the binary designs the nodes of void elements, are removed from the system rather than regularized. Pressures become the consistent nodal forces of Eq. (2.24). The problem is linear, so all load cases share one stiffness matrix, which is factorized once and solved for each $\boldsymbol{F}_c$.

3.4 Utilization Index

Each solved design is judged by its stresses and displacements. Element stresses are evaluated at the centroid, $\boldsymbol{\sigma}_e = \boldsymbol{C}_0 \boldsymbol{B}(0,0)\boldsymbol{u}_e$, as defined in Section 2.6.2, and the von Mises stress follows from Eq. (2.19) as $\sigma_{\mathrm{VM},e} = \sqrt{3J_2}$. The largest element stress is mesh-dependent at singular corners, as discussed in Section 2.5.2. The yield check therefore uses the percentile $\sigma_{\mathrm{VM},p99.5}$ of Eq. (2.20), computed over the solid elements only. Section 5.2 compares the maximum and the percentile under mesh refinement.

For each binary design and load case c, the three limits of Section 3.1 are combined into the utilization index

$$U_c = \max\left(\frac{\max_{i\in\Gamma_{\mathrm{crit}}} |\boldsymbol{u}_{i,c}|}{u_{\mathrm{adm,crit}}},\ \frac{\max_{i\in\Gamma_{\mathrm{rem}}} |\boldsymbol{u}_{i,c}|}{u_{\mathrm{adm,rem}}},\ \frac{\mathrm{FS}_{\min}\,\sigma_{\mathrm{VM},p99.5,c}}{S_y} \right), \qquad U = \max_c U_c, \tag{3.8}$$

where $|\boldsymbol{u}_{i,c}|$ is the displacement magnitude of node i in load case c. The passive shell is never removed, so the control nodes always belong to solid elements. The yield term uses the von Mises criterion with the compressive yield strength S_y across the layers. The verification adds a directional check with the in-plane tensile strength $S_{y,T}$, as reported in Section 5.9.

Each term of Eq. (3.8) is the ratio of a response to its admissible value. The utilization is therefore the fraction of the most restrictive limit that the design consumes, and the term that attains the maximum identifies the governing constraint. For example, $U = 0.48$ means that the governing constraint is at 48% of its limit. A design is feasible if $U \leq 1$. By Eq. (2.21), $1/U_c$ is also the factor by which the loads of case c could grow before that limit is reached.

3.5 SIMP Topology Optimization

The inner level of the strategy is a SIMP optimization at a fixed volume fraction. The stiffness of each element follows Eq. (2.25) with $p = 3$ and $E_{\min} = 10^{-9}$ [17, 76]. Elements whose centroids lie within a distance t_{s} of the contour form the passive shell, fixed at $\bar{\rho}_e = 1$.

The shell has three roles. It preserves the outer geometry, including the supported and critical surfaces. It keeps the loaded surfaces fixed, so the loads do not depend on the design and no load is lost at binarization. Its thickness is also of the order of the perimeters that the printer deposits along the walls, as detailed in Section 4.2.2. As a consequence, the target V_f must exceed the volume fraction of the shell. Figure 3.5.1 summarizes the loop described below.

3.5.1 Filtering and Projection

The loop works with the three density fields of Section 2.7.4. The design densities $\boldsymbol{\rho}$ are updated by the optimizer. The filtered densities $\tilde{\boldsymbol{\rho}}$ impose the length scale $R_{\min}$. The nearly binary projected densities $\bar{\boldsymbol{\rho}}$ enter the stiffness.

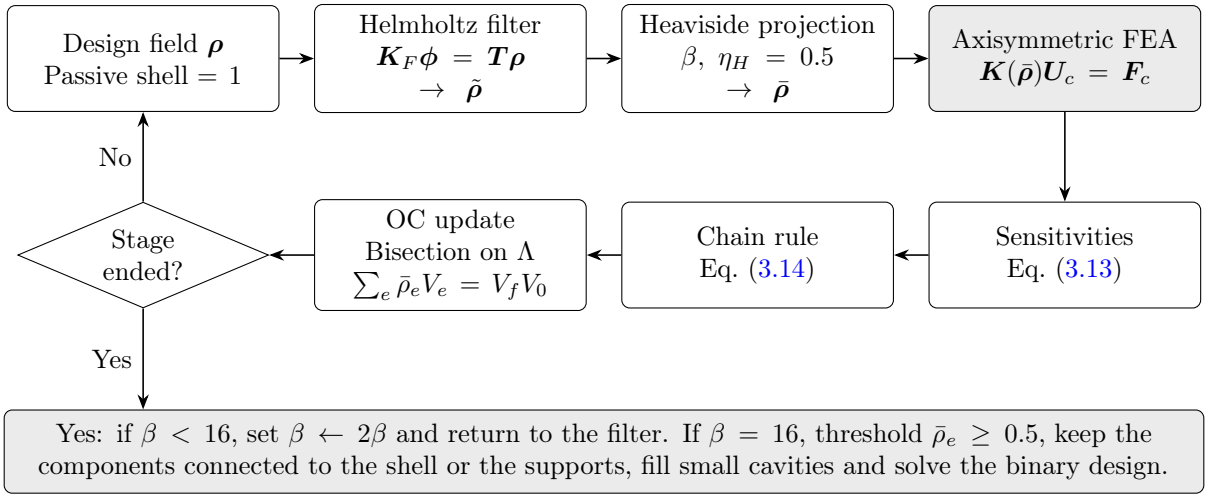


Figure 3.5.1: SIMP loop with axisymmetric Helmholtz filter, Heaviside projection with β-continuation and volume-weighted OC update.

The filter is the axisymmetric Helmholtz equation of Eq. (2.29) [78], with r_H from Eq. (2.28). The filtered field is interpolated with the bilinear nodal functions, $\tilde{\rho}(\boldsymbol{x}) \approx \boldsymbol{N}(\boldsymbol{x})\,\boldsymbol{\phi}$, while the design density is constant within each element. Taking $v = N_i$ in the weak form gives one equation per node,

$$\boldsymbol{K}_F\boldsymbol{\phi} = \boldsymbol{T}\boldsymbol{\rho}, \qquad \tilde{\boldsymbol{\rho}} = \boldsymbol{D}^{-1}\boldsymbol{T}^{\top}\boldsymbol{\phi}, \tag{3.9}$$

with

$$\boldsymbol{K}_F = \sum_e \int_{A_e} \left(r_H^2 \nabla \boldsymbol{N}^{\top} \nabla \boldsymbol{N} + \boldsymbol{N}^{\top}\boldsymbol{N}\right) 2\pi r \,\mathrm{d}A, \qquad T_{ie} = \int_{A_e} N_i \, 2\pi r \,\mathrm{d}A, \qquad \boldsymbol{D} = \mathrm{diag}(V_e). \tag{3.10}$$

Here $\boldsymbol{K}_F$ collects the gradient and mass terms, and $\boldsymbol{T}$ distributes each element density to its nodes with the weight $2\pi r$. The second relation of Eq. (3.9) returns one density per element, the volume average of the nodal field over it. Since $\sum_i T_{ie} = V_e$, a uniform density passes through the filter unchanged. $\boldsymbol{K}_F$ does not depend on the design, so it is factorized once, and each filter evaluation costs one forward and one backward substitution.

The filtered density is then projected with the smooth Heaviside function of Wang et al. [37], with threshold $\eta_H = 0.5$

$$\begin{aligned} \bar{\rho}_e &= \frac{\tanh(\beta\eta_H) + \tanh(\beta(\tilde{\rho}_e - \eta_H))}{\tanh(\beta\eta_H) + \tanh(\beta(1-\eta_H))}, \\ \frac{\partial \bar{\rho}_e}{\partial \tilde{\rho}_e} &= \frac{\beta\left[1 - \tanh^2(\beta(\tilde{\rho}_e - \eta_H))\right]}{\tanh(\beta\eta_H) + \tanh(\beta(1-\eta_H))}. \end{aligned} \tag{3.11}$$

The steepness follows the continuation $\beta = 1, 2, 4, 8, 16$. The first stage is close to the identity Eq. (2.30) and behaves like a filtered SIMP problem. Each doubling of β then sharpens the design towards 0-1. After each projection, shell elements are reset to $\bar{\rho}_e = 1$ with zero derivative.

The remaining gray level is monitored with the measure of non-discreteness [80],

$$M_{\mathrm{nd}} = \frac{\sum_e 4\bar{\rho}_e(1-\bar{\rho}_e)V_e}{\sum_e V_e}, \tag{3.12}$$

which is 0 for a binary design and 1 when every density equals 0.5.

3.5.2 Sensitivities and Optimality Criteria Update

Each iteration computes the gradients in two steps. Compliance is self-adjoint, as shown in Section 2.7.5, so the sensitivity of J_w with respect to the physical densities is the weighted sum of Eq. (2.32) over the load cases, with $w_c = 1$ for a single case

$$\frac{\partial J_w}{\partial \bar{\rho}_e} = -p\,\bar{\rho}_e^{\,p-1}(1-E_{\min}) \sum_c w_c\, \boldsymbol{u}_{e,c}^{\top} \boldsymbol{k}_e^0\, \boldsymbol{u}_{e,c}. \tag{3.13}$$

This gradient, and the volume gradient that starts from the element volumes $\boldsymbol{D1}$, are then carried back to the design variables through the projection and the filter

$$\frac{\partial J_w}{\partial \boldsymbol{\rho}} = \boldsymbol{T}^{\top} \boldsymbol{K}_F^{-1} \boldsymbol{T} \boldsymbol{D}^{-1} \left(\frac{\partial J_w}{\partial \bar{\boldsymbol{\rho}}} \odot \frac{\partial \bar{\boldsymbol{\rho}}}{\partial \tilde{\boldsymbol{\rho}}} \right), \qquad \frac{\partial V}{\partial \boldsymbol{\rho}} = \boldsymbol{T}^{\top} \boldsymbol{K}_F^{-1} \boldsymbol{T} \boldsymbol{D}^{-1} \left(\boldsymbol{D1} \odot \frac{\partial \bar{\boldsymbol{\rho}}}{\partial \tilde{\boldsymbol{\rho}}} \right), \tag{3.14}$$

where $\odot$ is the element-wise product. Because the volume gradient carries V_e, material at large radius is more expensive.

The design variables are updated with the OC scheme of Section 2.7.6 [10, 23]

$$\begin{aligned} \rho_e^{(k+1)} &= \mathrm{clip}\left(\rho_e^{(k)} \sqrt{\frac{-\partial J_w/\partial \rho_e}{\Lambda\, \partial V/\partial \rho_e}},\ \rho_e^{\mathrm{lo}},\ \rho_e^{\mathrm{up}} \right), \\ \rho_e^{\mathrm{lo}} &= \max\!\left(0, \rho_e^{(k)} - \Delta\rho_{\lim}\right), \qquad \rho_e^{\mathrm{up}} = \min\!\left(1, \rho_e^{(k)} + \Delta\rho_{\lim}\right). \end{aligned} \tag{3.15}$$

The move limit $\Delta\rho_{\lim} = \max(0.05,\, 0.15\min(1, 2/\beta))$ decreases as the projection becomes sharper. The multiplier Λ is found by bisection, evaluating the filter and the projection for each trial value. The volume constraint $\sum_e \bar{\rho}_e V_e = V_f V_0$ therefore holds on the projected densities from the second iteration on.

3.5.3 Binarization and Engine Settings

At the end of the $\beta = 16$ stage, the projected field is thresholded at $\bar{\rho}_e \geq 0.5$ and the shell is added. The binary image is labeled with 4-connectivity, so elements that share only a corner are not connected, as explained in Section 2.7.7. Only the components connected to the shell or the supports are kept. Enclosed voids smaller than 20 elements are filled, which does not change the thickness of the members.

The void elements and their nodes are then removed, and the binary design is solved to compute all reported quantities. Its volume fraction differs slightly from the target. Reported fractions are therefore the targets, while masses come from the binary designs.

The engine is implemented to be fast and robust. $\boldsymbol{K}_F$ and $\boldsymbol{T}$ are integrated with 2×2 Gauss points, and $\boldsymbol{K}_F$ is factorized once with a sparse direct solver after checking that the filter preserves a uniform field. The element energies of all load cases are evaluated in one vectorized operation. For numerical safety, the compliance and volume sensitivities are kept away from zero, below -10^{-30} and above 10^{-30}, and those of the shell elements are set to zero. Table 3.5.1 lists the parameters of the engine and of the search.

Table 3.5.1: Parameters of the optimization engine and of the search for the minimum-mass design.

Parameter	Value	Reference
SIMP penalization p / void stiffness $E_{\min}$	3 / 10^{-9}	Eq. (2.25)
Filter radius $R_{\min}$ / Helmholtz length r_H	2.6 mm / 0.7506 mm	Eq. (2.28)
Passive shell thickness t_s	2.0 mm	Sec. 3.5
Projection threshold η_H / steepness β	0.5 / $1, 2, 4, 8, 16$	Eq. (3.11)
Move limit $\Delta\rho_{\text{lim}}$	$\max(0.05,\ 0.15 \min(1, 2/\beta))$	Eq. (3.15)
Bisection on Λ	Geometric mean in $[10^{-12}, 10^{12}]$, relative tolerance 10^{-4}	Eq. (3.15)
End of a β-stage	At least 6 iterations, and $\max_e \lvert\Delta\rho_e\rvert < 0.01$ or relative change of J_w over 5 iterations $< 2 \times 10^{-4}$	Sec. 3.5.2
Iteration cap per β-stage	40 in Block 3, 30 in Blocks 5 and 6 with at most 150 per run	Sec. 3.5.2
Initial design	Uniform density that satisfies the volume constraint, warm start from the previous run within a search	Sec. 3.6
Binarization	$\bar{\rho}_e \geq 0.5$, 4-connectivity, components connected to the shell or supports kept, voids below 20 elements filled	Sec. 3.5.3
Adaptive weights, multi-load	$w_c^{(0)} \propto 1/J_{c,\text{solid}}$, damping 1/2, exponent $q = 2$	Eq. (3.17)
Sweep of V_f, multi-load	0.90, 0.80, 0.70, 0.60, 0.50, 0.45, 0.40, 0.35, 0.30	Sec. 3.6
Sweep of V_f, single-case	From $V_{f,\text{multi}} + 0.10$ rounded up to a multiple of 0.05, at most 0.90, in steps of 0.05	Sec. 3.6
Fractions skipped	$V_f\, V_0 \leq 1.02\, V_{\text{shell}}$	-
Early stop / bisection steps	2 consecutive infeasible designs / 3	Sec. 3.6

3.5.4 Benchmark of the Engine

Before the design searches, Block 3 verifies the engine with a single-load benchmark. The PA6 component is loaded by an axial reference pressure on its top face and optimized with $V_f = 0.60$. The benchmark is judged with its own thresholds, not with the design criteria of Section 3.4. Its maximum displacement must not exceed 1.25 times that of the solid, and $\mathrm{FS}_{p99.5}$ must be at least 2.0.

The gain in stiffness per unit mass is measured with the efficiency index

$$\mathrm{IE} = \frac{J_{\mathrm{solid}}\, m_0}{J_{\mathrm{opt}}\, m_{\mathrm{opt}}}, \qquad \mathrm{IE}_{\mathrm{val}} = \mathrm{IE} \cdot \min\left(1,\ \frac{1.25\, u_{\mathrm{solid}}}{u_{\mathrm{opt}}}\right) \cdot \min\left(1,\ \frac{\mathrm{FS}_{p99.5}}{2.0}\right), \tag{3.16}$$

where J_{solid}, m_0 and u_{solid} are the compliance, mass and maximum displacement of the solid. A value $\mathrm{IE} > 1$ means more stiffness per gram than the solid, and $\mathrm{IE}_{\mathrm{val}}$ penalizes the index when a threshold is missed. The index is used only in this benchmark, because the designs of the component are selected by minimum mass subject to $U \leq 1$. The results are reported in Section 5.3.

3.6 Search for the Minimum-Mass Design

The outer level searches the volume fraction for the lightest feasible design. In the multi-load problem, it also adapts the weights of $J_w = \sum_c w_c\, \boldsymbol{F}_c^{\top} \boldsymbol{U}_c$ between runs.

The initial weights are $w_c \propto 1/J_{c,\mathrm{solid}}$, normalized to add up to one, following Section 2.8. Compliance, however, is a global measure and does not directly control the displacements of the critical walls. After each run, the weights are therefore updated with a damped rule that favors the cases with the highest utilization

$$w_c \leftarrow \frac{1}{2} w_c + \frac{1}{2} \frac{U_c^{\,q} / J_{c,\mathrm{solid}}}{\sum_{k=1}^{n_c} U_k^{\,q} / J_{k,\mathrm{solid}}}, \qquad q = 2, \tag{3.17}$$

where U_c is the utilization of case c in the binary design just obtained. The weights evolve along the whole search, including the bisection steps.

The search itself is summarized in Figure 3.6.1. It starts with a descending sweep over $V_f = 0.90, 0.80, 0.70, 0.60, 0.50, 0.45, 0.40, 0.35$ and 0.30, in which each run starts from the converged design variables of the previous one. The sweep stops after two consecutive infeasible designs, with $U > 1$.

Three bisection steps then refine the interval between the lowest feasible fraction $V_{f,\mathrm{ok}}$ and the highest infeasible fraction $V_{f,\mathrm{fail}}$ below it. Each step starts from the current feasible design. The final resolution is one eighth of the initial interval, 0.0125 for an interval of 0.10 and 0.00625 for 0.05. Among all evaluated binary designs, the feasible one of minimum mass is selected, and the non-dominated pairs of mass and U form the Pareto front.

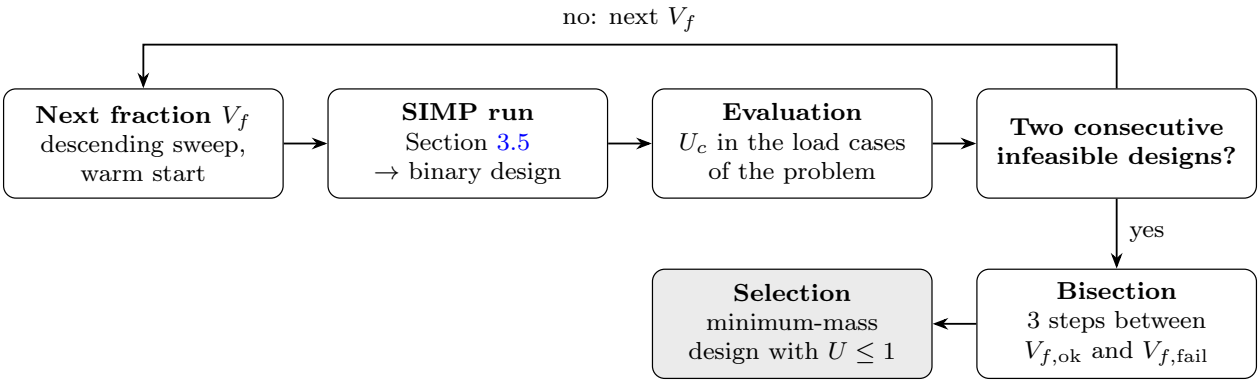


Figure 3.6.1: Search for the minimum-mass feasible design along the volume fraction.

3.7 Geometric Reconstruction and STL Export

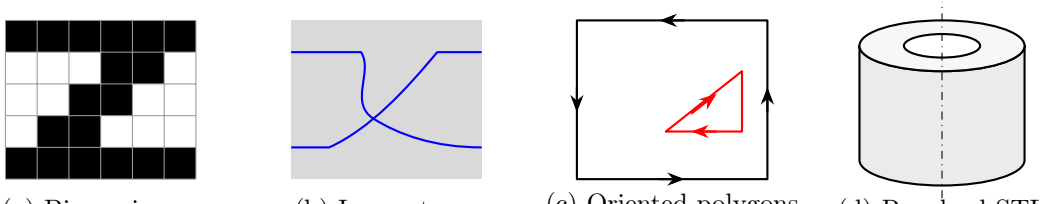
The selected designs exist only as binary images of the meridian section. Blocks 4 and 7 turn them into closed STL files, for the complete ring and for the test sectors, using the relations of Section 2.10.

3.7.1 Meridian Contours

Each binary image is first converted into smooth closed polygons (Figure 3.7.1). A Gaussian smoothing with a standard deviation of 0.7 cells removes the staircase of inclined members. The image is then padded with a one-cell border of zeros, so that contours touching the domain boundary close.

Marching squares at the level 0.5 [87] extracts the iso-contours with the mapping $z = (\mathrm{row} - 0.5)\Delta z$ and $r = R_{\mathrm{in}} + (\mathrm{col} - 0.5)\Delta r$, where the offset of 0.5 compensates the padding. The Douglas-Peucker algorithm [88] simplifies the contours with a tolerance of 0.03 mm, about $0.1\,\Delta r$. Duplicated and collinear vertices are then removed.

Each polygon is finally oriented by its nesting depth. Polygons enclosed by an even number of others are outer boundaries and run counterclockwise. Those enclosed by an odd number are cavities and run clockwise. The solid therefore lies to the left of each edge, as required by Eq. (2.37).



(a) Binary image (b) Iso-contours (c) Oriented polygons Outer CCW, cavity CW (d) Revolved STL

Figure 3.7.1: Reconstruction of a printable solid from a binary design, schematic.

3.7.2 Revolution and Closed Meshes

The oriented polygons are revolved by indexed copying. Each vertex (r_i, z_i) is copied to N_θ angular positions θ_k, giving the points $(r_i \cos\theta_k, r_i \sin\theta_k, z_i)$. Each contour edge produces two triangles between consecutive positions, with shared vertex indices. For a full revolution the indices are taken modulo N_θ, so the lateral surface is closed by construction.

The angular step follows from a chordal tolerance t_{ch}, the largest distance allowed between a flat facet and the exact circle at R_{out}, as derived in Section 2.10

$$R_{\mathrm{out}}\left(1-\cos\frac{\Delta\theta}{2}\right) \le t_{\mathrm{ch}} \quad \Longrightarrow \quad \Delta\theta = 2\arccos\left(1-\frac{t_{\mathrm{ch}}}{R_{\mathrm{out}}}\right), \qquad N_\theta = \left\lceil \frac{\Theta}{\Delta\theta} \right\rceil . \tag{3.18}$$

The actual step, Θ/N_θ, is slightly smaller than this bound.

For a sector with $\Theta < 360°$, the two planar cut faces are triangulated by ear clipping, using the vertex indices of the first and last angular positions. Their orientation is set from the signed area of the cap. Collinear vertices must be removed beforehand, because the cap triangulation discards them while the lateral surface keeps them, which would leave the seam open. If a mesh is not closed, the export is repeated with a larger collinearity tolerance. Table 3.7.1 lists the exports.

Table 3.7.1: Angular discretization and purpose of the STL exports.

Export	Chord tol. t_{ch}	N_θ	$\Delta\theta$	Purpose
Complete ring, $\Theta = 360°$, all selected designs	5.0 µm	345	1.043°	CAD record of the optimized components, 12.7 to 32.7 MB per file
Cut-away view, $\Theta = 270°$, Block 3 benchmark only	5.0 µm	259	1.043°	Visualization of the internal structure
Test sector, $\Theta = 7.5°$, all selected designs	0.5 µm	23	0.326°	Compression test specimens printed at full scale, 0.9 to 2.3 MB per file

Every exported mesh is checked for closure. Each directed edge must appear exactly once, and its opposite edge must also be present, which guarantees a closed, consistently oriented surface. The signed volume orients the normals outward. It is compared with the finite element volume scaled by $\Theta/360°$ and with the volume of revolution of the polygons from Eq. (2.37).

3.8 Virtual Compression Test

The exported sectors are tested in compression, as described in Section 4.7. To predict these tests, Block 8 simulates the same experiment on the full ring.

3.8.1 Model and Solution

Each design is compressed between two rigid platens at $z = 0$ and $z = H$ under quasi-static displacement control, without the supports of the design problem, as sketched in Figure 3.8.1. The lower platen imposes $u_z = 0$ on the nodes at $z = 0$. The upper platen imposes $u_z = -\delta$ on the nodes Γ_{top} at $z = H$. The contact is bonded, $u_r = 0$ on the platen nodes, which represents high friction. The reaction of the upper platen, $F(\delta) = \sum_{i \in \Gamma_{\text{top}}} R_{z,i}$, is the virtual load-cell signal.

For equivalent geometry, material properties and imposed platen displacement, bonded contact and axisymmetric constraints restrict the admissible displacement field. By Eq. (2.9), the stiffness scaled by $\Theta/360^\circ$ is therefore a conditional upper bound for an otherwise equivalent sector with relaxed constraints.

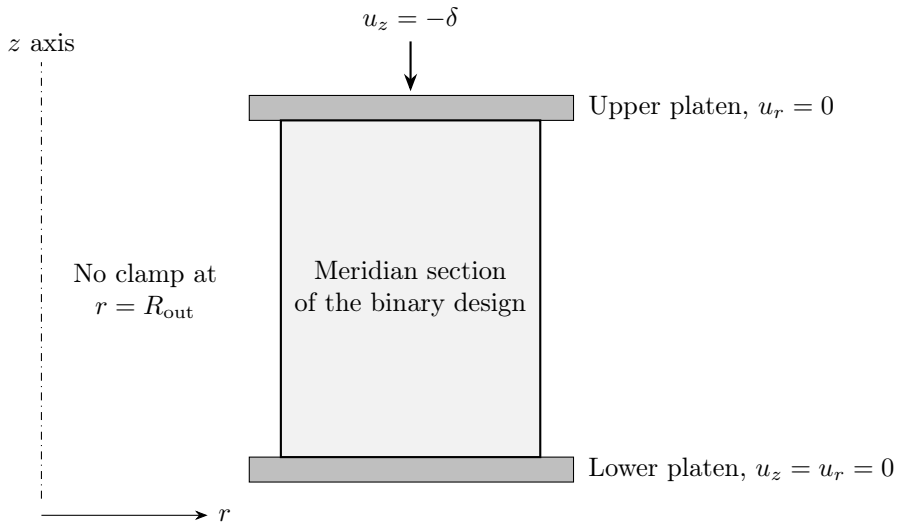


Figure 3.8.1: Boundary conditions of the virtual compression test.

Yielding is introduced with the elastic compensation method of Section 2.9.2, through a secant stiffness factor d_e per element, bounded below by 10^{-9}. For given factors the problem is linear. The system is therefore solved once for a unit platen displacement and scaled by δ. The first solution, with $d_e = 1$, gives the elastic response.

In the secant iterations, only a few factors change between solutions. The systems are therefore solved by preconditioned conjugate gradients, using the sparse LU factorization of an earlier state as preconditioner. The factorization is recomputed only when the solver fails or needs more than 80 iterations.

3.8.2 Metrics

Four quantities are extracted from each curve $F(\delta)$. The initial stiffness K_{ini}, in kN/mm, is the slope of the elastic solution. The yield onset (δ_y, F_y) is the point at which $\sigma_{\text{VM},p99.5} = S_y$. The collapse point $(\delta_{\text{col}}, F_{\text{col}})$ is the first step at which the tangent stiffness $K_{\text{t}} = \Delta F/\Delta\delta$

falls below 5% of K_{ini}, which indicates a mechanism. Finally, the collapse load per unit mass F_{col}/m, in N/g, and the energy to collapse $\mathcal{E}_{\mathrm{col}} = \int_0^{\delta_{\mathrm{col}}} F \, \mathrm{d}\delta$, in J, are stored.

The model has two known limitations, which decide which metric is compared with the experiment. It has no contact formulation, since only the nodes at $z = H$ follow the platen. Nodes of the top surface slightly below $z = H$ can therefore cross the platen plane at large displacements. It is also geometrically linear, as discussed in Section 2.9.2, and every collapse at $\delta/H > 5\%$ is flagged.

For these reasons, K_{ini} is the main quantity compared with the physical test, and F_{col} is only indicative. The test also loads the designs through the platens instead of their service supports. It therefore checks the optimized geometry and the material model rather than the service performance, as clarified in Section 4.7.1.

3.9 Verification Procedures

The last two blocks verify the stored results without generating new designs. Block 9 runs seven independent checks, and Block 10 regenerates the figures and tables of Chapter 5.

Block 9 first re-implements the element matrices and the evaluation of the designs. It compares the utilization, displacements, stress percentile and compliance of every design in every load case with the stored values. It also checks the residual of the linear systems, the energy identity of Eq. (2.7), the axial equilibrium and the fraction of load applied on removed or restrained nodes.

It then evaluates a directional strength criterion. The in-plane tensile stress, $\max(\sigma_r, \sigma_\theta) > 0$, is compared with $S_{y,T}$, and the axial compressive stress, $-\sigma_z > 0$, with S_y. The interlaminar tensile stress $\sigma_z > 0$ and the shear stress $|\tau_{rz}|$ are only reported, because their strengths were not characterized.

The remaining checks address the inputs of the model. Block 9 summarizes the robustness of the volume-fraction searches and re-evaluates the selected designs with $\alpha_G = 0.32$, 0.41 and Huber's value, as defined in Section 2.4.5. It repeats the analysis of the solid with meshes of 0.5, 1.0 and 1.5 times the reference density. It also recomputes K_{ini} of the virtual test and measures the penetration of the material into the platens. Its outputs are the tables T01 to T09 and the figures F01 and F02.

Block 10 solves again the twelve designs, the solid and the five selected designs of each material, in the four load cases. From these results it generates the figures `F10_01` to `F10_06` and the tables `T10_01` to `T10_08`, which Chapter 5 cites by file name.

Chapter 4

Application to the Selected Aeronautical Component

This chapter applies the framework of Chapter 3 to the selected component. Section 4.1 defines the component and its requirements. Section 4.2 describes how the parts are printed, and Sections 4.3 and 4.4 characterize the two printed polyamides. Section 4.5 discretizes the component, and Section 4.6 lists the optimization studies. Section 4.7 describes the physical tests with full-scale sectors, and Section 4.8 closes with the assumptions on which the results depend.

4.1 Component Definition and Design Requirements

4.1.1 Reference Component and Supports

The studied part is a solid of revolution inspired by a turbine shaft component taken from an aeronautical drawing. The reference part is made of polyamide 6. The geometry is not an exact copy. It keeps the main features of the drawn part, a ring with a U-shaped meridian section, a central bore and a radial cavity, so the method is applied to a representative aeronautical solid of revolution. As in the drawing, the component consists of two identical 180° halves split along a meridian plane.

The geometry is provided as a closed STL surface mesh of one half, `Solido_half.stl`, shown in Figure 1.1.1. Its inner radius is $R_{\text{in}} = 57.63\,\text{mm}$, its outer radius $R_{\text{out}} = 120.00\,\text{mm}$ and its height $H = 81.00\,\text{mm}$.

The drawing specifies neither service loads nor mounting conditions, so four assumptions are made from the geometry and the probable function of the part. A rigid housing restrains the outer wall and supports the flat base. The two halves behave as a continuous ring whose joint transmits the hoop forces. Uniform pressures on the exposed surfaces form four static load cases. Finally, drawing tolerances motivate the nodal-displacement limits adopted as functional design criteria.

The first assumption gives two kinematic conditions, used in all design load cases and shown

in Figure 4.1.1. The base at $z = 0$, with 201 nodes, slides with $u_z = 0$ and u_r free, like a flat seat that does not restrain radial expansion. The outer wall at $r = R_{\text{out}}$, with 244 nodes, is clamped with $u_r = u_z = 0$, like the rigid housing.

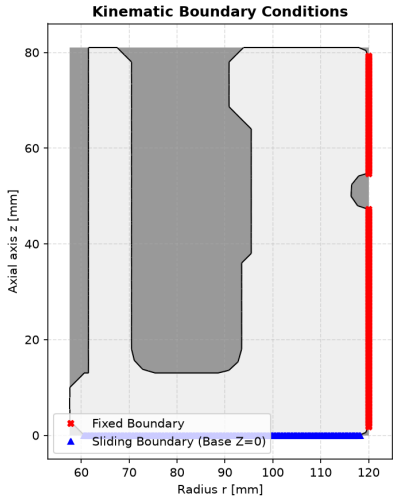


Figure 4.1.1: Supports of the component on the meridian section.

4.1.2 Design Load Cases

The other assumptions define the loads. Table 4.1.1 lists the four static load cases and Figure 4.1.2 draws them. C1 compresses the top face axially, C2 pressurizes the U-shaped cavity, C3 combines both, and C4 applies an outward radial pressure on the inner walls.

The pressures are nominal values, because no service loads are available for the inspiring component. The same loads are used for both materials, so PA6 and PA6-CF are compared under identical forces.

Table 4.1.1: Design load cases and utilization of the solid component, from Table `T10_02`.

Load case	Loaded surfaces	Direction	Pressure [MPa]	Net axial force [N]	Solid U, PA6 / CF
C1, axial	Top annular face, $z = H$	$-z$	$P = 0.390$	-7746.6	0.478 / 0.204
C2, internal	U-cavity: bottom, $70 \le r \le 96$ mm, left wall, $68 \le r \le 76$ mm, and right wall, $88 \le r \le 97$ mm	Normal to each face	$P_i = 0.344$	-4457.2	0.470 / 0.193
C3, combined	Top face and U-cavity, surfaces of C1 and C2	$-z$ and normal	$P = 0.225$, $P_i = 0.375$	-9328.1	0.477 / 0.195
C4, radial	Inner bore, $r \le 63$ mm, and right cavity wall, $88 \le r \le 97$ mm	$+r$	$P_r = 0.368$	0	0.470 / 0.193

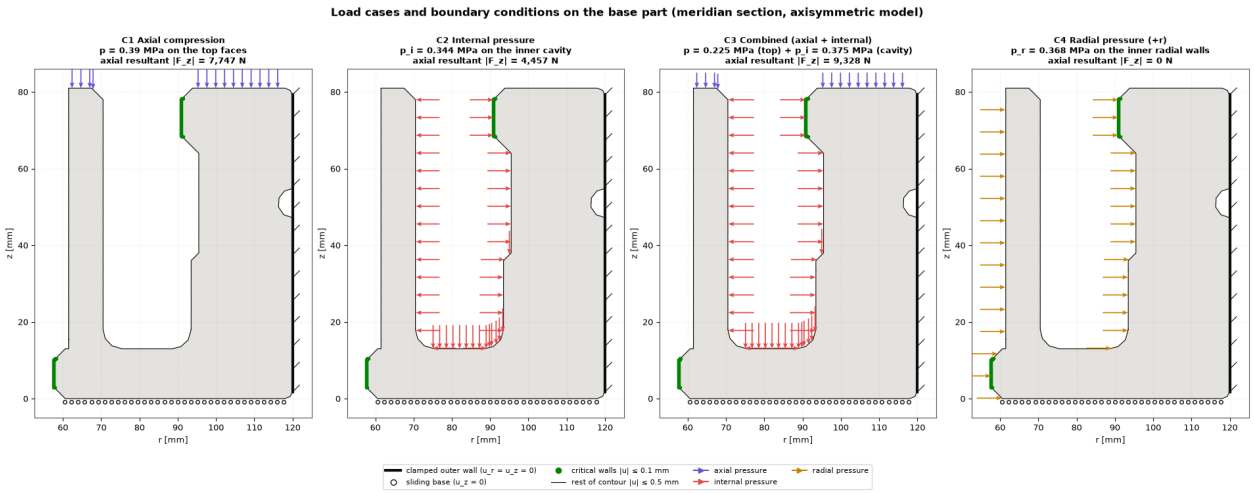


Figure 4.1.2: Supports and loaded surfaces of the four design load cases on the meridian section.

The loaded edges are found on the voxelized boundary and converted into the consistent nodal forces of Eq. (2.24). The voxelized top face follows the grid, so its loaded area, 19 863.0 mm^2, is 7.4% larger than the exact flat area of the STL contour, 18 494.3 mm^2. The axial resultants of Table 4.1.1 are therefore 7.4% higher than the nominal pressure times the exact area. Because the pressures act on the passive shell, the applied loads do not change during optimization, as described in Section 3.5.

4.1.3 Critical Dimensions and Admissible Displacements

The drawing specifies ±0.1 mm for critical dimensions and ±0.5 mm elsewhere. These values motivate the nodal-displacement limits adopted in Section 3.1; they are operational design criteria rather than a direct verification of every toleranced dimension.

The critical walls Γ_{crit} have 70 boundary nodes and $u_{\text{adm,crit}} = 0.10\,\text{mm}$. They form two zones, marked in green in Figure 4.1.2. The first is the lower part of the inner bore, at $r \leq 58.6\,\text{mm}$ and $2.8\,\text{mm} \leq z \leq 10.6\,\text{mm}$. The second is the upper part of the right cavity wall, at $90.2\,\text{mm} \leq r \leq 91.6\,\text{mm}$ and $68.0\,\text{mm} \leq z \leq 78.6\,\text{mm}$.

The rest of the contour forms the remaining boundary Γ_{rem}, with 1450 nodes and $u_{\text{adm,rem}} = 0.50\,\text{mm}$. The strength limit is $\text{FS}_{\text{min}} = 1.5$. These values complete the utilization index of Eq. (3.8).

The criterion controls nodal movement under load. Dimensional acceptance would additionally require relative-displacement checks for the specified dimensions and an allowance for manufacturing deviations.

4.2 Additive Manufacturing Setup

The characterization specimens and the test sectors are printed by FFF on a low-cost desktop printer. This section describes the printer, its cost, and the printing parameters shared by both materials.

4.2.1 Printer, Modifications and Cost

All specimens and test sectors were printed on a Flashforge Adventurer 5M (AD5M), an open-frame CoreXY FFF printer with a build volume of $220 \times 220 \times 220$ mm [89]. Its stock limits are 280 °C for the nozzle and 100 °C for the bed.

To print short-carbon-fiber polyamide, the stock hotend was replaced by a Phaetus Conch hotend designed for the AD5M and for fiber-filled filaments, with a 0.4 mm wear-resistant nozzle [90]. The stock plate was replaced by a textured BIQU CryoGrip Pro Glacier plate to improve first-layer adhesion [91]. The printer is also operated above its stock temperature limits, as required by Table 4.2.2. Figure 4.2.1 shows the modified printer, and the printing parameters were configured and sliced in OrcaSlicer.

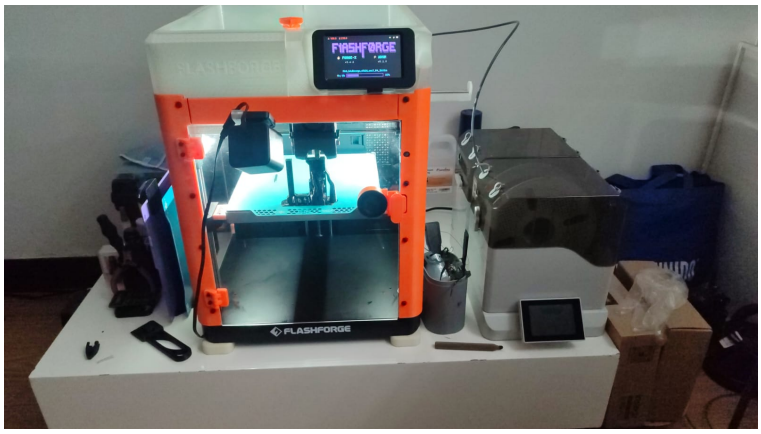


Figure 4.2.1: Modified Flashforge Adventurer 5M used for all printed specimens and sectors.

A goal of the work is to show that the method can be carried out with low-cost equipment. Table 4.2.1 lists approximate prices of the setup and the filaments, and compares them with two commercial printers marketed for fiber-reinforced polyamides. Prices were consulted on 28 October 2026 in the online stores of manufacturers or distributors, in US dollars, without shipping, duties or taxes, so they are indicative only.

The modified setup costs roughly 13 to 19 times less than the UltiMaker Method X Carbon Fiber and 43 to 65 times less than the Markforged Mark Two. The trade-offs are the absence of a heated chamber and the need to run the printer above its stock temperature limits.

4.2.2 Materials and Printing Parameters

Two INSLOGIC filaments are used. The unreinforced filament, sold as Nylon PA6/66 [93], is denoted PA6 in this work. The reinforced filament is a polyamide 6 with 20 % short carbon

Table 4.2.1: Approximate cost of the manufacturing setup, in USD, consulted on 2026-10-28.

Item	Approx. price [USD]	Source
Flashforge Adventurer 5M printer	249 sale to 399 list	[89]
Phaetus Conch hotend for AD5M, 0.4 mm	27	[90]
BIQU CryoGrip Pro Glacier build plate	34	[91]
Total printing setup	310 to 460	
INSLOGIC PA6-CF filament, 1 kg	60, about 0.060 USD/g	[92]
INSLOGIC Nylon PA6/66 filament, 1 kg	46, about 0.046 USD/g	[93]
Reference: printers marketed for fiber-reinforced polyamides		
UltiMaker Method X Carbon Fiber, heated chamber	from 5 999	[94]
Markforged Mark Two, Gen 2	from 20 000	[95]

fiber, denoted PA6-CF [92].

Printed polyamides absorb moisture, which degrades their mechanical properties [96]. The spools were therefore dried at 60 °C for 6 h and fed from a dry enclosure. Both materials share the same process and filament profiles, so the coupons and the sectors of both materials have the same deposition strategy. Table 4.2.2 lists the parameters.

The main choices follow from the material. The disabled part-cooling fan slows cooling and favors interlayer diffusion, as explained in Section 2.2.1. The high nozzle temperature lowers the viscosity of the fiber-filled melt, and top-surface ironing produces flat faces for contact with the compression platens.

The five perimeters, $5 \times 0.44\,\text{mm} = 2.2\,\text{mm}$ in total, are of the same order as the 2.0 mm passive shell described in Section 4.5. The gyroid infill does not concentrate the lines along one or two directions of the layer, so the model assumes $E_\theta = E_r$. This remains an assumption, because only one in-plane direction was tested and the concentric perimeters may stiffen the outer shell circumferentially.

4.3 Experimental Material Characterization

The properties of an FFF part depend on its mesostructure and infill, as discussed in Section 2.2.2. The constants of the model are therefore measured on specimens printed with the parameters of Table 4.2.2.

4.3.1 Specimens and Tests

Each material is characterized with two standard tests, with five specimens per test, which gives 20 curves in total.

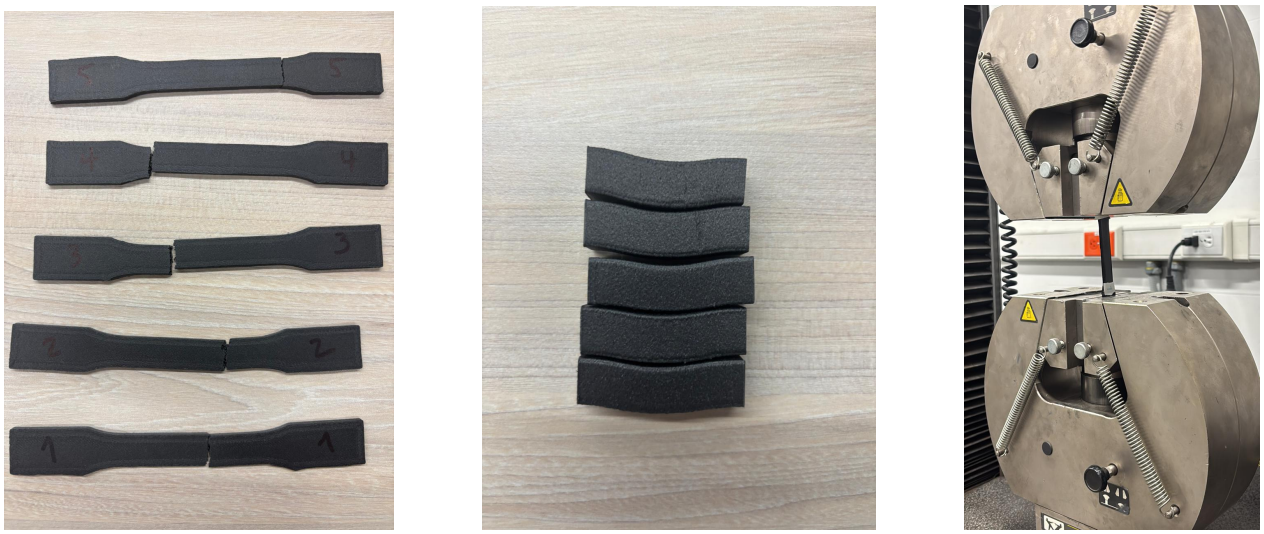
Table 4.2.2: FFF printing parameters, common to PA6 and PA6-CF.

Category	Process parameter	Configured value
Pre-processing	Filament drying	60 °C for 6 h
Thermal	Nozzle temperature, initial / other layers	285 °C / 280 °C
	Build plate temperature	105 °C
	Hot plate temperature, initial / other layers	80 °C to 70 °C / 70 °C
	Part cooling fan, minimum / slow-down layers	0 %, disabled / 2 layers
Geometry	Layer height	0.16 mm
	Line width, initial layer / outer, inner and infill	0.48 mm / 0.44 mm
	Perimeters / top shell layers	5 / 3
	Sparse infill pattern and density	Gyroid, 35 %
	Infill-wall overlap / top-bottom overlap	25 % / 15 %
	Brim type and brim-object gap	Outer only, 0.021 mm
Kinematics	Initial layer and bridge speed	10 mm/s
	Wall, infill and top speed	60 mm/s
	Travel speed / skirt speed	300 mm/s / 15 mm/s
	Acceleration, outer / inner / internal solid / default	2000 / 3000 / 4000 / 5000 mm/s^2
Extrusion	Extruder type and pressure advance	Direct drive, 0.038
	Top solid infill flow ratio / bridge flow ratio	0.95 / 1.50
	Top-surface ironing, angle / flow / inset / spacing / speed	45° / 30 % / 0.21 mm / 0.15 mm / 60 mm/s
	Retraction on layer change, wipe, crossing walls	Disabled, disabled, reduced

Uniaxial tension following ASTM D638 [97] loads the specimen parallel to the deposition plane. It gives the in-plane modulus $E_r = E_\theta$, the tensile yield strength $S_{y,T}$ and the ultimate tensile strength $S_{u,T}$. The dog-bone specimens are printed flat, with a gauge section of 13.0 mm × 3.20 mm and a gauge length of 50.0 mm.

Uniaxial compression following ASTM D695 [98] loads the specimen along the build direction z. It gives the out-of-plane modulus E_z and the compressive yield strength S_y. The prisms are printed upright, with a section of 12.7 mm × 12.7 mm and a height of 50.8 mm.

All coupon test the universal testing machine of the Universidad EAFIT. Figure 4.3.1 shows the PA6-CF specimens after testing and a tensile specimen mounted in the machine. The process for PA6 was the same.



(a) (b) (c)

Figure 4.3.1: PA6-CF tensile specimens (a) and compression prisms (b) after testing, and tensile specimen mounted in the testing machine (c).

4.3.2 Data Reduction and Apparent Density

After taring the initial offsets, the force-displacement records give the engineering stress and strain with the nominal section and gauge length of each specimen. Figure 4.3.2 shows the resulting curves. The elastic modulus is the slope of a least-squares fit to the linear part of each curve.

The measured moduli are $E_r = 839.44$ MPa and $E_z = 294.32$ MPa for PA6, and $E_r = 2056.59$ MPa and $E_z = 684.46$ MPa for PA6-CF.

The yield strength is determined by the 0.2% offset method. In tension, a specimen that broke below 80% of the highest ultimate strength of its batch is excluded from the strength averages, because its failure started at the fillet and not in the gauge length. Its elastic slope still counts for E_r. Specimens 2, 3 and 5 of PA6 failed in this way and appear dashed in Figure 4.3.2, so the PA6 strengths are averages of two specimens.

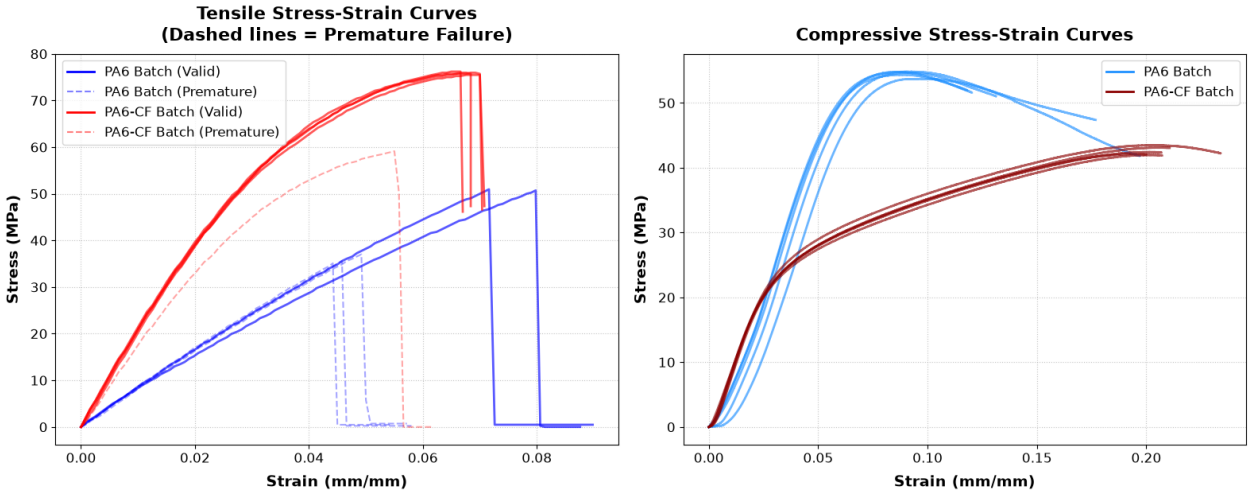


Figure 4.3.2: Engineering stress-strain curves in tension (left) and compression (right). Dashed: premature fillet failures.

The results are $S_{u,T} = 50.84\,\text{MPa}$ and $S_{y,T} = 20.07\,\text{MPa}$ for PA6, and $S_{u,T} = 75.90\,\text{MPa}$ and $S_{y,T} = 42.19\,\text{MPa}$ for PA6-CF. In compression, $S_y = 53.29\,\text{MPa}$ for PA6 and $25.84\,\text{MPa}$ for PA6-CF.

The apparent density ρ_{mat} comes from the same compression prisms, which share the 35 % gyroid infill and settings of the parts. Each prism is weighed with a scale with a resolution of $\pm 0.1\,\text{g}$, as shown in Figure 4.3.3, and its external volume follows from the nominal dimensions. The ratio of mass to external volume, averaged over the five prisms, is $0.7933\,\text{g/cm}^3$ for PA6 and $0.7445\,\text{g/cm}^3$ for PA6-CF.

The PA6 value is 70% of the density of bulk PA6, $1.13\,\text{g/cm}^3$ [99]. Moduli and density are measured on specimens with the same infill, so they are used as homogenized properties without porosity corrections.



Figure 4.3.3: Weighing of a PA6 compression prism.

4.4 Material Constants

The measured moduli fix two of the five constants, and the other three follow from the closure described in Section 3.2. The in-plane Poisson's ratio is $\nu_{r\theta} = 0.39$ for both materials, the value of bulk PA6 [99]. Measurements on FDM-printed thermoplastics show that the layup orientation has only a minor influence on the effective Poisson's ratio [100], which supports the bulk value. The out-of-plane ratios follow from Eq. (3.3).

The interlaminar shear modulus is $G_{rz} = \alpha_G E_z$ with $\alpha_G = 1/3$. This value lies below the isotropic reference $1/[2(1+\nu_{r\theta})] = 0.36$ and Huber's estimates of 0.61 for PA6 and 0.62 for PA6-CF. Since G_{rz} was not measured, Section 5.10 evaluates the designs with $\alpha_G = 0.32$, 0.41 and Huber's estimate. Table 4.4.1 lists all the constants.

Table 4.4.1: Material constants used in the model, from Table `T10_01`.

Property	PA6	PA6-CF	Source
In-plane modulus $E_r = E_\theta$ [MPa]	839.44	2056.59	ASTM D638, $\parallel$ layers
Out-of-plane modulus E_z [MPa]	294.32	684.46	ASTM D695, $\perp$ layers
Anisotropy ratio E_r/E_z [-]	2.852	3.005	Derived
Compressive yield strength S_y [MPa]	53.29	25.84	ASTM D695, 0.2% offset
Tensile yield strength $S_{y,T}$ [MPa]	20.07	42.19	ASTM D638, 0.2% offset
Ultimate tensile strength $S_{u,T}$ [MPa]	50.84	75.90	ASTM D638
Apparent density ρ_{mat} [g/cm^3]	0.7933	0.7445	Gravimetric, 35 % gyroid
In-plane Poisson's ratio $\nu_{r\theta}$ [-]	0.39	0.39	Bulk PA6 [99]
Poisson's ratio ν_{rz} [-]	0.6586	0.6760	Huber, Eq. (3.3)
Poisson's ratio ν_{zr} [-]	0.2309	0.2250	Reciprocity, Eq. (2.11)
Interlaminar shear modulus G_{rz} [MPa]	98.11	228.15	$\alpha_G E_z$, $\alpha_G = 1/3$

Both materials satisfy the conditions of Eq. (2.16). With Huber's closure and $\nu_{r\theta} = 0.39$, $\Delta = 0.4251$ in both cases, as given by Eq. (2.18). Substituting the constants in Eq. (2.15) gives the positive-definite stiffness matrices, in the order of Eq. (2.1),

$$\boldsymbol{C}_{0,\mathrm{PA6}} = \begin{bmatrix} 1674.49 & 633.92 & 1070.57 & 0 \\ 633.92 & 587.10 & 633.92 & 0 \\ 1070.57 & 633.92 & 1674.49 & 0 \\ 0 & 0 & 0 & 98.11 \end{bmatrix} \text{ MPa}, \tag{4.1}$$

$$\boldsymbol{C}_{0,\mathrm{PA6\text{-}CF}} = \begin{bmatrix} 4102.42 & 1513.13 & 2622.86 & 0 \\ 1513.13 & 1365.34 & 1513.13 & 0 \\ 2622.86 & 1513.13 & 4102.42 & 0 \\ 0 & 0 & 0 & 228.15 \end{bmatrix} \text{ MPa}. \tag{4.2}$$

With these constants, the linear model reaches the compressive yield strength at a strain S_y/E_z of 3.8% in PA6-CF and 18.1% in PA6. The second value is well beyond the small-

strain range discussed in Section 2.9.2. The linear model is therefore used only at service level, where strains are small, and the collapse loads of PA6 in the virtual test are indicative.

4.5 Discretization of the Component

With geometry, loads and material defined, the component is meshed as described in Section 3.3.1. The meridian section of the STL file is a closed polygon of 352 segments. The structured mesh has $n_r \times n_z = 216 \times 280 = 60\,480$ elements, 60 977 nodes and 121 954 DOFs.

The element size is $\Delta r = 0.2888\,\text{mm}$ by $\Delta z = 0.2893\,\text{mm}$, so $\Delta r / R_{\text{in}} = 0.5\%$, which meets the condition stated in Section 3.3.2. The voxelization gives 38 238 solid elements that form the design domain Ω, with 38 999 active nodes and 77 309 free DOFs once the supports are applied.

For the voxelized solid, the planar area fraction of the bounding rectangle is 63.22%, whereas the volume fraction is 67.10%, because the solid places more area at large radius. The voxelized volume is $V_0 = 1\,891\,751.4\,\text{mm}^3$, 0.50% above the STL volume of the two halves, $1\,882\,332.4\,\text{mm}^3$.

All volumes and masses in this work refer to the complete component. The solid component weighs $m_0 = 1500.73\,\text{g}$ in PA6 and 1408.41 g in PA6-CF, and its percentile $\sigma_{\text{VM},p99.5}$ discards about 190 elements.

The passive shell has a thickness $t_s = 2.0\,\text{mm}$, of the same order as the five printed perimeters. Its 9 954 elements hold 24.2% of V_0, and the remaining 28 284 elements are design variables. Any volume fraction must therefore satisfy $V_f > 0.242$.

The filter radius $R_{\min} = 2.6\,\text{mm}$ spans about 9 elements and gives $r_H = 0.7506\,\text{mm}$ by Eq. (2.28). With this mesh, the cavities filled at binarization have an area below about $1.7\,\text{mm}^2$.

4.6 Optimization Studies

On this discretization, the framework runs the studies of Table 4.6.1. The benchmark verifies the engine under a single axial load. The multi-load searches give the designs proposed for the component. The single-case searches give one design per load case and material, used for the cross-utilization. All searches use the load cases of Table 4.1.1 and the limits stated in Section 4.1.3.

The virtual compression test is applied to twelve designs, the solid component and the five selected designs of each material. The ten optimized designs are exported as complete rings and as 7.5° sectors for the physical tests.

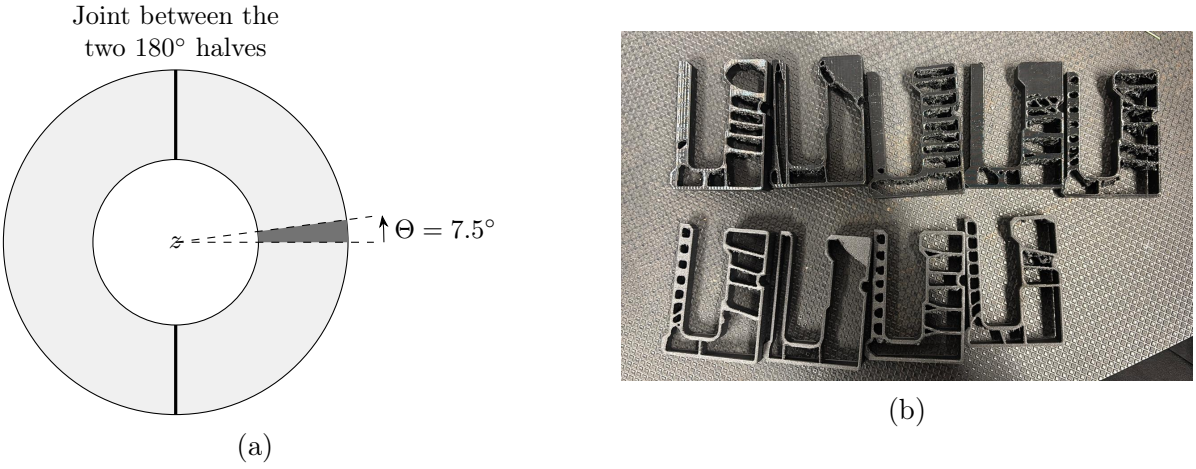
Table 4.6.1: Optimization studies and tests carried out on the component.

Study	Block	Material	Loads and settings	Output
Benchmark of the engine	3	PA6	Axial reference pressure on the top face, $V_f = 0.60$	One design, Section 5.3
Multi-load searches	5	PA6, PA6-CF	C1 to C4 with adaptive weights, sweep from $V_f = 0.90$	One design per material, 8 and 12 SIMP runs
Single-case searches	6	PA6, PA6-CF	One case at a time, sweep from $V_f = 0.75$ for PA6 and 0.50 for PA6-CF	Four designs per material, 47 and 36 SIMP runs
Virtual compression tests	8	PA6, PA6-CF	Rigid platens, bonded contact	12 force-displacement curves
Verification	9	PA6, PA6-CF	Re-evaluation of all selected designs	Tables T01 to T09, figures F01 and F02

4.7 Experimental Validation with Full-Scale Sectors

4.7.1 Purpose of the Test and Conversion of the Predictions

The physical test is a structural check of the optimized geometry, not a reproduction of the service behavior of the component. It verifies that the topologies print as closed, connected solids and carry an axial load through the predicted load paths. It also checks that the differences in stiffness between designs and between materials follow the trends of the simulation.



(a) (b)

Figure 4.7.1: Top view of the ring with the tested 7.5° sector (a) and printed sectors (b).

Figure 4.7.1 shows the tested sector, which retains the ring's height and meridian section. For a sector constrained to follow the axisymmetric solution, displacement, strain and stress

remain unchanged at the same platen displacement, while extensive quantities scale with the angle. The sector reference predictions are

$$K_{\mathrm{pred}} = \frac{\Theta}{360^\circ} K_{\mathrm{ini}}, \qquad F_{\mathrm{pred}} = \frac{\Theta}{360^\circ} F_{\mathrm{col}}, \qquad \frac{\Theta}{360^\circ} = \frac{1}{48}. \tag{4.3}$$

Mass and energy scale by the same factor. For equivalent geometry, material properties and imposed platen displacement, releasing the cut-face and platen constraints cannot increase elastic stiffness. Thus, K_{pred} is a conditional upper bound. The angular factor cancels in design ratios, although relaxation and contact may affect each topology differently. In the complete component, hoop continuity also requires a joint capable of transmitting circumferential forces. The factor is exact for the axisymmetric solution, which corresponds to a sector whose cut faces cannot move circumferentially. In the printed sector the cut faces are free, so the hoop stress that confines the ring is lost. By Eq. (2.9), the measured stiffness cannot exceed K_{pred}, which is therefore an upper bound, as shown in Section 2.3.4. The bonded platens of the model add a second overestimation with respect to a real frictional contact.

The same argument applies to the real component, where hoop continuity holds only if the joint between the two halves transmits circumferential forces. The test therefore verifies the optimized geometry and the relative behavior of the designs.

4.7.2 Test Procedure and Data Reduction

The sectors are compressed axially between flat platens in the universal testing machine of the university, at a constant crosshead speed of 2 mm/min, 24 h after printing. Figure 4.7.2 shows the setup and a tested sector. Each force-displacement curve is tared as described in Section 4.3.2, and a least-squares fit to its linear part gives the experimental stiffness K_{exp}.

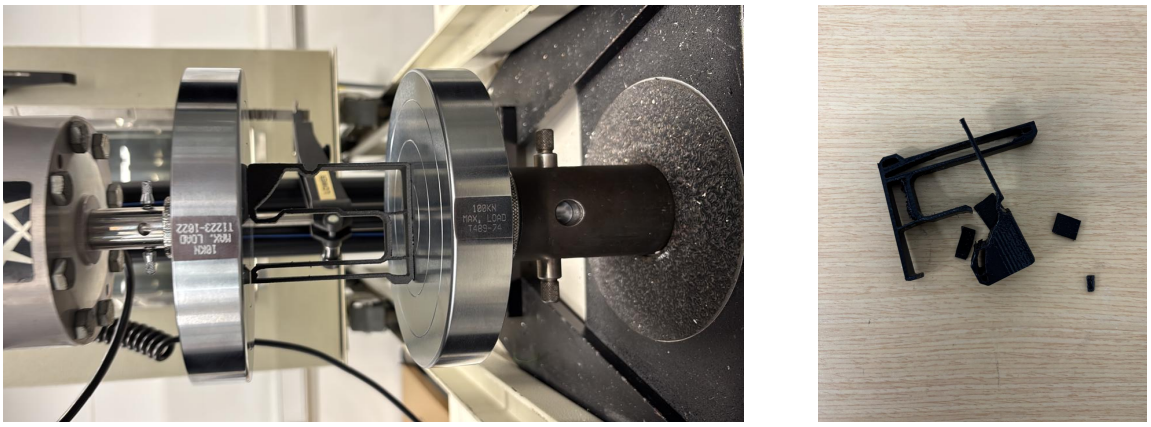
The comparison evaluates absolute stiffness and stiffness ratios between designs and materials. The upper-bound interpretation requires equivalent geometry, material properties and loading conditions. Maximum force is compared with F_{pred} only as an indication, as discussed in Section 3.8.2.

The integrity of the printed geometry and the location of the first visible damage are compared with the plastic zones of the virtual test.

4.8 Summary of Modeling Assumptions

The results of Chapter 5 depend on the following assumptions, which that chapter discusses with each result:

1. Loads and supports. The pressures of Table 4.1.1 and the supports described in Section 4.1.1 are assumed. They are not service requirements of the inspiring component.
2. Material symmetry. The deposition plane is assumed isotropic, with $E_\theta = E_r$, although only one in-plane direction was tested and the concentric perimeters may stiffen the shell circumferentially. $\nu_{r\theta}$ comes from bulk PA6, G_{rz} is estimated, and the same moduli are used in tension and compression.



(a) (b)

Figure 4.7.2: Test setup (a) (the figure is rotated 90 degrees) and sector after testing (b).

3. Homogenization. Properties and density are those of the printed material with 35 % gyroid infill. Members and shell regions as thin as the five perimeters print mostly solid, so their local stiffness and density may exceed the homogenized values, as discussed in Section 2.2.2.
4. Strength criterion. Yielding is checked with the isotropic von Mises criterion and the compressive yield strength, on the 99.5th percentile of the element stresses. Interlaminar tensile and shear strengths are not characterized.
5. Optimization. The displacement and stress constraints are checked on the binary designs after each run, not inside the optimizer. The problem is non-convex, the results depend on the warm-start path and the weights, several runs end at the iteration cap, and several selected designs are close to the feasibility limit.
6. Loads on the voxelized boundary. The loaded area of the top face is 7.4% larger than the exact area, and the axial resultants are correspondingly larger.
7. Manufacturability. The revolved internal voids form closed annular cavities, and no overhang or support constraint is included in the optimization.
8. Two-piece component. The two 180° halves are modeled as one continuous ring whose joint transmits hoop forces. The joint is not modeled.
9. Virtual and physical tests. The virtual test uses bonded platens without a contact formulation and is geometrically linear. Its prediction for the 7.5° sectors is a conditional elastic upper bound requiring equivalent geometry, material properties and loading. Physical tests assess response under platen compression.

Chapter 5

Verification and Results

This chapter follows the framework from numerical verification to experimental assessment. It first checks the implementation and the discretization, then presents the original component and the lightest feasible designs found under the four load cases. The virtual compression test provides a separate reference for the printed sectors, whose measured response is compared with the predictions. Throughout, numerical agreement is distinguished from physical validation under the assumptions of Section 4.8.

The numerical results come from the stored designs and reports of Blocks 1 to 8 of `Final.ipynb`, their independent re-evaluation in Block 9, and the figures and tables of Block 10. The experimental results come from `experimental_compression.ipynb`. Masses are calculated with the measured apparent densities of Table 4.4.1. The printed sectors were not weighed.

5.1 Verification of the Implementation

Table 5.1.1 summarizes the numerical checks. The constitutive matrices and consistent loads agree with their reference expressions to machine precision. The independent implementation in Block 9 reproduces the responses of twelve designs in four load cases with a largest relative difference of 7.85×10^{-11}. The residual, energy and equilibrium checks also remain small, and no force is applied to removed or restrained nodes.

These results support the consistency of the assembly, solution and post-processing. Reproducing the same model independently does not establish that its material properties, supports or loads represent the printed part accurately. Those assumptions are assessed separately below.

The axisymmetric solid has 77 309 free degrees of freedom. On the computer used for these runs, a SIMP iteration took about 0.4 s to 1.2 s, and a complete run took 14 s to 158 s. The two multi-load searches required 20 runs and 36.2 min, while the eight single-case searches required 83 runs and 150.7 min. These are recorded execution times rather than a hardware-independent performance benchmark.

Table 5.1.1: Verification of the implementation. Relative quantities are dimensionless.

Item	Check	Value	Source
Constitutive matrix	Relative asymmetry of $\boldsymbol{C}_0$	1.36×10^{-16}	Block 2
	Inversion error $\max \lvert \boldsymbol{C}_0 \boldsymbol{S} - \boldsymbol{I} \rvert$	2.40×10^{-16}	Block 2
	Isotropic input against the Lamé matrix	1.69×10^{-16}	Block 2
Loads	Sum of axial nodal forces against $P \sum \pi (r_b^2 - r_a^2)$	2.9×10^{-16}	Block 2
Baseline solution	Residual $\lVert \boldsymbol{K}\boldsymbol{U} - \boldsymbol{F} \rVert / \lVert \boldsymbol{F} \rVert$	1.18×10^{-12}	Block 2
	Equilibrium $\sum R_z - F_{\text{total}}$	-1.2×10^{-7} N	Block 2
	Energy $\lvert \boldsymbol{F}^\top \boldsymbol{U} - \boldsymbol{U}^\top \boldsymbol{K} \boldsymbol{U} \rvert / \boldsymbol{F}^\top \boldsymbol{U}$	6.28×10^{-14}	Block 2
Helmholtz filter	Error on a uniform field (partition of unity)	1.3×10^{-15}	Block 3
Re-implementation	$\boldsymbol{C}_0$ and $\boldsymbol{k}_e^0$ of all 38 238 elements against stored	$\leq 4.3 \times 10^{-16}$	Block 9
	12 designs $\times$ 4 cases: U, displacements, $\sigma_{\text{VM},p99.5}$, J	7.85×10^{-11}	Block 9, T02, F01
All designs and cases	Largest residual of the linear systems	1.33×10^{-11}	Block 9
	Largest energy inconsistency	3.96×10^{-12}	Block 9
	Largest axial imbalance $\lvert \sum_z (\boldsymbol{K}\boldsymbol{U}) \rvert / \sum \lvert F_z \rvert$	2.68×10^{-11}	Block 9
	Load on removed / restrained nodes	0.000% / 0.000%	Block 9
Mesh study	Reproduction of Block 2 at 216×280 (J, $\sigma_{\max}$, $\sigma_{p99.5}$)	$\leq 2.9 \times 10^{-13}$	Block 9, T07
Virtual test	K_{ini} recomputed for the 12 designs	$\leq 4.5 \times 10^{-11}$	Block 9, T08
Benchmark	Load lost at thresholding	0.000%	Block 3
STL exports	Closed surfaces / volume error against FE	20 of 20 / -0.21 to -0.67%	Block 7

The geometric checks complete the computational workflow. All 20 exports, comprising a complete ring and a 7.5° sector for each optimized design, have closed surfaces. Their volumes differ from the finite element volumes by −0.21% to −0.67%. Reconstruction therefore preserves the computed material volume closely. These checks establish geometric consistency, while the fabrication of sectors assesses only part of the manufacturability of a complete ring.

5.2 Mesh Study

The original solid was evaluated under the axial reference load with meshes of 108 × 140, 216×280 and 324×420 elements, corresponding to element sizes of 0.578, 0.289 and 0.193 mm. The middle mesh is used for the design searches. Figure 5.2.1 compares the resulting stresses and compliance, using the data of Table `T07_mesh_study_baseline`.

The maximum element stress increases from 21.72 to 27.34 and 31.27 MPa. It remains at the junction of the clamped outer wall and the top face, consistent with the singularity discussed in Section 2.5.2. In contrast, the element-count 99.5th percentile decreases from 7.966 to 7.738 and 7.624 MPa. The volume-weighted percentile follows a similar trend, although the design criterion uses the element-count percentile defined in Section 2.5.2.

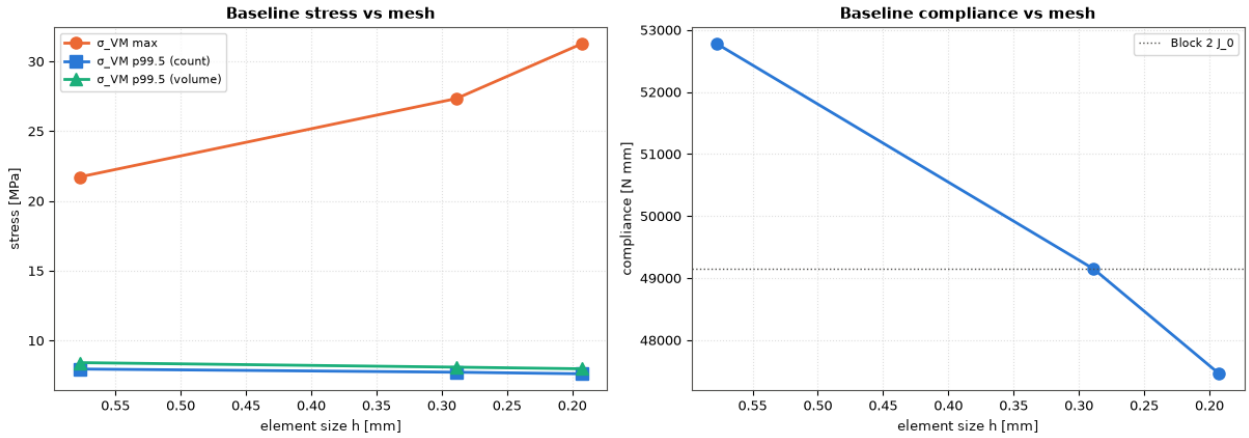


Figure 5.2.1: Mesh refinement of the original solid under the axial reference load. Left: maximum von Mises stress and 99.5th percentiles weighted by element count and volume. Right: compliance.

Refinement also changes the applied force because the boundary is voxelized. The loaded area exceeds the exact top-face area by 11.1%, 7.4% and 4.2% on the three meshes. After normalization by the force, the element-count percentile changes by about 2.1%, whereas the maximum increases by more than 50%. Compliance decreases from 52 780 to 49 150 and 47 458 N mm, but its load-normalized value varies by less than 2.6%.

The comparison supports the percentile as a more stable stress measure and shows modest changes in the load-normalized stiffness of the original solid. It does not establish mesh independence for every optimized topology. The percentile also retains its stated limitation: it excludes the highest-stressed 0.5% of the meridian area and therefore does not bound every local stress.

5.3 Benchmark of the Optimization Engine

The PA6 benchmark at $V_f = 0.60$ ended after 112 iterations and 92.2 s, with a non-discreteness measure of 1.09%. The first projection stage reached its iteration cap, while the others ended by the compliance criterion. The density-change criterion was not reached. Binarization removed no islands, filled no cavities and lost no load.

Figure 5.3.1 shows how the engine removes material mainly below the cavity and from the lower outer block, while retaining the inner wall and an upper connection to the clamped wall. The binary design weighs 898.8 g, a reduction of 40.1%. Relative to the solid, its compliance, maximum displacement and stress percentile are 1.184, 1.063 and 1.319, respectively. Thus, mass reduction increases deformation and stress, but the benchmark thresholds remain satisfied: the displacement ratio is below 1.25 and the safety factor is 5.22, above 2.0.

Both efficiency indices are 1.41, corresponding to a 41% gain in compliance-based stiffness per unit mass under this reference load. The persistent stress concentration near the clamp shows that this gain does not imply a uniform stress field. The benchmark verifies the engine's operation under its own thresholds. The component designs below are selected instead by minimum mass subject to the utilization limits of Section 3.4.

5.4 Response of the Original Component

The original component satisfies the four assumed load cases for both materials, as shown in Table 5.4.1. Its calculated mass is 1500.73 g in PA6 and 1408.41 g in PA6-CF. The largest utilization is 0.478 and 0.204, respectively, leaving room for material removal under the adopted criteria.

Table 5.4.1: Response of the solid component to the design load cases: largest displacement of the critical walls and of the remaining boundary, 99.5th-percentile von Mises stress, safety factor, utilization and governing limit (crit.: critical walls; rem.: remaining boundary) (F10_02; T10_03).

Case	u_{crit} [mm]	u_{rem} [mm]	$\sigma_{\mathrm{VM},p99.5}$ [MPa]	$\mathrm{FS}_{p99.5}$	U_c	Governs
PA6						
C1	0.0478	0.0767	0.60	88.3	0.478	crit.
C2	0.0243	0.2351	2.82	18.9	0.470	rem.
C3	0.0287	0.2385	3.00	17.7	0.477	rem.
C4	0.0345	0.2350	2.91	18.3	0.470	rem.
PA6-CF						
C1	0.0204	0.0329	0.61	42.5	0.204	crit.
C2	0.0100	0.0965	2.82	9.2	0.193	rem.
C3	0.0118	0.0974	3.00	8.6	0.195	rem.
C4	0.0141	0.0963	2.91	8.9	0.193	rem.

Figure 5.4.1 identifies the displacement contribution to this margin. The critical walls govern

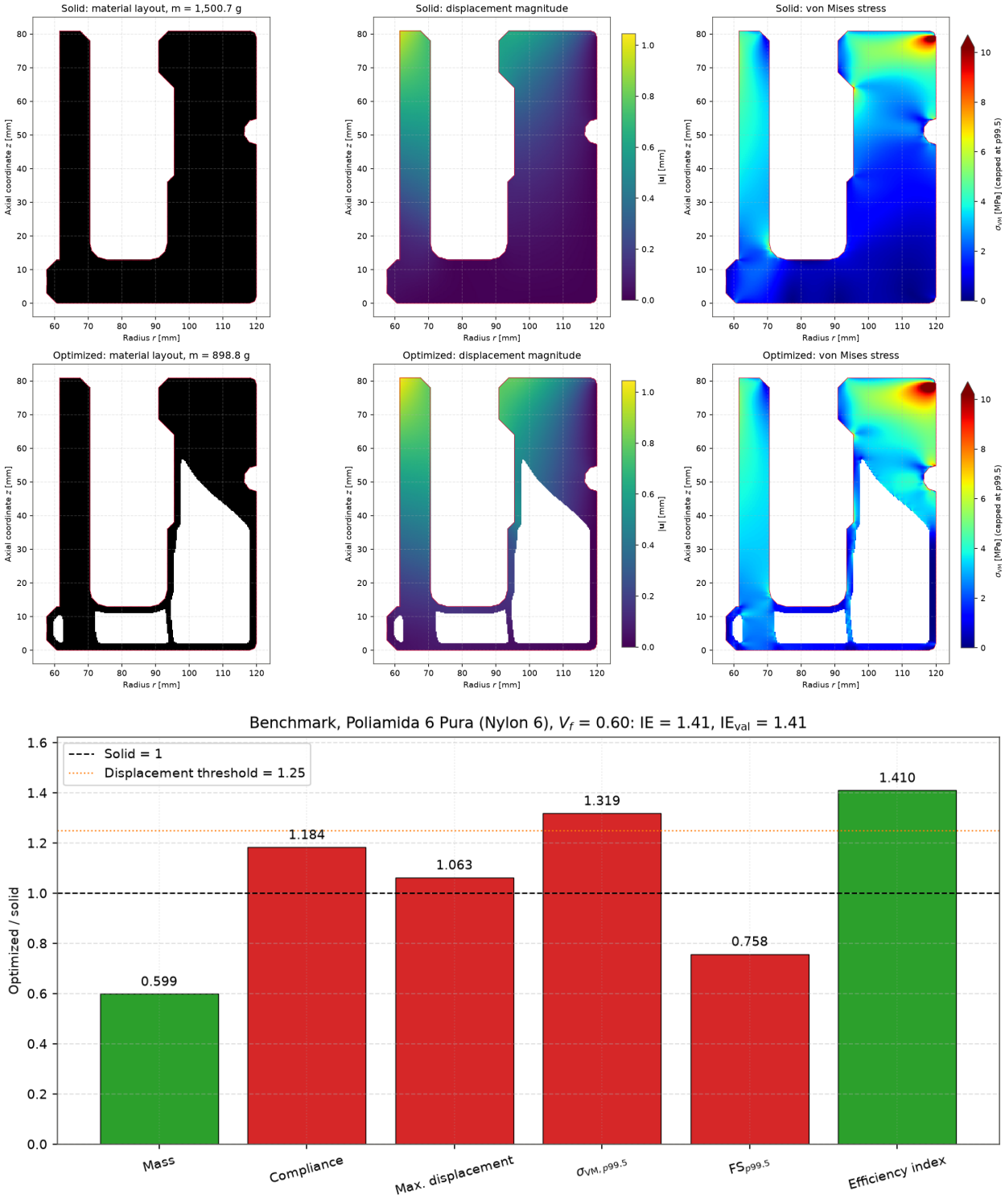


Figure 5.3.1: Benchmark at $V_f = 0.60$: material layouts, displacement and stress fields, and ratios to the solid. The dotted threshold applies to the displacement ratio only.

C1, whereas the remaining boundary governs C2 to C4. In those cases, the free top of the inner wall moves radially under pressure. PA6-CF gives displacement ratios about 0.41 to 0.43 times those of PA6, consistent with its higher measured moduli.

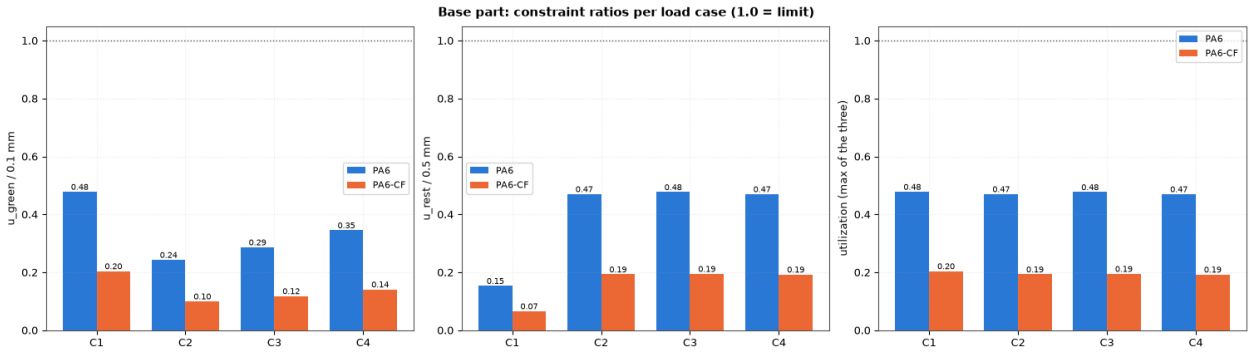


Figure 5.4.1: Original component under the four load cases: normalized critical-wall displacement, normalized remaining-boundary displacement and total utilization.

The stress percentiles are nearly equal between materials, reaching about 3.0 MPa, although their safety factors differ because their compressive yield strengths differ. The similar elastic anisotropy ratios help explain this stress agreement under identical forces. The displacement difference, rather than a stress difference, provides the main opportunity for further mass removal in PA6-CF.

The assumed support also shapes the response. Under the axial reference load, the outer clamp carries 71.6% of the axial resultant and the base carries 28.4%. This load transfer is important when interpreting both the optimized layouts and the later platen tests, which remove the outer clamp.

5.5 Minimum-Mass Designs

Table 5.5.1 lists the lightest feasible designs found in the ten searches. The multi-load designs reduce the calculated mass by 35.7% in PA6 and 61.9% in PA6-CF. Both satisfy the displacement and percentile-based stress criteria in all four cases, with governing utilizations of 0.985 and 0.994. C4 governs through the displacement of the lower critical bore wall, which reaches 0.0985 mm and 0.0994 mm against the 0.10 mm limit.

The layouts in Figure 5.5.1 explain the mass difference. The PA6 multi-load design keeps the inner wall nearly solid and retains a broad upper outer region with several connections below it. PA6-CF uses a perforated inner wall and fewer, thinner internal connections. In both materials, the passive shell preserves the original boundary and critical surfaces. The white regions are voids in the meridian section and become annular cavities upon revolution.

The single-case designs save more mass, from 53.9% to 64.5% in PA6 and 65.7% to 68.9% in PA6-CF. C1 retains an upper wedge that connects the loaded face to the outer support. The pressure-driven designs use mainly radial or inclined connections between the cavity

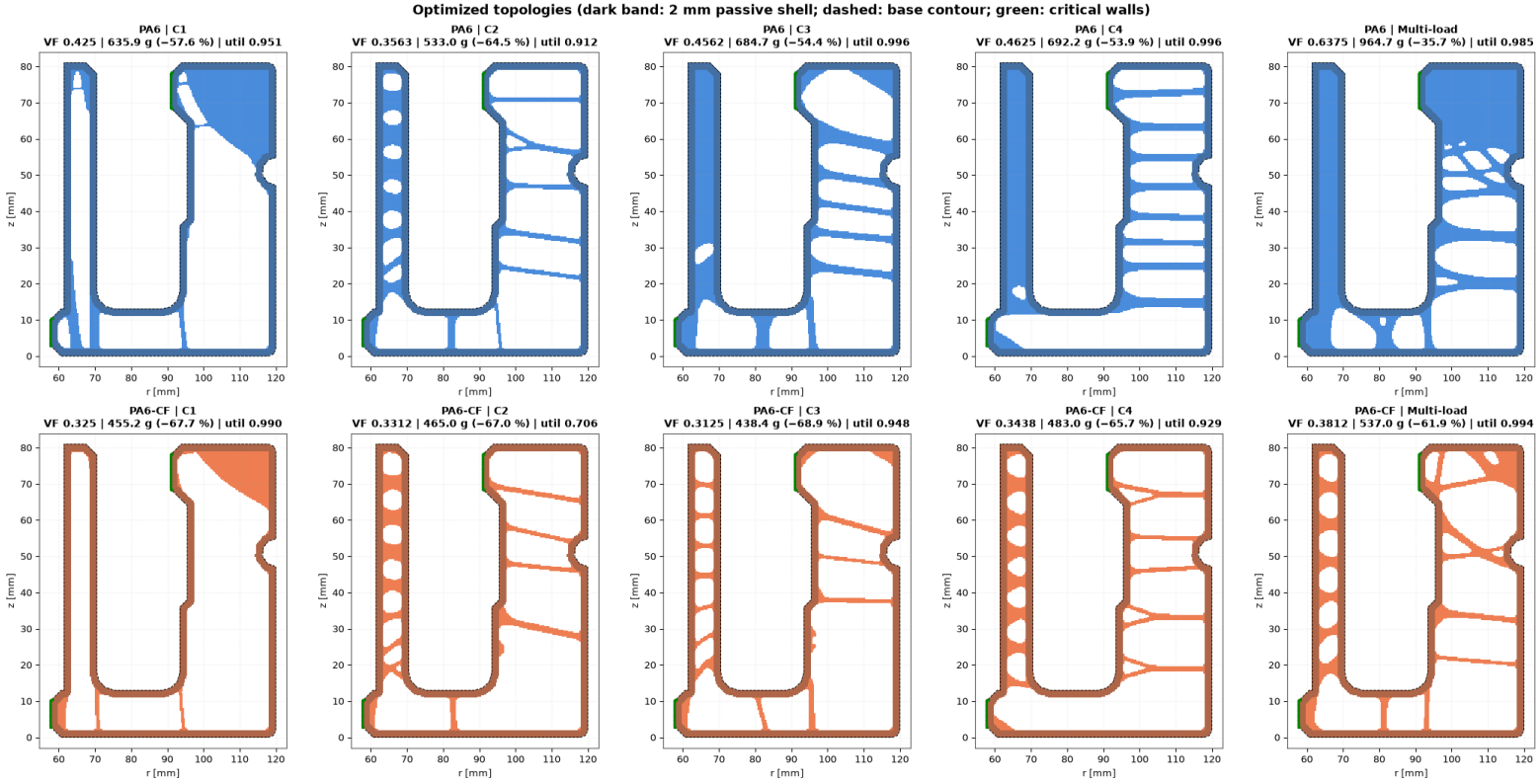


Figure 5.5.1: Selected topologies in PA6 and PA6-CF. The dark band is the 2 mm passive shell, the dashed line is the original contour, and green marks the critical walls.

Table 5.5.1: Minimum-mass feasible designs. Displacements and safety factor in the governing case, which is C4 for the multi-load designs and the design case for the single-case designs.

Design	V_f	Mass [g]	Saving [%]	u_{crit} [mm]	u_{rem} [mm]	$\text{FS}_{p99.5}$	U
PA6							
Multi-load	0.6375	964.7	35.7	0.0985	0.2438	18.31	0.985
C1	0.4250	635.9	57.6	0.0951	0.1718	46.94	0.951
C2	0.3563	533.0	64.5	0.0912	0.2859	15.01	0.912
C3	0.4562	684.7	54.4	0.0996	0.2299	17.64	0.996
C4	0.4625	692.2	53.9	0.0996	0.2519	18.31	0.996
PA6-CF							
Multi-load	0.3812	537.0	61.9	0.0994	0.1539	5.84	0.994
C1	0.3250	455.2	67.7	0.0990	0.0992	12.87	0.990
C2	0.3312	465.0	67.0	0.0706	0.1796	6.30	0.706
C3	0.3125	438.4	68.9	0.0948	0.3400	3.42	0.948
C4	0.3438	483.0	65.7	0.0929	0.1416	6.38	0.929

wall and the outer wall. These layouts reflect different load paths, which explains why their admissibility must also be checked under the other cases.

Every selected design is governed by critical-wall displacement in its own design problem. The largest stress utilization is only 0.44, in PA6-CF C3, whose percentile safety factor is 3.42. Stiffness therefore governs the reported selections. Several feasibility margins are small, including 0.4% for PA6 C3 and C4 and 0.6% for the PA6-CF multi-load design. These are numerical margins under the assumed properties, without an allowance for manufacturing deviations.

5.6 Search on the Volume Fraction

Figure 5.6.1 shows the mass-utilization pairs evaluated in each search. Filled markers are feasible, open markers are infeasible, diamonds identify the original solids, and stars identify the selected designs. Lines connect only the non-dominated evaluated points. Increasing mass generally reduces utilization, but the nearly horizontal regions in PA6 C2 and C3 include points that do not improve it. The complete sets of runs should therefore not be described as strictly monotonic.

The PA6 multi-load search selects $V_f = 0.6375$ after eight runs. Its nearest lower infeasible fraction is 0.625, with $U = 1.043$. PA6-CF selects $V_f = 0.38125$ after twelve runs, next to an infeasible design at 0.375 with $U = 1.017$. The final volume-fraction interval is 0.0125 for the PA6 multi-load search and 0.00625 for the others. It corresponds approximately to 19 g for the first search and 9 g to 10 g for the others, rather than an exact bound on the global minimum mass.

Some boundaries are abrupt. PA6 C2 changes from $U = 0.912$ at $V_f = 0.35625$ to 2.87 at

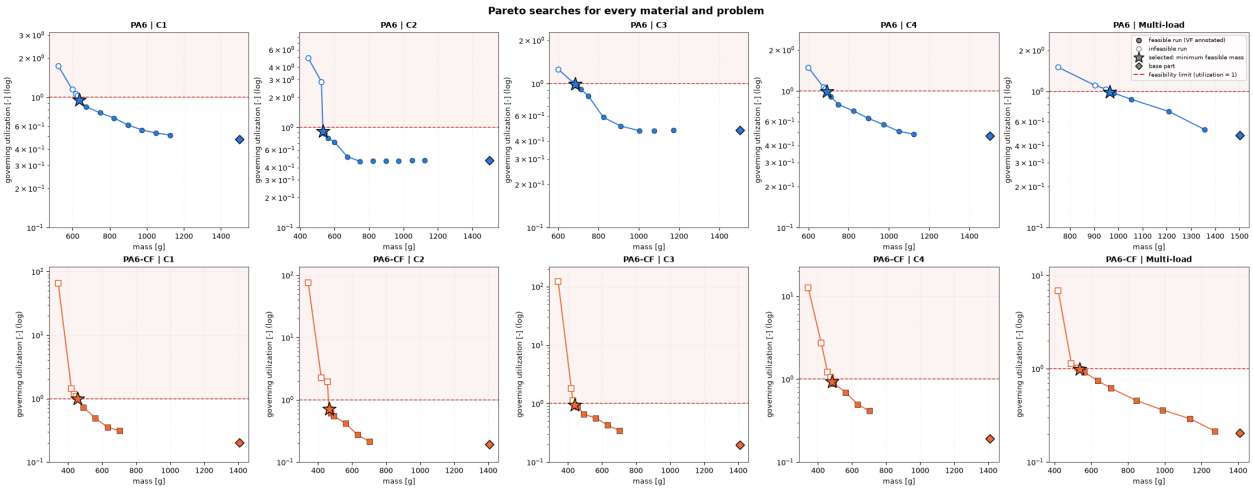


Figure 5.6.1: Ten volume-fraction searches. The dashed line marks $U = 1$. Lines connect the non-dominated evaluated points, while stars mark the lightest feasible designs found.

0.35, and PA6-CF C2 changes from 0.706 at 0.33125 to 1.968 at 0.325. These jumps show that small volume reductions can produce a large loss of functional stiffness in the resulting binary layout. They are consistent with changes in load-carrying connections, but the search curves alone do not establish the mechanism. This explains why some selected designs retain a larger utilization margin than others.

Seven selected runs reached the cap of 150 iterations. The PA6 multi-load, C3 and C4 runs ended after 92, 141 and 147 iterations. The adaptive weights increased the contribution of C4, which governs both multi-load selections, but the procedure remains dependent on its initialization and search path. The reported results are consequently the lightest feasible designs found, without a proof of global optimality.

5.7 Comparison Between PA6 and PA6-CF

PA6-CF gives a lighter selected design in every problem. Figure 5.7.1 shows the largest absolute difference for the multi-load problem: 537.0 g against 964.7 g, or 44.3% less mass than the PA6 selection. The single-case mass ratio ranges from 0.640 in C3 to 0.872 in C2. Thus, the benefit of changing material also depends on the loading problem and resulting topology.

The comparison in Figure 5.7.2 persists when mass is normalized by each material's original solid. The sampled PA6-CF fronts allow lower utilization at comparable relative mass, so its lower selected masses are not explained only by its 6% lower apparent density. Its measured moduli are 2.45 times higher in-plane and 2.33 times higher axially. The axial modulus per apparent density is therefore 2.48 times that of PA6.

Figure 5.7.3 confirms why these elastic properties matter: critical-wall displacement governs every selected design, while the stress term remains below the limit. PA6-CF C3 has the largest stress contribution, about 0.44, and a remaining-boundary displacement ratio of 0.68.

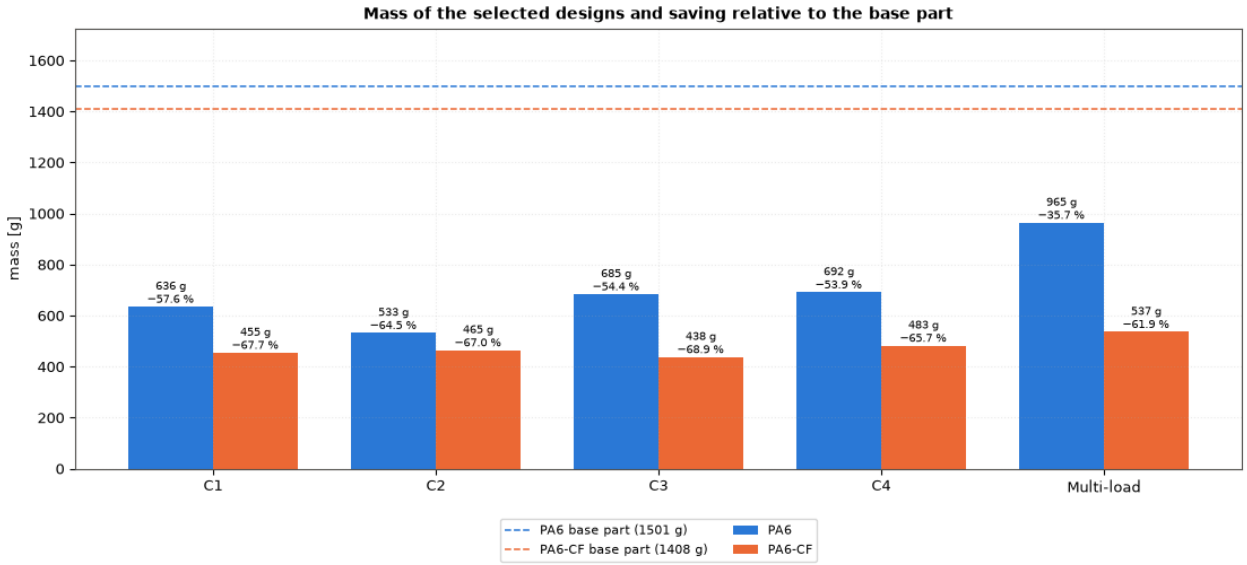


Figure 5.7.1: Calculated masses of the selected designs and savings relative to the original solid of each material.

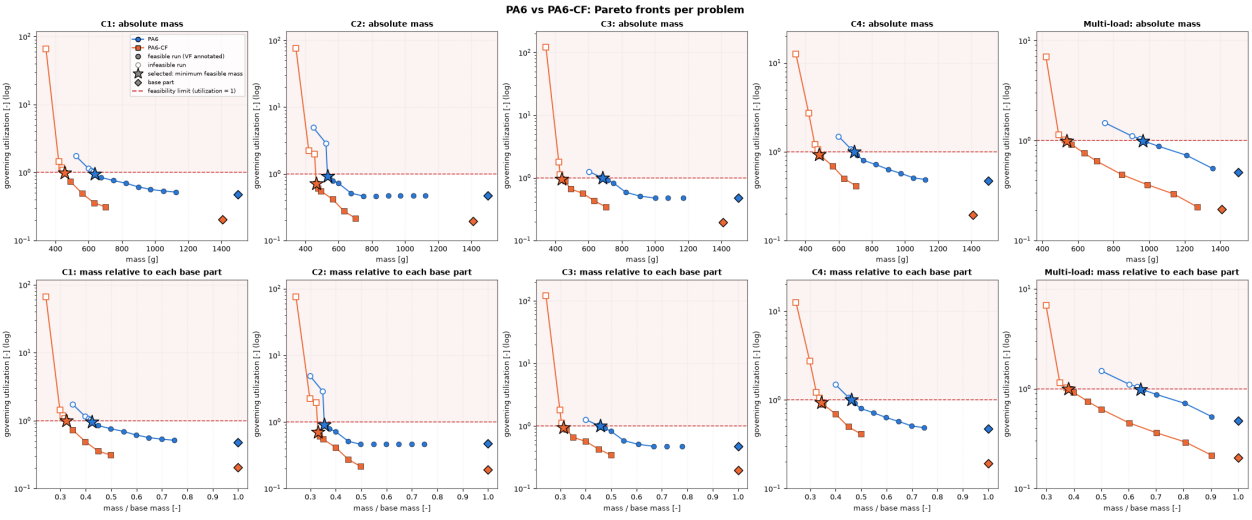


Figure 5.7.2: PA6 and PA6-CF search results for each problem, using absolute mass in the upper row and mass normalized by each material's original solid in the lower row.

Its lower compressive yield strength therefore does not control the present selection.

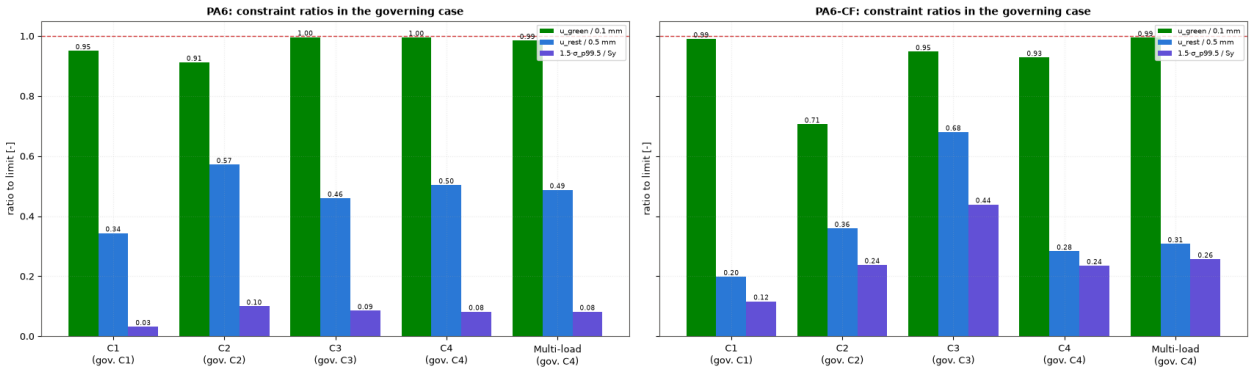


Figure 5.7.3: Constraint ratios of the selected designs in their governing load case. Critical-wall displacement governs all selections.

The numerical advantage of PA6-CF follows from the measured properties under identical assumed loads. It remains conditional on those properties representing the printed members. The physical tests below assess that transfer, using different optimized geometries for the two materials.

5.8 Cross-Utilization

Figure 5.8.1 evaluates every selected design under all four cases. Every off-diagonal entry of the single-case designs exceeds 1, ranging from 1.68 to 11.63. Each single-case design therefore violates the adopted limits in all three other cases. The largest violations occur for C4 designs evaluated under C3, with $U = 9.61$ in PA6 and 11.63 in PA6-CF.

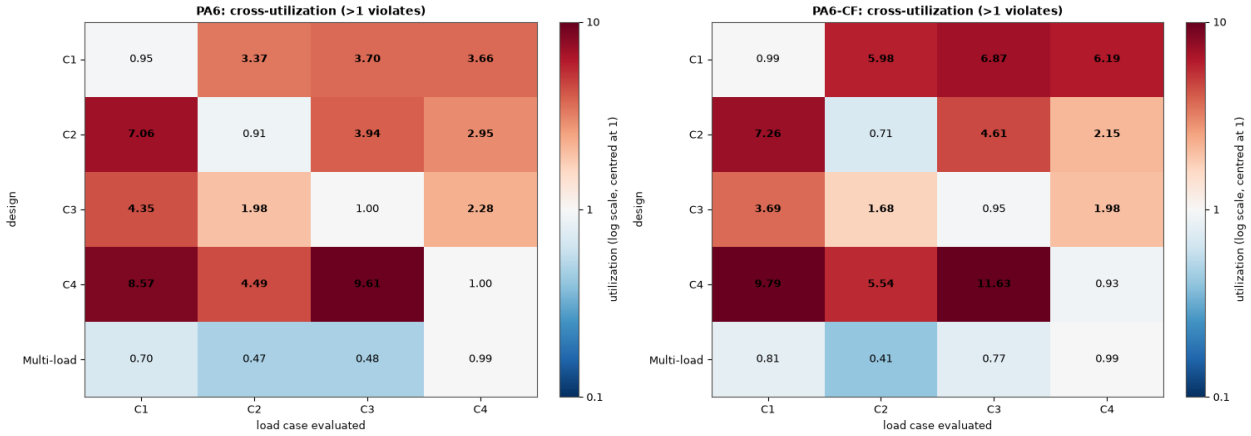


Figure 5.8.1: Cross-utilization of the optimized designs. Rows identify designs and columns identify evaluated load cases. Values above 1 violate a design limit.

Only the multi-load selections satisfy all cases, with C4 governing both. This demonstrates that minimizing mass for one loading condition does not preserve the displacement limits

under the others. The additional coverage costs mass: the PA6 multi-load design is 39% heavier than its heaviest single-case selection, while the corresponding increase is 11% for PA6-CF. The mass savings of the single-case designs cannot therefore be adopted for the component under the complete assumed load set.

5.9 Strength Checks

The directional checks of Table `T04_directional_strength` do not change the feasibility of the 16 evaluated design-case pairs: the two multi-load designs under four cases and the eight single-case designs under their own case. The lowest directional safety factors are 5.30 in in-plane tension and 6.57 in axial compression, both for PA6-CF C3. These results support the conclusion that the measured strength limits do not govern the reported designs.

The assessment remains incomplete for interlaminar failure. Neither axial tensile strength nor interlaminar shear strength was measured. The corresponding stress percentiles reach 2.77 and 1.86 MPa in PA6-CF C3. For the multi-load designs, they reach 0.51 and 0.55 MPa in PA6 and 1.92 and 1.16 MPa in PA6-CF. These values can be reported, but their safety factors cannot be established from the available characterization. The von Mises and directional checks therefore verify the adopted criteria without establishing resistance to every printed-material failure mode.

5.10 Sensitivity to the Interlaminar Shear Modulus

Table 5.10.1 re-evaluates the stored geometries without re-optimizing them. Reducing $\alpha_G = G_{rz}/E_z$ from 1/3 to 0.32, a change of about 4%, makes PA6 C3 and PA6-CF C1 infeasible. It also brings the PA6-CF multi-load design to $U = 0.99995$. Increasing α_G to 0.41 or Huber's value lowers all reported utilizations.

Table 5.10.1: Governing utilization of the selected designs for different values of $\alpha_G = G_{rz}/E_z$. Values above 1 are infeasible.

Material	Design	$\alpha_G = 1/3$	$\alpha_G = 0.32$	$\alpha_G = 0.41$	Huber
PA6	Solid	0.478	0.485	0.476	0.475
	Multi-load	0.985	0.990	0.963	0.928
	C1	0.951	0.971	0.860	0.724
	C2	0.912	0.926	0.861	0.815
	C3	0.996	1.017	0.900	0.758
	C4	0.996	0.9996	0.976	0.949
PA6-CF	Solid	0.204	0.207	0.194	0.194
	Multi-load	0.994	0.99995	0.967	0.918
	C1	0.990	1.007	0.910	0.779
	C2	0.706	0.716	0.660	0.588
	C3	0.948	0.967	0.864	0.736
	C4	0.929	0.936	0.893	0.833

This sensitivity gives practical meaning to the small margins in Table 5.5.1. The critical displacements depend on shear deformation in the internal connections, and the shear modulus is an estimated input. Measuring it or selecting a utilization target below 1 would strengthen confidence in the design. The present study establishes the sensitivity of the selected geometries, rather than the mass that a new optimization would obtain for each alternative modulus.

5.11 Virtual Compression Tests

Table 5.11.1 summarizes the virtual compression tests between bonded rigid platens. The PA6-CF solid is 2.32 times stiffer than the PA6 solid, close to the measured axial modulus ratio of 2.33. Initial stiffness is the main quantity used for the experimental comparison, while the collapse estimates are interpreted within the limitations of Section 3.8.2.

Table 5.11.1: Virtual compression test between rigid platens with bonded contact. Collapse values are indicative.

Design	m [g]	K_{ini} [kN/mm]	K_{ini}/m [N/mm/g]	F_y [kN]	F_{col} [kN]	δ_{col} [mm]	δ_{col}/H [%]
PA6							
Solid	1500.7	82.9	55.2	792.1	1069.9	20.32	25.1
Multi-load	964.7	27.0	28.0	227.1	312.3	21.04	26.0
C1	635.9	14.4	22.7	118.5	168.9	19.48	24.1
C2	533.0	10.7	20.1	80.7	123.7	24.48	30.2
C3	684.7	16.6	24.2	130.6	189.0	24.58	30.3
C4	692.2	9.3	13.4	59.9	85.6	23.38	28.9
PA6-CF							
Solid	1408.4	192.3	136.6	385.1	517.9	4.25	5.25
Multi-load	537.0	30.3	56.5	56.8	77.4	5.15	6.36
C1	455.2	22.5	49.5	41.7	58.9	5.08	6.28
C2	465.0	20.0	42.9	28.6	47.4	5.72	7.07
C3	438.4	22.4	51.1	36.9	56.7	5.56	6.87
C4	483.0	14.1	29.1	23.5	34.5	5.84	7.21

The multi-load PA6 design retains about 33% of the solid's stiffness with 64% of its mass, while PA6-CF retains 16% with 38% of its mass. Their stiffness per unit mass is therefore 49% and 59% lower. Removing the outer clamp changes the load path and requires the entire force to pass from the top to the base, so this test measures a different response from the design cases.

Every reported collapse occurs above 5% nominal strain, from 24.1% to 30.3% in PA6 and from 5.3% to 7.2% in PA6-CF. These displacements exceed the adopted small-strain screening threshold. Together with the absence of updated contact, this limits the collapse forces to indicative reference values and supports using initial stiffness for the quantitative comparison.

5.12 Predictions for the Sector Tests

Table 5.12.1 applies the angular factor 1/48 of Eq. (4.3) to the virtual ring results. The optimized-sector masses come from their STL volumes and the measured apparent density. The solid-sector values are numerical references, calculated from the full-ring results.

Table 5.12.1: Predictions for the 7.5° sectors. K_{pred} is a conditional elastic upper bound, and F_{pred} is indicative.

Design	m_{sec} [g]	K_{pred} [kN/mm]	F_{pred} [kN]	δ_{col} [mm]
PA6				
Solid	31.27[a]	1.726	22.29	20.32
Multi-load	20.06	0.562	6.51	21.04
C1	13.19	0.301	3.52	19.48
C2	11.06	0.223	2.58	24.48
C3	14.21	0.346	3.94	24.58
C4	14.37	0.193	1.78	23.38
PA6-CF				
Solid	29.34[a]	4.007	10.79	4.25
Multi-load	11.15	0.632	1.61	5.15
C1	9.42	0.470	1.23	5.08
C2	9.64	0.416	0.99	5.72
C3	9.09	0.466	1.18	5.56
C4	10.02	0.293	0.72	5.84

[a] Finite element mass divided by 48, since no sector STL of the solid was exported.

The conversion is exact for a sector constrained to follow the axisymmetric solution. Free cut faces and relaxed platen restraint enlarge the admissible displacement set, giving the upper-bound argument of Section 2.3.4. That comparison presumes equivalent geometry, material properties and prescribed contact surfaces. It is therefore a conditional bound for the printed sector, rather than a guarantee against differences in manufacturing or material characterization.

The predicted optimized-sector collapse forces reach about 6.5 kN in PA6 and 1.6 kN in PA6-CF. These values indicate the anticipated force scale, while the stiffness predictions are used for the quantitative comparison. Although the factor 1/48 cancels in ratios between designs, changes in circumferential relaxation and contact need not affect every topology equally.

5.13 Experimental Results

Nine test records are available from the Instron 3369: five PA6 designs and four single-case PA6-CF designs. The multi-load PA6-CF sector could not be printed because of moisture-related printing problems. The tested sectors had the intended 7.5° geometry, full-scale dimensions and vertical build axis, with the printing conditions described in Chapter 4.

The processing fits $F = K_{\text{exp}}\delta + b$ between 20% and 50% of the measured maximum force, using points before the peak or first detected load drop. No band overrides or machine-compliance correction are applied. The displacement origin is shifted by $\delta_0 = -b/K_{\text{exp}}$, giving the corrected displacement $\delta - \delta_0$. This translation leaves the fitted stiffness unchanged and does not remove nonlinear seating from a fit that includes it. Table 5.13.1 uses the saved notebook outputs for all nine records.

Table 5.13.1: Sector tests and predictions. Stiffness and force are given in kN/mm and kN. Displacements are shifted by the fitted intercept. One record is available per design.

Design	K_{exp} [kN/mm]	K_{pred} [kN/mm]	$K_{\text{exp}}/K_{\text{pred}}$	$F_{\text{max,exp}}$ [kN]	$F_{\text{max,exp}}/F_{\text{pred}}$	δ_{max} [mm]	δ_{max}/H [%]
PA6							
Multi-load	2.126	0.562	3.78	4.436	0.68	2.31	2.86
C1	0.957	0.301	3.18	0.569	0.16	1.32	1.62
C2	0.702	0.223	3.15	0.950	0.37	1.54	1.90
C3	1.452	0.346	4.20	2.156	0.55	2.10	2.60
C4	0.658	0.193	3.40	0.717	0.40	1.14	1.41
PA6-CF							
Multi-load	Not printed because of moisture-related printing problems						
C1[a]	0.160	0.470	0.34	0.160	0.13	0.94	1.15
C2	0.417	0.416	1.00	0.525	0.53	1.78	2.20
C3	0.397	0.466	0.85	0.522	0.44	2.22	2.74
C4[a]	0.113	0.293	0.38	0.225	0.31	2.35	2.91

[a] Strong sensitivity to the fit interval. The steepest local slopes are 0.315 and 0.186 kN/mm for C1 and C4, respectively.

Figure 5.13.1 shows distinct responses. PA6 reaches larger forces and exhibits abrupt drops. The multi-load sector reaches 4.436 kN, followed by C3 at 2.156 kN. C1 develops a plateau before its main drop, while C2 and C4 eventually lose nearly all recorded load. Audible cracking and broken specimens confirm that fracture occurred in PA6, although the initiation and preceding local mechanism were not observed.

PA6-CF C2 and C3 follow similar rising branches and reach nearly equal maxima, about 0.525 kN. C2 then loses force more sharply, whereas C3 retains more load. C1 and C4 reach lower maxima and show extended residual-load regions. These curves establish different structural responses, but they do not identify a unique local failure mechanism.

The measured maxima are 16% to 68% of the indicative virtual collapse forces in PA6 and 13% to 53% in PA6-CF. Fracture and manufacturing variability are not represented by the virtual elastic-compensation model, so these ratios are descriptive comparisons. All maxima occur below 3% nominal shortening. That global value does not establish small local strains or rotations in the thin members.

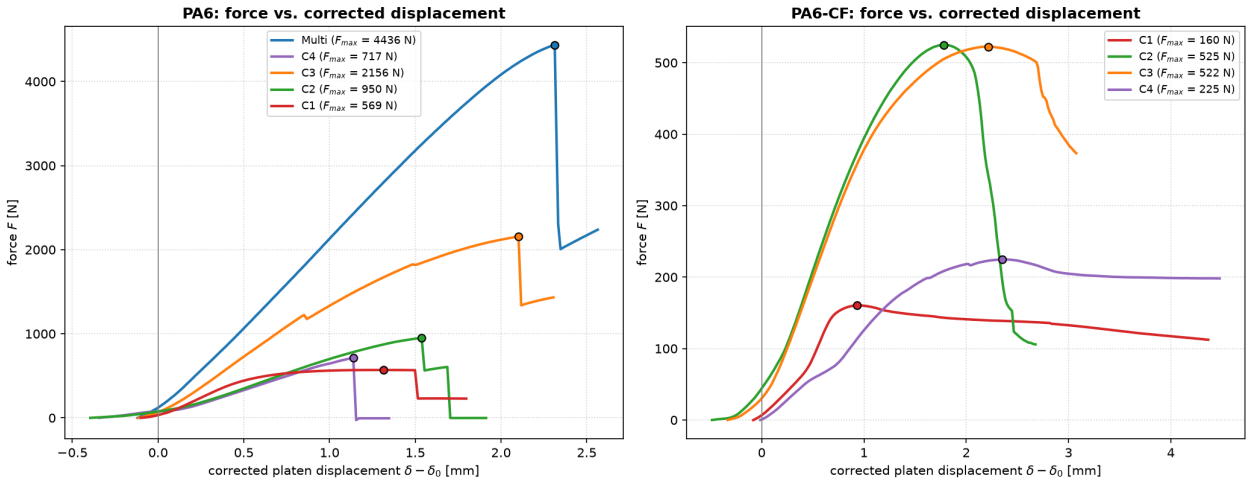


Figure 5.13.1: Experimental force-displacement curves of the PA6 and PA6-CF sectors after shifting the displacement origin by the stiffness-fit intercept. Circles identify the measured maxima.

5.13.1 Stiffness and Relative Design Response

Figure 5.13.2 separates a systematic discrepancy in PA6 from a more variable response in PA6-CF. The PA6 stiffness ratios range from 3.15 to 4.20, with a mean of 3.54 and an across-design coefficient of variation of 12.6%. The experimental order is exactly the predicted one: Multi-load, C3, C1, C2 and C4. The Spearman coefficient is 1.0 for these five designs, supporting the predicted ordering within the tested set despite the absolute mismatch.

The common PA6 offset suggests a systematic difference between the modeled continuum and the tested structures. The model uses properties homogenized from coupons with 35% gyroid infill, while thin topology members may be filled largely by perimeters and solid layers. Their local stiffness can therefore differ from the coupon-based representation. The characterization description also does not establish an independent local-strain measurement or a machine-compliance correction. These explanations cannot be separated with the available records, and no measured sector masses are available to check the predicted material content.

The processing provides a further check. Changing the force band to 10–40% and 30–60% produces a stiffness spread, normalized by the main fit, of 4.1% to 31.6% across the PA6 designs. The discrepancy of more than a factor of three remains even with the lower-band estimates. Choosing the main fit interval alone therefore does not explain the PA6 offset, although it affects its magnitude.

PA6-CF C2 is close to the prediction and C3 reaches 85% of it, while C1 and C4 reach only 34% and 38%. For the latter two, the alternative-band spread is 36.3% and 58.0%. C4 also has $R^2 = 0.967$ in the main fit. Their reported slopes are therefore fit-dependent apparent stiffnesses rather than well-isolated initial elastic stiffnesses. The across-design stiffness-ratio variation is 51.5%, and the predicted ranking is not preserved, with a Spearman coefficient of 0.2 for four designs. These statistics describe different designs and do not estimate specimen-to-specimen repeatability.

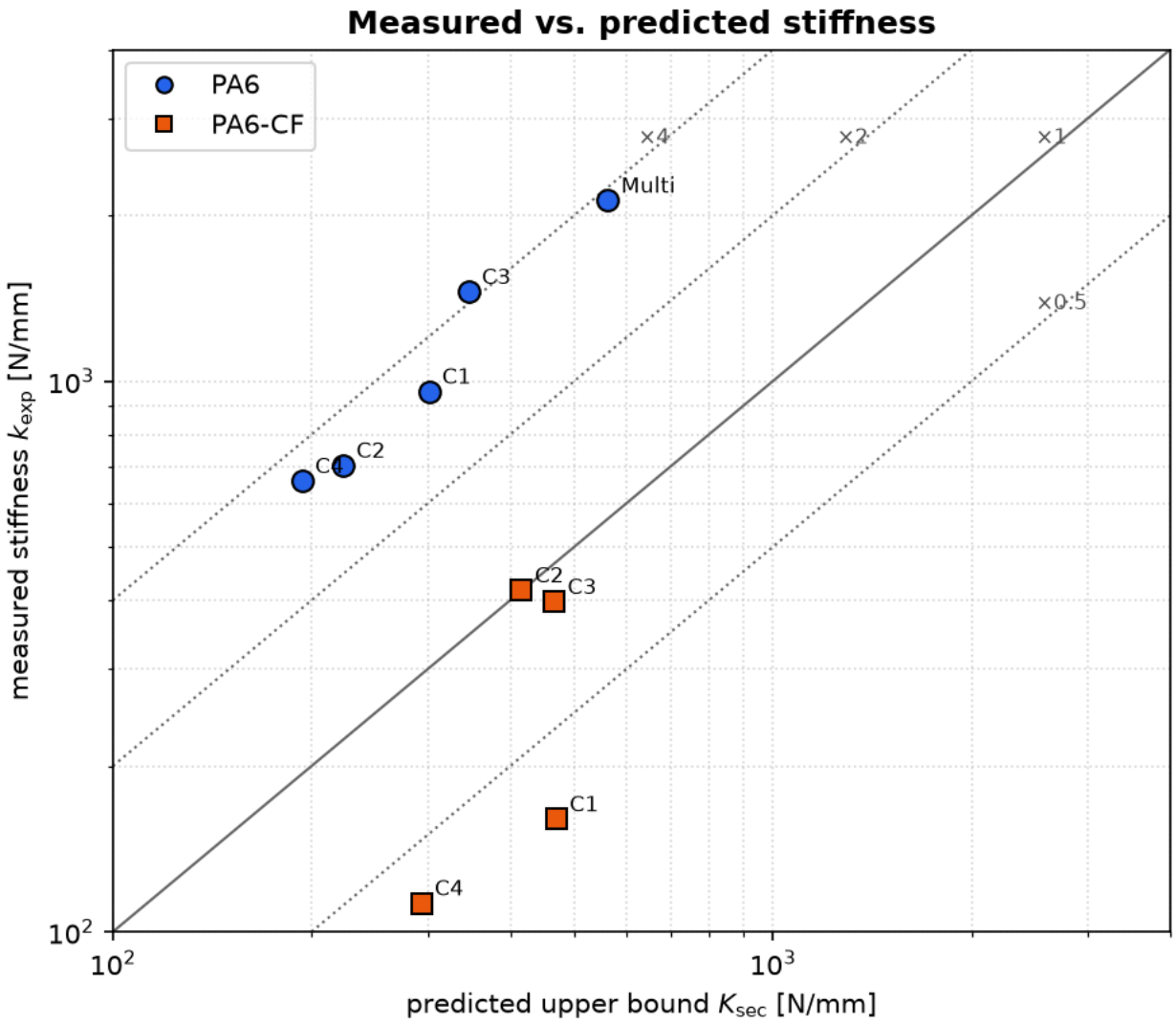


Figure 5.13.2: Measured stiffness against the sector prediction on logarithmic axes. The solid line indicates equality and dotted lines indicate constant ratios. The predicted upper bound presumes corresponding material, geometry and loading conditions.

5.13.2 Manufacturing Conditions and Scope of the Comparison

Figure 5.13.3 documents PA6-CF prints rejected before testing because of moisture-related printing defects. These pieces were excluded from the material characterization. The moduli, strengths and apparent density were obtained from other accepted prisms. The image therefore records a manufacturing problem, rather than the condition of the specimens used to determine the model constants.

The sectors were printed from the same spool in a separate campaign a few days later, with differences in moisture conditions during printing. Moisture was not measured. The rejected prints and the unsuccessful multi-load print establish a process-control limitation, but they do not quantify its effect on the accepted sectors. Moisture-related variability remains a

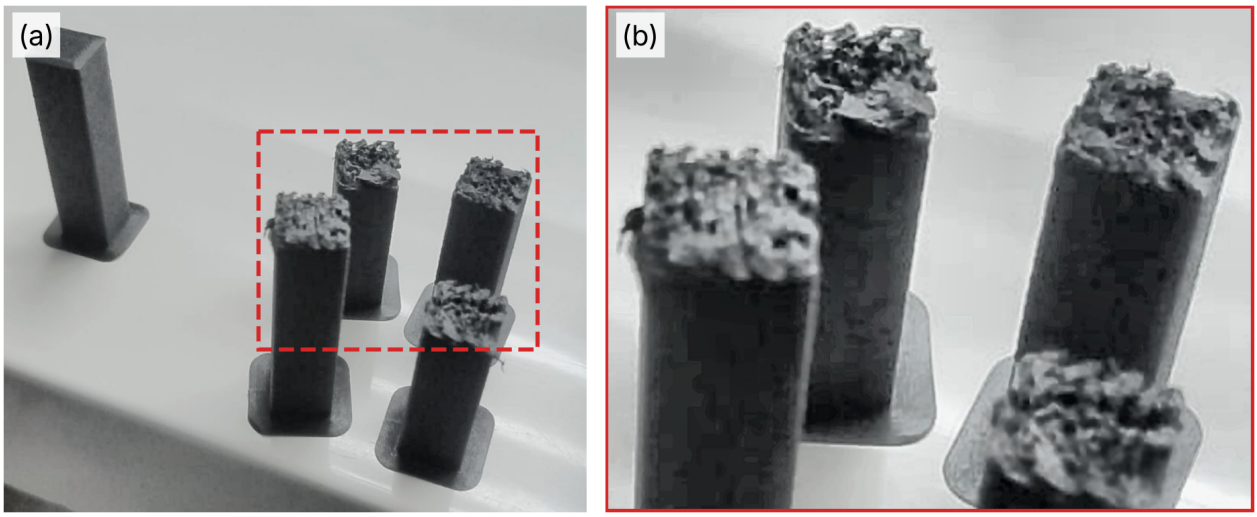


Figure 5.13.3: PA6-CF prints showing moisture-related defects: (a) overview and (b) detail. The defective pieces were rejected before testing and were not used to determine the material properties.

plausible contributor to the PA6-CF discrepancy alongside fit sensitivity and the limits of the material and contact representation.

For C1 to C4, the model predicts PA6-CF-to-PA6 stiffness ratios of 1.35 to 1.86, while the measurements give 0.17 to 0.59. Each material has its own optimized topology, so this reversal combines material, geometry and manufacturing effects. It does not measure the effect of carbon-fiber reinforcement at fixed geometry. The small test set also provides no repeatability estimate. The experiments support the PA6 design ranking and demonstrate load-bearing printed sectors, while the absolute stiffness and the PA6-CF design ranking remain unvalidated.

Chapter 6

Conclusions

This work developed a procedure for reducing the mass of a representative aeronautical solid of revolution while preserving its functional surfaces and checking its mechanical response. It addresses the design question introduced in Chapter 1 through a sequence of material characterization, topology optimization, numerical verification, geometric reconstruction and physical testing. The resulting framework provides a basis for studying how material can be redistributed under several loading conditions with accessible computational and manufacturing resources.

The theoretical foundations of Chapter 2 made this approach possible. Axisymmetry retains the hoop strain and the volume of revolution while reducing the analysis to the meridian section. Transverse isotropy incorporates the difference between the deposition plane and the build direction. Together, these choices allow the model to represent important features of the printed component at a lower computational cost than a general three-dimensional discretization, within the stated symmetry assumptions.

Specific Objective 1 was fulfilled through the Python framework developed in Chapter 3. The inner SIMP loop minimizes compliance at a prescribed volume fraction, while the outer search identifies the lightest feasible binary design evaluated under displacement and stress limits. Axisymmetric filtering and volume-weighted updates keep the optimization consistent with the solid of revolution. Independent re-evaluation supports the computed responses, and the closed STL exports connect the numerical design with fabrication. This integration allows candidate designs to be generated, compared and checked within the same workflow, although the search does not prove global optimality.

The application in Chapter 4 established the link between the model and the manufacturing process. Measured properties of printed PA6 and PA6-CF supplied the elastic moduli, strengths and apparent densities, while the build orientation was kept consistent between characterization and sector fabrication. Using these properties, finite element characterization of the original component fulfilled Specific Objective 2 by establishing baseline mass, maximum displacement and von Mises stress distribution. The later discrepancies also showed where coupon-based properties may not transfer directly to thin printed members.

The principal numerical result in Chapter 5 is the calculated mass reduction of 35.7% in PA6 and 61.9% in PA6-CF for the multi-load designs, which weigh 964.7 g and 537.0 g, respectively. Specific Objective 3 was fulfilled within the adopted numerical model: both designs satisfy the nodal-displacement and percentile-based von Mises stress limits under the four assumed load cases. Critical-wall displacement governs the selections, so the higher measured stiffness of PA6-CF permits greater material removal despite its lower compressive yield strength. These savings describe the modeled components and remain conditional on the adopted loads, supports and material representation.

Specific Objective 4 was fulfilled in its fabrication component through additive manufacturing of the selected Nylon sectors, whose designs satisfied the numerical displacement and stress limits for their intended load cases. Future work can extend the minimum-mass selection procedure by incorporating an efficiency index that explicitly quantifies stress uniformity and completing fabrication of the PA6-CF multi-load design.

Specific Objective 5 was fulfilled through comparison of numerical predictions with experimental response and assessment of manufacturing effects. Nine full-scale 7.5° sectors were tested. For PA6, the predicted stiffness ordering was preserved, although the measured stiffness exceeded the prediction by a mean factor of 3.54. PA6-CF showed greater variability and did not preserve the predicted ordering. These findings support the relative PA6 design response within the tested set, while the absolute stiffness predictions remain unvalidated. The tests also exposed the influence of fabrication conditions and the sensitivity of some fitted slopes to initial seating.

The main contribution is the resulting capability to study the complete path from material properties to a manufactured structural layout. The representative geometry and desktop fabrication establish a proof of concept for a broader investigation. Other solids of revolution can be studied by adapting the geometry, material constants, loads and functional limits within the same formulation. More demanding components would require further characterization and verification, but the present framework already provides a structured way to identify promising layouts, compare assumptions and define the experiments needed for their assessment.

6.1 Limitations and Future Work

The component was studied under assumed loads and mounting conditions because service requirements were unavailable. The two halves were represented as a continuous ring, and drawing tolerances were adopted as nodal-displacement allowances. Future applications should introduce measured or specified loads, realistic joints and supports, and a displacement budget that includes manufacturing deviations. Complete-component tests under those conditions would extend the evidence beyond the platen compression of free sectors.

The material model also requires further characterization. Poisson's ratios and the interlaminar shear modulus were estimated, while interlaminar tensile and shear strengths were not measured. The sensitivity study showed that small changes in the shear modulus can alter the feasibility of designs near their limits. Future work should measure these proper-

ties, document local strain and machine compliance, and examine how perimeters and infill affect thin members. These measurements would help separate material-model errors from manufacturing and test effects.

The experimental campaign was constrained by the high filament cost, long printing times and restricted access to the modified printer. Printing 7.5° sectors reduced material consumption and fabrication time while retaining the full-scale meridian geometry. One record per design, the absence of measured sector masses and the different supports of the physical test limit the conclusions about repeatability and service behavior. Additional specimens, solid reference sectors, mass measurements and dimensional inspection should therefore accompany future comparisons.

Manufacturing access was a further practical limitation. Commercial printing options for these specialized polyamides were limited in Medellín during the study, so fabrication depended on the modified desktop machine. Moisture caused rejected prints and prevented fabrication of the PA6-CF multi-load sector. Support removal also produced uneven surfaces. Changing the build orientation to simplify fabrication would change the material directions relative to the loads and require corresponding characterization and modeling. Future campaigns should therefore control filament conditioning and support removal while maintaining a documented relation between printing orientation and material properties.

Numerical improvements should address the small feasibility margins, search-path dependence and manufacturability of the reconstructed layouts. Conservative utilization targets and repeated searches would help assess robustness. Overhang and support-access constraints would make fabrication requirements part of the design process. Updated contact and geometric nonlinearity would improve the virtual compression test, whose present collapse estimates are indicative. Predicting the observed fracture would also require a suitable failure model and the strength measurements needed to calibrate it.

Within GRID at Universidad EAFIT, this work contributes to continuing research on technical materials and additive manufacturing. Funding secured for additional printers provides a basis for expanding local fabrication capacity and assessing the feasibility of reopening the student rocketry research group. This infrastructure would support more controlled campaigns and allow the present methodology to be developed alongside other projects in the group.

A longer-term direction is to pursue collaborations with air forces in different countries to evaluate representative components manufactured with these techniques. Such collaborations could provide defined service requirements, candidate parts and access to broader testing programs. Progress toward that application should connect the numerical framework with controlled manufacturing and component-level assessment. The present academic study supplies an initial method and experimental evidence for that development, while further work is needed to establish industrial performance.

6.1.1 Outputs of the Work

The two outputs of this work are this thesis and its associated project repository. The thesis documents the theoretical basis, implementation, application and results, together with the assumptions and limitations that determine their interpretation. It provides a reference for reproducing the design strategy and planning subsequent investigations.

The repository complements the document by providing a shared location for the computational work and its continued development. It supports review and reuse of the implementation and provides a basis for extending the framework to further studies within GRID. Together, the written record and the code preserve the methodology beyond the specific component examined here.

Bibliography

[1] M. P. Bendsøe and O. Sigmund, *Topology optimization: theory, methods, and applications*. Berlin ; New York: Springer, 2003.

[2] M. P. Bendsøe and N. Kikuchi, "Generating optimal topologies in structural design using a homogenization method," *Computer Methods in Applied Mechanics and Engineering*, vol. 71, no. 2, pp. 197–224, Nov. 1988. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/0045782588900862

[3] T. Tang, L. Wang, M. Zhu, H. Zhang, J. Dong, W. Yue, and H. Xia, "Topology Optimization: A Review for Structural Designs Under Statics Problems," *Materials*, vol. 17, no. 23, p. 5970, Dec. 2024. [Online]. Available: https://www.mdpi.com/1996-1944/17/23/5970

[4] M. P. Bendsøe, "Optimal shape design as a material distribution problem," *Structural Optimization*, vol. 1, no. 4, pp. 193–202, Dec. 1989. [Online]. Available: http://link.springer.com/10.1007/BF01650949

[5] Y. Xie and G. Steven, "A simple evolutionary procedure for structural optimization," *Computers & Structures*, vol. 49, no. 5, pp. 885–896, Dec. 1993. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/004579499390035C

[6] M. Y. Wang, X. Wang, and D. Guo, "A level set method for structural topology optimization," *Computer Methods in Applied Mechanics and Engineering*, vol. 192, no. 1-2, pp. 227–246, Jan. 2003. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0045782502005595

[7] G. Allaire, F. Jouve, and A.-M. Toader, "Structural optimization using sensitivity analysis and a level-set method," *Journal of Computational Physics*, vol. 194, no. 1, pp. 363–393, Feb. 2004. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S002199910300487X

[8] O. Sigmund and K. Maute, "Topology optimization approaches: A comparative review," *Structural and Multidisciplinary Optimization*, vol. 48, no. 6, pp. 1031–1055, Dec. 2013. [Online]. Available: http://link.springer.com/10.1007/s00158-013-0978-6

[9] D. Yago, J. Cante, O. Lloberas-Valls, and J. Oliver, "Topology Optimization Methods for 3D Structural Problems: A Comparative Study," *Archives of Computational*

Methods in Engineering, vol. 29, no. 3, pp. 1525–1567, May 2022. [Online]. Available: https://link.springer.com/10.1007/s11831-021-09626-2

[10] O. Sigmund, "A 99 line topology optimization code written in Matlab," *Structural and Multidisciplinary Optimization*, vol. 21, no. 2, pp. 120–127, Apr. 2001. [Online]. Available: http://link.springer.com/10.1007/s001580050176

[11] I. El Khadiri, M. Zemzami, N.-Q. Nguyen, M. Abouelmajd, N. Hmina, and S. Belhouideg, "Topology optimization methods for additive manufacturing: a review," *International Journal for Simulation and Multidisciplinary Design Optimization*, vol. 14, p. 12, 2023. [Online]. Available: https://www.ijsmdo.org/10.1051/smdo/2023015

[12] R. C, R. Shanmugam, M. Ramoni, and G. Bk, "A review on additive manufacturing for aerospace application," *Materials Research Express*, vol. 11, no. 2, p. 022001, Feb. 2024. [Online]. Available: https://iopscience.iop.org/article/10.1088/2053-1591/ad21ad

[13] X. Gao, S. Qi, X. Kuang, Y. Su, J. Li, and D. Wang, "Fused filament fabrication of polymer materials: A review of interlayer bond," *Additive Manufacturing*, vol. 37, p. 101658, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214860420310307

[14] M. Somireddy and A. Czekanski, "Mechanical Characterization of Additively Manufactured Parts by FE Modeling of Mesostructure," *Journal of Manufacturing and Materials Processing*, vol. 1, no. 2, p. 18, Nov. 2017. [Online]. Available: https://www.mdpi.com/2504-4494/1/2/18

[15] L. Zhu, N. Li, and P. Childs, "Light-weighting in aerospace component and system design," *Propulsion and Power Research*, vol. 7, no. 2, pp. 103–119, Jun. 2018. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2212540X18300191

[16] Jobanpreet Singh, K. Srivastawa, S. Jana, C. Dixit, and R. S, "Advancements in Lightweight Materials for Aerospace Structures: A Comprehensive Review," *Acceleron Aerospace Journal*, vol. 2, no. 3, pp. 173–183, Mar. 2024. [Online]. Available: https://acceleron.org.in/index.php/aaj/article/view/AAJ.11.2106-2409

[17] M. P. Bendsøe and O. Sigmund, *Topology Optimization*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2004. [Online]. Available: http://link.springer.com/10.1007/978-3-662-05086-6

[18] J.-S. Liu, G. Parks, and P. Clarkson, "Topology/shape optimisation of axisymmetric continuum structures ? a metamorphic development approach," *Structural and Multidisciplinary Optimization*, vol. 29, no. 1, pp. 73–83, Jan. 2005. [Online]. Available: http://link.springer.com/10.1007/s00158-004-0445-5

[19] C. Wang, C. Zhang, X. He, Y. Luo, H. Liu, and C. Yan, "Topology optimization of additively manufactured axisymmetric thermomechanical structures with self-supporting cavities," *Computers & Structures*, vol. 330, p. 108361, Sep. 2026. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0045794926002658

[20] Jailson Moraes Bastos Júnior, Julia Lopes Da Costa Abreu, Rodolfo Giacomim Mendes De Andrade, and Rafael Marin Ferro, "Development and Validation of Optimized Multiscale Structures in 3D-Printed Polymers: An Integrated Simulation and Experimental Approach," *Ibero-Latin American Congress on Computational Methods in Engineering (CILAMCE)*, vol. 5, Mar. 2026. [Online]. Available: https://publicacoes.softaliza.com.br/cilamce-2025/article/view/14519

[21] M. Kyun Sung, M. Schwerin, Y. Badhe, and D. Porter, "Influence of topology optimization parameters on the mechanical response of an additively manufactured test structure," *Journal of the Mechanical Behavior of Biomedical Materials*, vol. 142, p. 105844, Jun. 2023. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S1751616123001972

[22] S. Zhang and D. Zhao, Eds., *Aerospace Materials Handbook*, 0th ed. CRC Press, Apr. 2016. [Online]. Available: https://www.taylorfrancis.com/books/9781439873304

[23] F. Ferrari and O. Sigmund, "A new generation 99 line Matlab code for compliance topology optimization and its extension to 3D," *Structural and Multidisciplinary Optimization*, vol. 62, no. 4, pp. 2211–2228, Oct. 2020. [Online]. Available: https://link.springer.com/10.1007/s00158-020-02629-w

[24] P. Duysinx and M. P. Bendsøe, "Topology optimization of continuum structures with local stress constraints," *International Journal for Numerical Methods in Engineering*, vol. 43, no. 8, pp. 1453–1478, Dec. 1998. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/(SICI)1097-0207(19981230)43:8<1453::AID-NME480>3.0.CO;2-2

[25] C. Le, J. Norato, T. Bruns, C. Ha, and D. Tortorelli, "Stress-based topology optimization for continua," *Structural and Multidisciplinary Optimization*, vol. 41, no. 4, pp. 605–620, Apr. 2010. [Online]. Available: http://link.springer.com/10.1007/s00158-009-0440-y

[26] R. W. Clough and Y. Rashid, "Finite Element Analysis of Axi-Symmetric Solids," *Journal of the Engineering Mechanics Division*, vol. 91, no. 1, pp. 71–85, Feb. 1965. [Online]. Available: https://ascelibrary.org/doi/10.1061/JMCEA3.0000585

[27] B. Rankouhi, K. Bertsch, G. Meric De Bellefon, M. Thevamaran, D. Thoma, and K. Suresh, "Experimental validation and microstructure characterization of topology optimized, additively manufactured SS316L components," *Materials Science and Engineering: A*, vol. 776, p. 139050, Mar. 2020. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0921509320301386

[28] Y. W. Chun and E. J. Haug, "Shape Optimization of a Solid of Revolution," *Journal of Engineering Mechanics*, vol. 109, no. 1, pp. 30–46, Feb. 1983. [Online]. Available: https://ascelibrary.org/doi/10.1061/%28ASCE%290733-9399%281983%29109%3A1%2830%29

[29] C. Yan, H. Guo, B. Pei, H. Liu, Y. Chen, C. Wang, and Z. Yin, "Concurrent optimization of building direction and structural topology for multi-axis additive manufacturing of rotary parts considering anisotropic strength," *Additive Manufacturing*, vol. 109, p. 104851, Jul. 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2214860425002155

[30] H. W. Guo and H. H. Chen, "Topology optimization of aero-engine disks considering additive manufacturing-induced anisotropy," vol. 46, 2026. [Online]. Available: https://hkxb.buaa.edu.cn/EN/abstract/article_21345.shtml

[31] S. Li, S. Yuan, J. Zhu, C. Wang, J. Li, and W. Zhang, "Additive manufacturing-driven design optimization: Building direction and structural topology," *Additive Manufacturing*, vol. 36, p. 101406, Dec. 2020. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2214860420307788

[32] C. Wang, D. Lou, Z. Duan, W. Du, J. Rong, and B. Xu, "Structural topology optimization considering material anisotropy induced by additive manufacturing processes," *Advances in Engineering Software*, vol. 211, p. 104021, Dec. 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0965997825001590

[33] S. Xu, J. Huang, J. Liu, and Y. Ma, "Topology Optimization for FDM Parts Considering the Hybrid Deposition Path Pattern," *Micromachines*, vol. 11, no. 8, p. 709, Jul. 2020. [Online]. Available: https://www.mdpi.com/2072-666X/11/8/709

[34] R. D. Kundu and X. S. Zhang, "Additive manufacturing of stiff and strong structures by leveraging printing-induced strength anisotropy in topology optimization," *Additive Manufacturing*, vol. 75, p. 103730, Aug. 2023. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2214860423003433

[35] S. Grinde, "TOPOLOGY OPTIMIZATION FOR ADDITIVE MANUFACTURING USING SIMP METHOD," Master's thesis, Montana Tech, 2018. [Online]. Available: https://digitalcommons.mtech.edu/grad_rsch/158/

[36] O. Sigmund and J. Petersson, "Numerical instabilities in topology optimization: A survey on procedures dealing with checkerboards, mesh-dependencies and local minima," *Structural Optimization*, vol. 16, no. 1, pp. 68–75, Aug. 1998. [Online]. Available: http://link.springer.com/10.1007/BF01214002

[37] F. Wang, B. S. Lazarov, and O. Sigmund, "On projection methods, convergence and robust formulations in topology optimization," *Structural and Multidisciplinary Optimization*, vol. 43, no. 6, pp. 767–784, Jun. 2011. [Online]. Available: https://link.springer.com/10.1007/s00158-010-0602-y

[38] S. Shin, D. Shin, and N. Kang, "Topology optimization via machine learning and deep learning: a review," *Journal of Computational Design and Engineering*, vol. 10, no. 4, pp. 1736–1766, Jul. 2023. [Online]. Available: https://academic.oup.com/jcde/article/10/4/1736/7223974

[39] G. D. Cheng and X. Guo, "?-relaxed approach in structural topology optimization," *Structural Optimization*, vol. 13, no. 4, pp. 258–266, Jun. 1997. [Online]. Available: http://link.springer.com/10.1007/BF01197454

[40] K. Svanberg, "The method of moving asymptotes—a new method for structural optimization," *International Journal for Numerical Methods in Engineering*, vol. 24, no. 2, pp. 359–373, Feb. 1987. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/nme.1620240207

[41] P. Li, Y. Wu, and J. Yvonnet, "A SIMP-phase field topology optimization framework to maximize quasi-brittle fracture resistance of 2D and 3D composites," *Theoretical and Applied Fracture Mechanics*, vol. 114, p. 102919, Aug. 2021. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0167844221000276

[42] M. Langelaar, "Topology optimization of 3D self-supporting structures for additive manufacturing," *Additive Manufacturing*, vol. 12, pp. 60–70, Oct. 2016. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2214860416301294

[43] M. Khorasani, E. MacDonald, D. Downing, A. Ghasemi, M. Leary, J. Dash, E. Sharabian, A. Almalki, M. Brandt, and S. Bateman, "Multi Jet Fusion (MJF) of polymeric components: A review of process, properties and opportunities," *Additive Manufacturing*, vol. 91, p. 104331, Jul. 2024. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S2214860424003774

[44] S. Rosso, R. Meneghello, L. Biasetto, L. Grigolato, G. Concheri, and G. Savio, "In-depth comparison of polyamide 12 parts manufactured by Multi Jet Fusion and Selective Laser Sintering," *Additive Manufacturing*, vol. 36, p. 101713, Dec. 2020. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S221486042031085X

[45] H. Yu, H. Hong, S. Cao, and R. Ahmad, "Topology Optimization for Multipatch Fused Deposition Modeling 3D Printing," *Applied Sciences*, vol. 10, no. 3, p. 943, Feb. 2020. [Online]. Available: https://www.mdpi.com/2076-3417/10/3/943

[46] S. Shin, B. Goh, Y. Oh, and H. Chung, "Topology optimization for multi-axis additive manufacturing considering overhang and anisotropy," *International Journal of Mechanical Sciences*, vol. 301, p. 110443, Sep. 2025. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0020740325005284

[47] J. Sáez-Landete, H. Vargas-Vargas, P. Siegmann, and C. Camacho-Bello, "New validation metric for solid mechanics models," *Optics and Lasers in Engineering*, vol. 180, p. 108306, Sep. 2024. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0143816624002859

[48] Q. Q. Liang, Y. M. Xie, and G. P. Steven, "Optimal selection of topologies for the minimum-weight design of continuum structures with stress constraints," *Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science*, vol. 213, no. 8, pp. 755–762, Aug. 1999. [Online]. Available: https://journals.sagepub.com/doi/10.1243/0954406991522374

[49] B. Briseghella, L. Fenu, Y. Feng, C. Lan, E. Mazzarolo, and T. Zordan, "Optimization Indexes to Identify the Optimal Design Solution of Shell-Supported Bridges," *Journal of Bridge Engineering*, vol. 21, no. 3, p. 04015067, Mar. 2016. [Online]. Available: https://ascelibrary.org/doi/10.1061/%28ASCE%29BE.1943-5592.0000838

[50] R. M. Jones, *Mechanics of composite materials*, 2nd ed. Philadelphia, PA: Taylor & Francis, 1999.

[51] O. C. Zienkiewicz and R. L. Taylor, *The finite element method*, 5th ed. Oxford ; Boston: Butterworth-Heinemann, 2000.

[52] B. N. Turner, R. Strong, and S. A. Gold, "A review of melt extrusion additive manufacturing processes: I. Process design and modeling," *Rapid Prototyping Journal*, vol. 20, no. 3, pp. 192–204, Apr. 2014. [Online]. Available: http://www.emerald.com/rpj/article/20/3/192-204/373514

[53] S. Ahn, M. Montero, D. Odell, S. Roundy, and P. K. Wright, "Anisotropic material properties of fused deposition modeling ABS," *Rapid Prototyping Journal*, vol. 8, no. 4, pp. 248–257, Oct. 2002. [Online]. Available: http://www.emerald.com/rpj/article/8/4/248-257/363187

[54] J. F. Rodriguez, J. P. Thomas, and J. E. Renaud, "Characterization of the mesostructure of fused-deposition acrylonitrile-butadiene-styrene materials," *Rapid Prototyping Journal*, vol. 6, no. 3, pp. 175–186, Sep. 2000. [Online]. Available: http://www.emerald.com/rpj/article/6/3/175-186/367909

[55] R. P. Wool and K. M. O'Connor, "A theory crack healing in polymers," *Journal of Applied Physics*, vol. 52, no. 10, pp. 5953–5963, Oct. 1981. [Online]. Available: https://pubs.aip.org/jap/article/52/10/5953/10727/A-theory-crack-healing-in-polymers

[56] Q. Sun, G. Rizvi, C. Bellehumeur, and P. Gu, "Effect of processing conditions on the bonding quality of FDM polymer filaments," *Rapid Prototyping Journal*, vol. 14, no. 2, pp. 72–80, Mar. 2008. [Online]. Available: http://www.emerald.com/rpj/article/14/2/72-80/365935

[57] H. L. Tekinalp, V. Kunc, G. M. Velez-Garcia, C. E. Duty, L. J. Love, A. K. Naskar, C. A. Blue, and S. Ozcan, "Highly oriented carbon fiber–polymer composites via additive manufacturing," *Composites Science and Technology*, vol. 105, pp. 144–150, Dec. 2014. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0266353814003716

[58] L. J. Gibson and M. F. Ashby, *Cellular Solids: Structure and Properties*, 2nd ed. Cambridge University Press, May 1997. [Online]. Available: https://www.cambridge.org/core/product/identifier/9781139878326/type/book

[59] X. Liu and V. Shapiro, "Homogenization of material properties in additively manufactured structures," *Computer-Aided Design*, vol. 78, pp. 71–82, Sep. 2016. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/S0010448516300379

[60] M. Tomanik, M. Żmudzińska, and M. Wojtków, "Mechanical and Structural Evaluation of the PA12 Desktop Selective Laser Sintering Printed Parts Regarding Printing Strategy," *3D Printing and Additive Manufacturing*, vol. 8, no. 4, pp. 271–279, Aug. 2021. [Online]. Available: https://journals.sagepub.com/doi/10.1089/3dp.2020.0111

[61] X. Zheng, J. Meng, and Y. Liu, "Strain Rate Dependence of Compressive Mechanical Properties of Polyamide and Its Composite Fabricated Using Selective Laser Sintering under Saturated-Water Conditions," *Micromachines*, vol. 13, no. 7, p. 1041, Jun. 2022. [Online]. Available: https://www.mdpi.com/2072-666X/13/7/1041

[62] K. Goryl, A. Vodilka, and M. Kočiško, "Residual Stresses and Distortion in Material Extrusion Additive Manufacturing of Reinforced Thermoplastic Composites: A Review," *Polymers*, vol. 18, no. 15, p. 1796, Jul. 2026. [Online]. Available: https://www.mdpi.com/2073-4360/18/15/1796

[63] L. D. Landau and E. M. Lifšic, *Theory of elasticity*, 3rd ed., ser. Course of theoretical physics / L. D. Landau and E. M. Lifshitz. Amsterdam Heidelberg: Elsevier, Butterworth-Heinemann, 2009, no. 7.

[64] O. C. Zienkiewicz, "Variational principles in FEM," in *Introductory Lectures on the Finite Element Method*. Vienna: Springer Vienna, 1972, pp. 15–16. [Online]. Available: http://link.springer.com/10.1007/978-3-7091-2973-9_3

[65] B. M. Lempriere, "Poisson's ratio in orthotropic materials." *AIAA Journal*, vol. 6, no. 11, pp. 2226–2227, Nov. 1968. [Online]. Available: https://arc.aiaa.org/doi/10.2514/3.4974

[66] M. T. Huber, "Die Theorie der kreuzweise bewehrten Eisenbetonplatten nebst Anwendungen auf mehrere bautechnisch wichtige Aufgaben über rechteckige Platten," *Der Bauingenieur*, vol. 4, no. 12–13, pp. 354–360, 392–395, 1923.

[67] W. Sun, C. Rasmussen, R. Vetter, and J. Paulose, "Geometric mapping from rectilinear material orthotropy to isotropy: Insights into plates and shells," *Physical Review E*, vol. 108, no. 6, p. 065003, 2023. [Online]. Available: https://doi.org/10.1103/PhysRevE.108.065003

[68] R. C. Hibbeler, *Mechanics of materials*, tenth edition in si units, global edition ed. Harlow: Pearson, 2018.

[69] R. Hill, "A theory of the yielding and plastic flow of anisotropic metals," *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, vol. 193, no. 1033, pp. 281–297, May 1948. [Online]. Available: https://royalsocietypublishing.org/rspa/article/193/1033/281/6687/A-theory-of-the-yielding-and-plastic-flow-of

[70] M. L. Williams, "Stress Singularities Resulting From Various Boundary Conditions in Angular Corners of Plates in Extension," *Journal of Applied Mechanics*, vol. 19, no. 4, pp. 526–528, Dec. 1952. [Online]. Available: https://asmedigitalcollection.asme.org/appliedmechanics/article/19/4/526/1106855/Stress-Singularities-Resulting-From-Various

[71] P. K. Kythe and D. Wei, "Plane Elasticity," in *An Introduction to Linear and Nonlinear Finite Element Analysis*. Boston, MA: Birkhäuser Boston, 2004, pp. 269–296. [Online]. Available: http://link.springer.com/10.1007/978-0-8176-8160-9_11

[72] ——, "One-Dimensional Shape Functions," in *An Introduction to Linear and Nonlinear Finite Element Analysis*. Boston, MA: Birkhäuser Boston, 2004, pp. 25–36. [Online]. Available: http://link.springer.com/10.1007/978-0-8176-8160-9_2

[73] K.-J. Bathe and K.-J. Bathe, *Finite element procedures*. Englewood Cliffs, N.J: Prentice Hall, 1996.

[74] G. Strang and E. Herman, *Calculus: volume 2*. Houston, Texas: OpenStax, Rice University, 2016.

[75] B. Bourdin, "Filters in topology optimization," *International Journal for Numerical Methods in Engineering*, vol. 50, no. 9, pp. 2143–2158, Mar. 2001. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/nme.116

[76] M. P. Bendsøe and O. Sigmund, "Material interpolation schemes in topology optimization," *Archive of Applied Mechanics (Ingenieur Archiv)*, vol. 69, no. 9-10, pp. 635–654, Nov. 1999. [Online]. Available: http://link.springer.com/10.1007/s004190050248

[77] M. Stolpe and K. Svanberg, "An alternative interpolation scheme for minimum compliance topology optimization," *Structural and Multidisciplinary Optimization*, vol. 22, no. 2, pp. 116–124, Sep. 2001. [Online]. Available: http://link.springer.com/10.1007/s001580100129

[78] B. S. Lazarov and O. Sigmund, "Filters in topology optimization based on Helmholtz-type differential equations," *International Journal for Numerical Methods in Engineering*, vol. 86, no. 6, pp. 765–781, May 2011. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/nme.3072

[79] J. K. Guest, J. H. Prévost, and T. Belytschko, "Achieving minimum length scale in topology optimization using nodal design variables and projection functions," *International Journal for Numerical Methods in Engineering*, vol. 61, no. 2, pp. 238–254, Sep. 2004. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/nme.1064

[80] O. Sigmund, "Morphology-based black and white filters for topology optimization," *Structural and Multidisciplinary Optimization*, vol. 33, no. 4-5, pp. 401–424, Feb. 2007. [Online]. Available: http://link.springer.com/10.1007/s00158-006-0087-x

[81] K. Miettinen, *Nonlinear Multiobjective Optimization*, ser. International Series in Operations Research & Management Science, F. S. Hillier, Ed. Boston, MA: Springer US, 1998, vol. 12. [Online]. Available: http://link.springer.com/10.1007/978-1-4615-5563-6

[82] I. Das and J. E. Dennis, "A closer look at drawbacks of minimizing weighted sums of objectives for Pareto set generation in multicriteria optimization problems," *Structural Optimization*, vol. 14, no. 1, pp. 63–69, Aug. 1997. [Online]. Available: http://link.springer.com/10.1007/BF01197559

[83] D. C. Drucker, W. Prager, and H. J. Greenberg, "Extended limit design theorems for continuous media," *Quarterly of Applied Mathematics*, vol. 9, no. 4, pp. 381–389, Jan. 1952. [Online]. Available: https://www.ams.org/qam/1952-09-04/S0033-569X-1952-45573-2/

[84] D. Mackenzie and J. Boyle, "A method of estimating limit loads by iterative elastic analysis. I—Simple examples," *International Journal of Pressure Vessels and Piping*, vol. 53, no. 1, pp. 77–95, Jan. 1992. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/0308016193901053

[85] H. Hencky, "Zur Theorie plastischer Deformationen und der hierdurch im Material hervorgerufenen Nachspannungen," *ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik*, vol. 4, no. 4, pp. 323–334, Jan. 1924. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/zamm.19240040405

[86] J. O'Rourke, *Computational Geometry in C*, 2nd ed. Cambridge University Press, Oct. 1998. [Online]. Available: https://www.cambridge.org/core/product/identifier/9780511804120/type/book

[87] W. E. Lorensen and H. E. Cline, "Marching cubes: A high resolution 3D surface construction algorithm," *ACM SIGGRAPH Computer Graphics*, vol. 21, no. 4, pp. 163–169, Aug. 1987. [Online]. Available: https://dl.acm.org/doi/10.1145/37402.37422

[88] D. H. Douglas and T. K. Peucker, "ALGORITHMS FOR THE REDUCTION OF THE NUMBER OF POINTS REQUIRED TO REPRESENT A DIGITIZED LINE OR ITS CARICATURE," *Cartographica*, vol. 10, no. 2, pp. 112–122, Dec. 1973. [Online]. Available: https://utppublishing.com/doi/10.3138/FM57-6770-U75U-7727

[89] Flashforge, "Adventurer 5M 3D Printer." [Online]. Available: https://flashforge.com/products/adventurer-5m-3d-printer

[90] Phaetus, "Conch Hotend M6 for Flashforge AD5M/AD5M Pro, 0.4 mm." [Online]. Available: https://www.3djake.com/phaetus/conch-hotend-m6-for-flashforge-ad5m-ad5m-pro

[91] BIQU, "CryoGrip Pro Glacier Buildplate, 235 x 235 mm, for Adventurer 5M." [Online]. Available: https://biqu.equipment/collections/heatbed/products/biqu-cryogrip-pro-glacier-buildplate-for-voron-300-anycubic-kobra-3

[92] Inslogic, "Inslogic PA6-CF Filament 1.75 mm." [Online]. Available: https://store.inslogic3d.com/products/pa6-cf-filament

[93] ——, "Inslogic Nylon PA6/66 Filament 1.75 mm." [Online]. Available: https://store.inslogic3d.com/products/nylon-pa6-filament

[94] UltiMaker, "MakerBot METHOD X 3D Printer, Carbon Fiber Edition." [Online]. Available: https://www.3dprintersdepot.com/products/makerbot-method-x-3d-printer-carbon-fiber-edition

[95] Aniwaa, "Markforged Mark Two (Gen 2)." [Online]. Available: https://www.aniwaa.com/product/3d-printers/markforged-mark-two/

[96] M. Shirinbayan, K. Benfriha, M. Ahmadifar, C. Penavayre, S. Nouira, and J. Fitoussi, "Effects of hygrothermal aging on the physicochemical and mechanical properties of 3D-printed PA6," *The International Journal of Advanced Manufacturing Technology*, vol. 131, no. 9-10, pp. 4811–4823, Apr. 2024. [Online]. Available: https://link.springer.com/10.1007/s00170-024-13323-2

[97] ASTM International, "Standard Test Method for Tensile Properties of Plastics," ASTM International, West Conshohocken, PA, Tech. Rep. ASTM D638-22, 2022.

[98] ——, "Standard Test Method for Compressive Properties of Rigid Plastics," ASTM International, West Conshohocken, PA, Tech. Rep. ASTM D695-26, 2026.

[99] Goodfellow, "Polyamide - Nylon 6, material properties." [Online]. Available: https://www.goodfellow.com/usa/nylon-6-powder-group

[100] I. Vindokurov, D. Sporysheva, and M. Tashkinov, "Effective poisson's ratio of FDM-printed thermoplastics: influence of layup orientation and comparative evaluation using optical measurements," *The International Journal of Advanced Manufacturing Technology*, Sep. 2026. [Online]. Available: https://link.springer.com/10.1007/s00170-026-19121-2